

1 The health-relevant architecture of the 2 everyday light exposome

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34 Abstract

35 Personal light exposure has been associated with sleep and non-communicable diseases,
36 yet how ocular exposure varies across the day, among people and between everyday
37 settings remains poorly understood. We analysed 191 adults at nine sites in seven countries;
38 184 contributed 1,478 participant-days (near eye: 141 adults, 816 days; complementary
39 chest: 154 adults, 902 days). The shared 24-hour near-eye pattern accounted for 78.9% of
40 full-model R^2 , participant patterns 12.9%, participant-day shifts 6.2% and site patterns 2.0%.
41 Reported light source and immediate setting produced large contrasts, accounting for 24.2%
42 and 30.7%, respectively, of variation in fitted hourly patterns, the next-largest shares after
43 time of day. Associations with site, day length and personal characteristics differed across
44 exposure measures. Only 24.0% of daytime minutes met the recommendation. By resolving
45 variation across sites, people, days and immediate settings, this study lays an empirical
46 foundation for targeted, harmonised exposure monitoring aligned with health-based light
47 recommendations.

48 Introduction

49 Light reaching the eye supports vision and signals circadian timing, sleep, alertness and
50 neuroendocrine responses^{1–5}. Melanopic equivalent daylight illuminance (melanopic EDI)
51 weights illuminance for melanopsin-related sensitivity and is most relevant to these effects^{6,7}.
52 Expert consensus recommends at least 250 lx melanopic EDI at the eye during daytime, no
53 more than 10 lx during the three hours before sleep and no more than 1 lx in the sleep
54 environment for healthy adults with regular daytime schedules⁸. These physiology-based
55 ranges make daily light a health-relevant exposure, but are not individual risk functions.

56 Population studies and reviews associate personal light patterns with sleep and major non-
57 communicable diseases⁹. Wrist-sensor studies link light timing and day-night distribution with
58 sleep timing and sleepiness, psychiatric disorders, incident type 2 diabetes, cardiovascular
59 disease and mortality^{10–15}. Other cohorts link evening, nocturnal or seasonal light with sleep
60 in bipolar disorder and with obesity, diabetes, hypertension and metabolic health^{16–19}. The
61 exposome framework characterises non-genetic exposures across the life course^{20,21}, yet
62 rarely includes time-resolved ocular melanopic light, the everyday component examined
63 here. These studies establish population relevance but mostly measured broad-spectrum
64 wrist or bedside light, providing only proxy evidence for melanopic ocular exposure.

65 An everyday exposure baseline against these recommendations is missing. Personal
66 exposure to natural and electric light follows an expected 24-hour rhythm, including in large
67 wrist-sensor cohorts²², but previous studies focus on particular populations, occupations or
68 settings^{23–25}. Its amplitude and timing across sites and countries, and the contributions of
69 time, site, individuals, day-to-day variation and immediate context, remain uncertain.
70 Daylight availability, satellite estimates and fixed-site measurements cannot resolve these
71 layers as people move through buildings, transport, outdoor spaces and sleep
72 environments²⁶. Sensor position also matters: near-eye sensing is closer to the incident
73 ocular field but is not a retinal measurement; wrist and chest sensors capture related fields
74 shaped by posture, wear and occlusion^{27–33}.

75 Using the harmonised MeLiDos protocol, we measured personal light across nine sites in
76 seven countries^{34,35}. The near-eye sensor characterised ocular light exposure during wear,
77 alongside diaries, light-source and immediate-setting reports, sleep information and
78 questionnaires. The complementary chest sensor provided non-ocular environmental
79 evidence and enabled participation among people reluctant to wear the glasses-mounted
80 sensor. We sought a multisite recommendation-based baseline, assessing temporal,
81 environmental and routine contributions alongside selected person-level correlates. We use

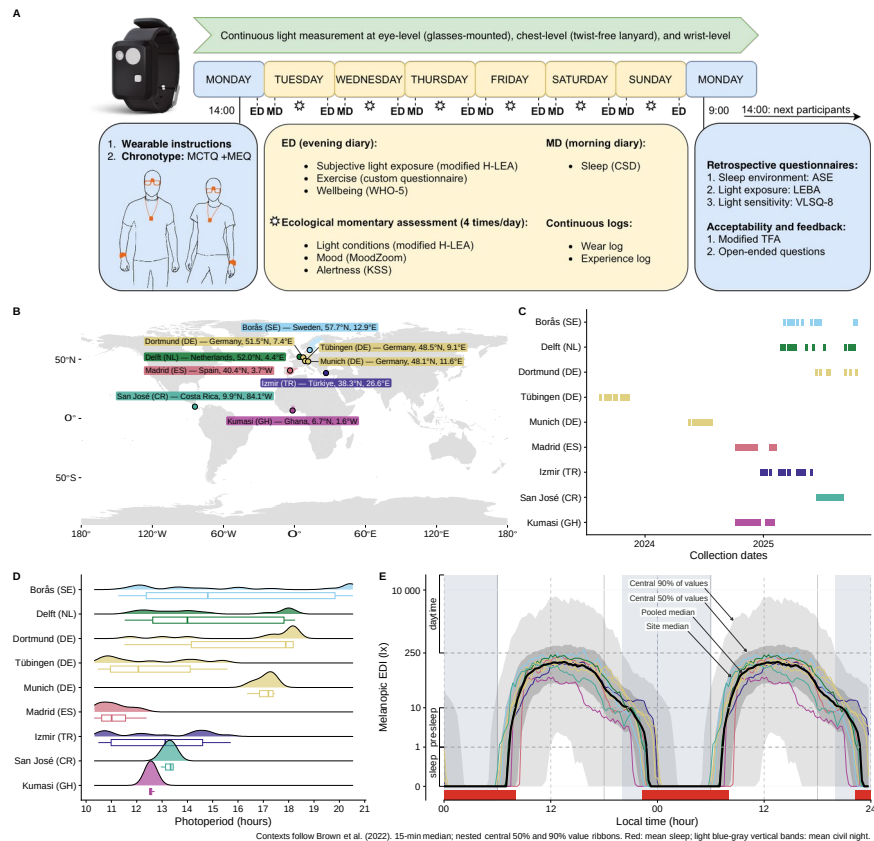
82 architecture to describe how everyday light exposure is organised across time, sites, people,
83 days and immediate settings, and benchmark this component of the exposome against
84 health-based light recommendations. Although not global, the protocol enables harmonised
85 measurement elsewhere, progressively closing gaps across locations, civil photoperiods,
86 latitudes and climates and building towards a global account of ocular light exposure.

87 Results

88 A multisite baseline and recommendation adherence

89 The normalised roster comprised 191 participants across Borås (SE), Delft (NL), Dortmund
90 (DE), Tübingen (DE), Munich (DE), Madrid (ES), Izmir (TR), San José (CR) and Kumasi
91 (GH). Of these, 184 contributed 1,478 participant-days with recorded, non-all-zero light data
92 at either position before quality screening. After screening, the near-eye dataset contained
93 141 participants and 816 participant-days, the complementary chest dataset 154 and 902,
94 and the common sample 112 participants and 643 participant-days. Near-eye
95 measurements provide ocular-exposure evidence during wear, whereas chest
96 measurements are non-ocular; both describe the bedside sleep environment during reported
97 sleep. [Supplementary Table S1](#) gives the sample flow.

98 The pooled near-eye record followed the expected 24-hour rhythm: low overnight exposure,
99 a morning rise, higher daytime values and an evening decline. Seventeen complementary
100 metrics captured thematic streams rather than one score. Level metrics described daily
101 geometric mean melanopic EDI and the brightest and darkest supported 10-hour means;
102 duration and timing metrics quantified exposure ranges, sustained bright periods, bright or
103 dark windows and threshold crossings. Dose integrated intensity, regularity captured stability
104 and fragmentation, and melanopic daylight efficacy (MDER) described spectral composition
105 relative to visual illuminance. These features are not interchangeable health-response
106 estimates. [Supplementary Table S2](#) and [Supplementary Figure S1](#) give the metric dictionary
107 and site distributions; [Supplementary Figure S2](#) shows a worked derivation. [Figure 1](#) and
108 [Table 1](#) summarise the design and data structure.



109

110 **Figure 1: Five-panel overview of the eight-day protocol, sensor positions, nine**
 111 **country-coded sites, collection windows, civil photoperiod, and the pooled near-eye**
 112 **daily profile. **A**, Eight-day field protocol and sensor positions. The wrist logger is**
 113 **shown as a protocol device but is not an exposure channel analysed in this**
 114 **manuscript. **B**, Nine sites in seven countries. **C**, Site collection periods. **D**, Observed**
 115 **civil photoperiod. **E**, Site and pooled 15-minute median near-eye melanopic**
 116 **equivalent daylight illuminance profiles from 816 participant-days contributed by 141**
 117 **participants, repeated over 48 hours for display only. Grey ribbons show the central**
 118 **50% and 90% of measured values, not confidence intervals; coloured lines are site**
 119 **medians, the black line is the pooled median, and red and blue shading show mean**
 120 **reported sleep and civil night (sun elevation < -6°). During reported sleep, the**
 121 **measurement represents the bedside sleep environment.**

122

123 Table 1: Participant and measurement characteristics overall and by country-coded study site. The table distinguishes the 191-person roster, 124 participants and participant-days with recorded light data, and the screened near-eye and complementary chest samples. Model-specific samples 125 are reported with each analysis. Of the 191 roster participants, 184 contributed the 1,478 recorded participant-days. Continuous participant 126 characteristics are median with the stated percentile range.

	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)
Site specifics										
Institution	Overall	RISE	THUAS	BAuA	MPI	TUM	FUSPCEU	IZTECH	UCR	KNUST
Country	—	Sweden	Netherlands	Germany	Germany	Germany	Spain	Türkiye	Costa Rica	Ghana
City	—	Borås	Delft	Dortmund	Tübingen	Munich	Madrid	Izmir	San José	Kumasi
Coordinates	—	57.7°N, 12.9°E	52.0°N, 4.4°E	51.5°N, 7.4°E	48.5°N, 9.1°E	48.1°N, 11.6°E	40.4°N, 3.7°W	38.3°N, 26.6°E	9.9°N, 84.1°W	6.7°N, 1.6°W
Data collection										
Participants ¹	191 roster (g:143; c:157; p:116)	17 roster (g:14; c:17; p:14)	20 roster (g:13; c:15; p:13)	24 roster (g:19; c:22; p:19)	26 roster (g:26; c:0; p:0)	10 roster (g:10; c:10; p:10)	23 roster (g:23; c:22; p:22)	17 roster (g:17; c:17; p:17)	39 roster (g:6; c:39; p:6)	15 roster (g:15; c:15; p:15)
Participant-days ¹	1478 roster (g:1134 c:1246; p:902)	137 roster (g:107 c:137; p:107)	125 roster (g:107 c:124; p:106)	176 roster (g:145 c:163; p:132)	208 roster (g:208 c:0; p:0)	80 roster (g:80 c:80; p:80)	182 roster (g:182 c:174; p:174)	138 roster (g:138 c:138; p:138)	312 roster (g:48 c:312; p:48)	120 roster (g:119 c:118; p:117)
Participant time ²	116w 4d	11w 23h	11w 23h	15w 2d	21w 3d	8w 4d	18w 3d	14w 3d	4w 4d	11w 4d
Declared non-wear ²	g:4w 1d (3.66%)	g:2d 14h (3.34%)	g:3d 1h (3.92%)	g:3d 15h (3.42%)	g:6d 10h (4.30%)	g:2d 17h (4.55%)	g:2d 20h (2.20%)	g:3d 3h (3.13%)	g:1d 11h (4.66%)	g:3d 21h (4.80%)
Screened days ^{1,3}	g:816 c:902	g:78 c:96	g:78 c:93	g:107 c:114	g:150 c:0	g:60 c:60	g:129 c:123	g:101 c:102	g:32 c:230	g:81 c:84
Civil photoperiod ^{4,2}	13.25 h (10.62–18.20) (n=1478 d)	14.81 h (11.80–20.49) (n=137 d)	14.00 h (11.88–18.09) (n=125 d)	17.88 h (11.76–18.22) (n=176 d)	12.07 h (10.55–15.41) (n=208 d)	17.20 h (16.45–17.45) (n=80 d)	11.01 h (10.35–12.22) (n=182 d)	13.13 h (10.52–15.57) (n=138 d)	13.34 h (13.01–13.47) (n=312 d)	12.54 h (12.49–12.68) (n=120 d)

Table continued

	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)
Participant information										
Age ⁴	28 y (21 y–52 y) (n=191 N)	38 y (22.6 y–64.8 y) (n=17 N)	30.5 y (19.9 y–57.1 y) (n=20 N)	35.5 y (20.1 y–58.5 y) (n=24 N)	27 y (22 y–37 y) (n=26 N)	27.5 y (22.2 y–29 y) (n=10 N)	31 y (20.2 y–54.3 y) (n=23 N)	24 y (21 y–29.6 y) (n=17 N)	34 y (22 y–48.1 y) (n=39 N)	23 y (20.1 y–25 y) (n=15 N)
Sex ⁵	F:105 / M:86 (n=191)	F:6 / M:11 (n=17)	F:8 / M:12 (n=20)	F:13 / M:11 (n=24)	F:14 / M:12 (n=26)	F:6 / M:4 (n=10)	F:15 / M:8 (n=23)	F:11 / M:6 (n=17)	F:24 / M:15 (n=39)	F:8 / M:7 (n=15)
Employment status ⁶	F:163 / P:26 / N:2 (n=191)	F:16 / P:0 / N:1 (n=17)	F:11 / P:9 / N:0 (n=20)	F:21 / P:3 / N:0 (n=24)	F:23 / P:3 / N:0 (n=26)	F:9 / P:1 / N:0 (n=10)	F:21 / P:2 / N:0 (n=23)	F:15 / P:2 / N:0 (n=17)	F:34 / P:5 / N:0 (n=39)	F:13 / P:1 / N:1 (n=15)
Chronotype group ⁷	m:62 / i:94 / e:30 (n=186)	m:8 / i:4 / e:5 (n=17)	m:3 / i:9 / e:3 (n=15)	m:8 / i:13 / e:3 (n=24)	m:9 / i:9 / e:8 (n=26)	m:0 / i:8 / e:2 (n=10)	m:5 / i:17 / e:1 (n=23)	m:2 / i:10 / e:5 (n=17)	m:16 / i:20 / e:3 (n=39)	m:11 / i:4 / e:0 (n=15)
Sleep-corrected midsleep on free days ^{4,8}	04:03 (02:08–06:15) (n=185 N)	02:55 (02:04–05:09) (n=17 N)	04:31 (03:05–05:37) (n=15 N)	03:47 (02:17–04:49) (n=24 N)	04:33 (02:41–06:04) (n=26 N)	05:08 (04:12–06:04) (n=10 N)	04:48 (03:10–06:52) (n=22 N)	05:29 (03:35–06:49) (n=17 N)	03:42 (02:17–05:31) (n=39 N)	02:19 (01:22–03:25) (n=15 N)

¹ g, near-eye glasses position; c, complementary chest position; p, paired common sample in which the same participant-day is available at both positions.

² w: weeks; d: days; h: hours; min: minutes. Participant time and declared non-wear use eligible real minutes on screened days; both are shown for near eye only.

³ Screened days are the final participant-days after the 80% completeness screen and exclusion of exact-all-zero melEDI days.

⁴ Median (5th percentile, 95th percentile).

⁵ Sex categories: Female and Male.

⁶ Employment categories: Full/studying, Part/marginal, and Not employed.

⁷ Chronotype groups follow the Morningness–Eveningness Questionnaire score.

⁸ MCTQ: Munich Chronotype Questionnaire; MEQ: Morningness–Eveningness Questionnaire. Clock summaries are circular.

129 Exposure was within the applicable recommendation during 24.0% of daytime minutes,
130 63.3% of pre-sleep minutes and 87.7% of sleep minutes. This represented 137,792 of
131 573,712 waking daytime minutes, 81,894 of 129,390 pre-sleep minutes and 336,052 of
132 383,366 minutes during reported sleep. The sleep quantity describes the bedside sleep
133 environment because the device was not worn at the eye; the Discussion addresses low-
134 light measurement accuracy. [Supplementary Table S3](#) gives exact numerators,
135 denominators and complementary ranges.

136 These pooled-minute percentages describe how often exposure lay within range, not
137 participant adherence. Model-based comparisons retained each window's adherent and
138 valid-minute counts.

139 The adherence model included 2,298 periods from 140 participants and 794 cycles
140 (1,043,192 valid minutes), linking preceding sleep, daytime and following pre-sleep to the
141 wake-start date's day type; windows qualified independently. Free-versus-work adherence
142 was 5.1 percentage points lower during daytime (95% confidence interval [95% CI] 2.4 to 7.8
143 lower), 5.9 higher pre-sleep (1.4 to 10.5) and 6.5 lower during sleep (3.8 to 9.1). At least
144 80% coverage preserved daytime and sleep interval conclusions, but pre-sleep was
145 inconclusive (4.1 points, 95% CI -0.9 to 9.1). Temporal dependence remained unresolved.

146 The window-by-site-by-day-type interaction was retained (FDR-adjusted $p = 0.008$), with site
147 variation during daytime and sleep (both $p = 0.002$), not detected pre-sleep ($p = 0.818$).
148 Relative to the site-average free-minus-work contrast, Dortmund (DE) was 14.8 points higher
149 during daytime (95% CI 6.7 to 22.9; FDR-adjusted $p = 0.004$), Madrid (ES) 9.2 lower (-15.6
150 to -2.8; $p = 0.044$), and Kumasi (GH) 6.3 higher during sleep (3.5 to 9.1; $p < 0.001$). These
151 departures met a separate 27-test FDR criterion and retained directions and interval
152 conclusions at 80% coverage.

153 On the design-standardized response scale, marginal R^2 was 58.0% and conditional R^2
154 60.2%; the participant increment was 2.3 percentage points and observation/distribution
155 variation 39.8%. Shapley allocated 94.2% of marginal R^2 to recommendation window, 4.8%
156 to site and 1.0% to day type. Site's 2.8-point allocation was about 1.2 times the participant-
157 intercept increment and about five times the 0.6-point day-type allocation. Allocations are
158 point-only; the window share partly reflects different thresholds. [Table 2](#) and [Supplementary](#)
159 [Figures S4](#) and [S5](#) give adherence estimates.

160 Table 2: Recommendation adherence by Brown et al. recommendation window and day type. Observed valid-minute fractions are descriptive;
 161 work-day and free-day estimates and their difference are model-based, with 95% confidence intervals and false-discovery-rate-adjusted p-values.
 162 Each of nine sites receives equal weight in the model estimates. Daytime, Pre-sleep and Sleep identify the recommendation windows: daytime
 163 excludes the three hours before reported sleep, pre-sleep comprises those three hours, and sleep describes the bedside sleep environment. The
 164 wake-start date assigns the day type to the preceding sleep, daytime and following pre-sleep windows; each window qualifies independently. Only
 165 the primary any-valid sample is shown: 2,298 periods, 140 participants, 794 cycles and 1,043,192 valid minutes. The 80% coverage sensitivity
 166 remains reported in the Results and Methods; its pre-sleep interval includes zero, and temporal dependence remains unresolved. Confidence-
 167 interval exclusion and FDR decisions are separate summaries. Descriptive pooled-minute fractions are detailed in Supplementary Table S3.
 168 Adherence does not classify participants.

Recommendation adherence by recommendation window and day type

Recommendation window	Recommendation	Observed pooled-minute adherence	Site-average recommendation-window model			
		Valid minutes meeting recommendation, n/N (%)	Work-day adherence % (95% CI)	Free-day adherence % (95% CI)	Free minus Work percentage points (95% CI)	FDR-adjusted p
Daytime	At least 250 lx melanopic EDI during wake, excluding the three hours before sleep	137,792/573,712 (24.0%)	26.6 (24.5 to 28.7)	21.5 (19.2 to 23.8)	-5.1 (-7.8 to -2.4)	<0.001
Pre-sleep	No more than 10 lx melanopic EDI during the three hours before reported sleep	81,894/129,390 (63.3%)	62.0 (58.8 to 65.2)	67.9 (64.2 to 71.6)	+5.9 (+1.4 to +10.5)	0.011
Sleep	No more than 1 lx melanopic EDI during reported sleep; bedside sleep environment	336,052/383,366 (87.7%)	90.4 (88.9 to 91.9)	83.9 (81.4 to 86.5)	-6.5 (-9.1 to -3.8)	<0.001

Observed pooled-minute fractions and fitted recommendation-window estimates are different quantities. Each of the nine sites contributes equally to the fitted estimates; the displayed model results use only the primary any-valid sample.

The recommendation boundaries are inclusive: daytime ≥250 lx, pre-sleep ≤10 lx and sleep ≤1 lx melanopic EDI. Near-eye measurements describe ocular exposure during wear; reported sleep describes the bedside sleep environment, not worn ocular exposure. Chest measurements are not pooled here.

Day type is assigned from the wake-start date to the preceding sleep, daytime and following pre-sleep windows; windows qualify independently. Temporal dependence remained unresolved. The 80% coverage sensitivity, including the inconclusive pre-sleep interval, is reported in the Results and Methods, not as additional rows here.

169

170 Exploratory cross-window within-participant contrasts per 10-percentage-point difference in
171 daytime adherence were -0.26 points for sleep (95% CI -0.95 to 0.43) and -1.09 for pre-
172 sleep (-2.75 to 0.56). Unresolved temporal dependence precluded a within-participant claim;
173 intervals including zero do not establish absence. Between participants, 10 points higher
174 average daytime adherence was associated with 2.52 points lower sleep adherence (1.48 to
175 3.55 lower; FDR-adjusted $p < 0.001$) and 3.59 lower pre-sleep adherence (1.25 to 5.94
176 lower; $p = 0.005$), consistent with higher evening and nighttime exposure during monitoring.
177 Both met the four-effect FDR criterion; directions and interval conclusions persisted at 80%
178 coverage (maximum change 0.60 points). [Supplementary Table S4](#) gives estimates and
179 [Supplementary Figure S6](#) participant profiles.

180 **Personal and day-to-day differences exceed site** 181 **differences**

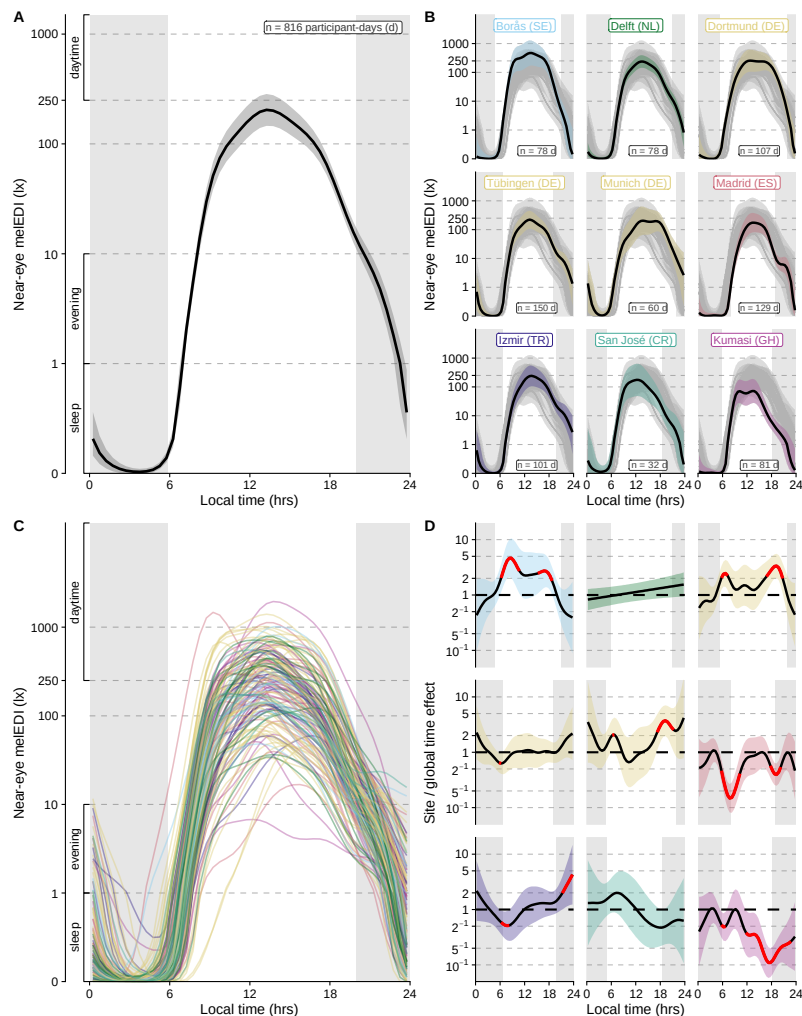
182 A nonlinear model decomposed personal light exposure into common local-clock, site and
183 participant curves, and participant-day shifts. Near-eye in-sample R^2 was 0.775. Participant
184 curves were 1.80 times as dispersed as site curves (95% CI 1.16 to 4.33); participant curves
185 plus day shifts had a dispersion ratio of 1.99 (1.29 to 4.77). After accounting for time of day,
186 fitted variation among people and their days within sites exceeded variation among sites.

187 Shares of full-model R^2 showed the same hierarchy: 78.9% for the shared local-clock
188 pattern, 12.9% for participant patterns, 6.2% for participant-day shifts and 2.0% for site
189 patterns. The participant-pattern share was 6.41 times the site-pattern share (3.85 to 15.14),
190 and the combined participant-pattern and participant-day-shift share was 9.49 times the site-
191 pattern share (5.77 to 21.81).

192 The shared rhythm did not imply uniform amplitude. Relative to the fitted site-average daily
193 pattern, Borås (SE) ratios were 2.04 to 4.64 from 06:00 to 11:00 and 1.80 to 2.73 from 15:00
194 to 19:00. Madrid (ES) ratios were 0.15 to 0.47 from 05:30 to 10:30 and 0.40 to 0.57 from
195 17:30 to 20:30. Izmir (TR) shifted from 0.51 to 0.59 at 06:00 to 08:30 to 1.98 to 4.28 at 21:00
196 to 24:00, while Kumasi (GH) was 0.11 to 0.38 from 11:30 to 23:00. These are clock-specific
197 departures from the average curve.

198 The chest model had an in-sample R^2 of 0.737. Of this full-model R^2 , common time
199 accounted for 79.1%, participant patterns 11.9%, participant-day shifts 7.0% and site
200 patterns 2.1%. Common-sample comparisons supported the combined participant-plus-day
201 ordering, but the participant-only contrast was less stable. Cross-position similarity is

202 complementary context, not replication, equivalence or interchangeability. [Figure 2](#) shows
203 the fitted layers; [Supplementary Tables S5](#) and [S6](#) give both decompositions.



204

205 **Figure 2: Multiscale daily pattern of near-eye melanopic equivalent daylight**
 206 **illuminance. A, Site-average daily curve. B, Fitted site curves. C, Fitted participant**
 207 **curves after the shared local-clock and site patterns were considered. D, Ratio of**
 208 **each fitted site curve to the site-average curve on the melanopic equivalent daylight**
 209 **illuminance plus 0.1 lx scale. Grey bands show mean civil night. Shaded bands are**
 210 **clock-specific 95% confidence intervals; red segments identify displayed half-hours**
 211 **whose interval did not include the site-average curve and are descriptive clock-**
 212 **specific contrasts. The analysis included 141 participants and 816 participant-days**
 213 **across nine sites. The label “global time effect” denotes the site-average local-clock**
 214 **curve (panel A).**

215 **Geography and photoperiod provide partial context**

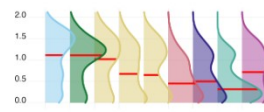
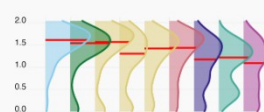
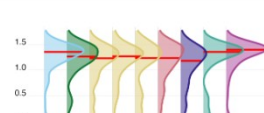
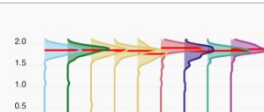
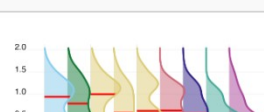
216 Site associations were stream-specific. Site differences met the FDR criterion for 8 of 17
 217 near-eye metrics: all three level metrics, waking time above 250 lx, the brightest- and
 218 darkest-10-hour midpoints, and the mean and last timing above 250 lx. They were not FDR-
 219 retained for dose, spectral composition, stability, fragmentation, sustained bright exposure,
 220 the other threshold-duration metrics or first timing above 250 lx. Across metric models,
 221 conditional R^2 averaged 36.4%; participant-associated R^2 averaged 25.6% among

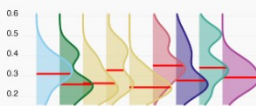
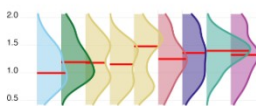
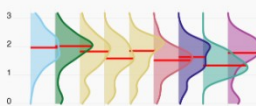
222 participant-day metrics. Among the eight metrics with site support, site-associated part R^2
223 averaged 9.0%. These non-additive components reinforce the stronger individual-level
224 contribution in the daily-pattern analysis.

225 Civil-photoperiod associations were selective. Per additional hour, the fitted daily geometric
226 mean was 1.18 times higher (95% CI 1.11 to 1.26), the brightest-10-hour mean 1.24 times
227 higher (1.13 to 1.36), waking time above 250 lx 1.14 times higher (1.07 to 1.20) and daily
228 dose 1.28 times higher (1.17 to 1.39). Each hour was also associated with 0.134 h less time
229 below 10 lx before sleep (0.066 to 0.201 h less) and the last exposure above 250 lx
230 occurring 0.323 h later (0.186 to 0.460 h). Civil photoperiod met the FDR criterion for 12 of
231 17 metrics but not stability, fragmentation, sleep time below 1 lx, the brightest-window
232 midpoint or first exposure above 250 lx. Among supported models, photoperiod-associated
233 part R^2 averaged 5.2%; the corresponding mean for the six supported latitude models was
234 4.0%, meaning that photoperiod had a stronger effect than latitude, when compared across
235 models.

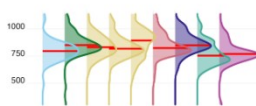
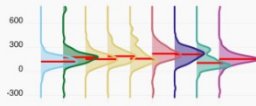
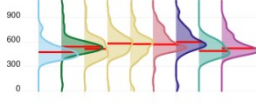
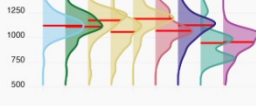
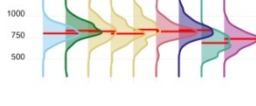
236 Across the recorded 10.33 to 20.52 h of civil photoperiod, six of nine level, exposure-history,
237 and duration-based metrics changed from increasing to a sustained near-flat tail between
238 14.24 and 16.20 hours of photoperiod: the daily mean, both 10-hour window means, time
239 above 1,000 lx, waking time above 250 lx and dose. Pre-sleep time below 10 lx, sleep-
240 environment time below 1 lx and the longest period above 250 lx did not. This transition
241 means that the effect of photoperiod becomes uncertain around 14 to 16 hours. The fitted
242 slope became compatible with zero and remained so through the observed maximum. Some
243 classifications were sensitive to site, sample or model form. Several limitations apply as
244 discussed below: latitude was confounded with site, collection periods and photoperiod
245 overlap differed, and weather, built environment, culture and routine could not be separated.
246 The metric-specific results are summarised in [Table 3](#), [Supplementary Figure S7](#) and
247 [Supplementary Tables S7](#) and [S8](#).

248 Table 3: Near-eye personal light-exposure metrics and their geographic and photoperiod context. Descriptive exposure values are medians with
249 interquartile ranges (25th to 75th percentiles). Density plots show site distributions and are descriptive. Metrics are grouped into thematic streams
250 and describe distinct features of the daily exposure pattern rather than interchangeable health-response estimates. Model cells report overall-site
251 evidence, civil-photoperiod associations with 95% confidence intervals, FDR decisions, participant-associated model information and model-
252 specific R² summaries where applicable. Site and photoperiod part-R² values may contain overlapping fitted information and must not be summed.
253 Associations are observational. Absolute latitude is fixed within and confounded with study site, so separate models containing either site or
254 latitude are reported; complete latitude and model-adequacy results are provided in the Supplementary Information.

Near-eye personal light-exposure metrics and their geographic and photoperiod context						
		Descriptive summary		Association evidence		Modelled variation
		Overall distribution	Site distribution	Overall site	Civil photoperiod	R ² summary
Duration						
Time above 1,000 lx meLEDI	Bright-light exposure duration; relevant to daytime alerting and circadian entrainment.	Unit h Median 0.683 IQR 0.183 to 1.558 Participants 141 Days 816		FDR not supported; adjusted p 0.071 Part R-squared 4.7% (2.6% to 12.1%)	Estimate Ratio per 1 h: 1.206 (1.128 to 1.288) FDR supported; adjusted p <0.001 Part R-squared 9.0% (3.7% to 15.7%)	Marginal 21.5% (16.3% to 32.3%) Conditional 50.1% (40.2% to 57.6%) Participant-associated 28.6% (17.7% to 34.2%)
Time above 250 lx meLEDI during wake	Waking time in recommended daytime light; relevant to alertness, entrainment, and subsequent sleep.	Unit h Median 2.417 IQR 0.850 to 4.617 Participants 141 Days 737		FDR supported; adjusted p 0.007 Part R-squared 8.6% (5.2% to 18.7%)	Estimate Ratio per 1 h: 1.136 (1.073 to 1.202) FDR supported; adjusted p <0.001 Part R-squared 6.4% (2.0% to 12.7%)	Marginal 18.0% (12.9% to 29.1%) Conditional 50.1% (41.6% to 58.0%) Participant-associated 32.1% (20.7% to 36.7%)
Time below 10 lx meLEDI before sleep	Low-light time before bed; limits evening melatonin suppression and circadian delay.	Unit h Median 1.883 IQR 1.025 to 2.583 Participants 139 Days 655		FDR supported; adjusted p 0.049 Part R-squared 5.4% (3.2% to 13.4%)	Estimate Difference per 1 h: -0.134 (-0.201 to -0.066) FDR supported; adjusted p <0.001 Part R-squared 4.9% (1.2% to 10.5%)	Marginal 7.6% (4.6% to 17.2%) Conditional 38.3% (31.8% to 48.5%) Participant-associated 30.7% (21.7% to 38.1%)
Time below 1 lx meLEDI during sleep	Darkness during sleep; supports nocturnal melatonin and an undisturbed sleep environment.	Unit h Median 7.133 IQR 5.967 to 8.250 Participants 141 Days 778		FDR not supported; adjusted p 0.058 Part R-squared 5.4% (3.2% to 14.2%)	Estimate Ratio per 1 h: 0.987 (0.968 to 1.006) FDR not supported; adjusted p 0.227 Part R-squared 0.6% (0.0% to 3.8%)	Marginal 8.0% (5.1% to 18.2%) Conditional 40.9% (32.8% to 48.5%) Participant-associated 32.9% (22.0% to 37.8%)
Longest period above 250 lx meLEDI	Longest sustained bright-light bout; captures continuity of daytime circadian stimulation.	Unit h Median 0.633 IQR 0.283 to 1.204 Participants 141 Days 816		FDR not supported; adjusted p 0.398 Part R-squared 2.2% (1.4% to 8.1%)	Estimate Ratio per 1 h: 1.125 (1.065 to 1.187) FDR supported; adjusted p <0.001 Part R-squared 4.9% (1.3% to 9.6%)	Marginal 10.0% (6.1% to 17.9%) Conditional 36.3% (29.9% to 45.3%) Participant-associated 26.3% (18.7% to 33.0%)

Near-eye personal light-exposure metrics and their geographic and photoperiod context						
Definition/relevance		Descriptive summary		Association evidence		Modelled variation
		Overall distribution	Site distribution	Overall site	Civil photoperiod	R ² summary
Dynamics						
Interdaily stability	Day-to-day regularity of the light–dark pattern; higher regularity supports circadian stability.	Unit dimensionless Median 0.308 IQR 0.248 to 0.380 Participants 141 Days 816		FDR not supported; adjusted p 0.228 Part R-squared 6.9% (4.1% to 20.4%)	Estimate Odds ratio per 1 h: 0.983 (0.944 to 1.024) FDR not supported; adjusted p 0.401 Part R-squared 0.4% (0.0% to 4.6%)	Marginal 15.4% (10.4% to 30.8%) Conditional 15.4% (10.4% to 30.8%) Participant-associated Not available
Intradaily variability	Within-day fragmentation of light exposure; higher values indicate less consolidated light–dark input.	Unit dimensionless Median 1.253 IQR 0.930 to 1.502 Participants 141 Days 816		FDR not supported; adjusted p 0.483 Part R-squared 5.1% (3.8% to 18.7%)	Estimate Difference per 1 h: -0.019 (-0.056 to 0.019) FDR not supported; adjusted p 0.327 Part R-squared 0.7% (0.0% to 5.7%)	Marginal 5.9% (4.6% to 21.2%) Conditional 5.9% (4.6% to 21.2%) Participant-associated Not available
Exposure history						
meLEDI dose	Intensity–duration-weighted melanopic exposure; summarizes cumulative non-visual retinal light input.	Unit klx·h Median 4.960 IQR 1.936 to 12.313 Participants 141 Days 761		FDR not supported; adjusted p 0.209 Part R-squared 2.8% (1.7% to 8.6%)	Estimate Ratio per 1 h: 1.275 (1.169 to 1.391) FDR supported; adjusted p <0.001 Part R-squared 7.5% (3.1% to 12.9%)	Marginal 11.8% (7.7% to 19.7%) Conditional 33.8% (26.7% to 42.8%) Participant-associated 22.0% (14.5% to 28.7%)

Near-eye personal light-exposure metrics and their geographic and photoperiod context						
Definition/relevance		Descriptive summary		Association evidence		Modelled variation
		Overall distribution	Site distribution	Overall site	Civil photoperiod	R ² summary
Level						
Mean melEDI	Geometric average of daily melEDI values, including zeros; summarizes overall exposure while reducing peak influence.	Unit lx Median 5.154 IQR 2.831 to 9.225 Participants 141 Days 816		FDR supported; adjusted p <0.001 Part R-squared 8.4% (4.8% to 16.6%)	Estimate Ratio per 1 h: 1.181 (1.109 to 1.257) FDR supported; adjusted p <0.001 Part R-squared 7.7% (2.9% to 13.9%)	Marginal 25.4% (18.6% to 34.4%) Conditional 57.7% (51.9% to 65.1%) Participant-associated 32.3% (24.5% to 39.4%)
Brightest 10 h geometric mean	Offset geometric mean melEDI in the brightest supported 10-hour window; describes the strength of the main sustained daily exposure.	Unit lx Median 110.566 IQR 41.513 to 243.212 Participants 141 Days 816		FDR supported; adjusted p 0.011 Part R-squared 5.7% (3.3% to 12.7%)	Estimate Ratio per 1 h: 1.243 (1.133 to 1.363) FDR supported; adjusted p <0.001 Part R-squared 5.8% (1.9% to 11.1%)	Marginal 17.6% (12.1% to 26.0%) Conditional 46.0% (39.6% to 54.3%) Participant-associated 28.4% (20.9% to 35.4%)
Darkest 10 h geometric mean	Offset geometric mean melEDI in the darkest supported 10-hour window; describes the sustained low-light period.	Unit lx Median 0.103 IQR 0.020 to 0.253 Participants 141 Days 816		FDR supported; adjusted p <0.001 Part R-squared 13.7% (9.2% to 23.9%)	Estimate Ratio per 1 h: 1.088 (1.034 to 1.144) FDR supported; adjusted p 0.002 Part R-squared 3.4% (0.5% to 8.3%)	Marginal 22.0% (15.8% to 31.8%) Conditional 61.1% (55.3% to 68.1%) Participant-associated 39.1% (30.4% to 46.2%)
Spectrum						
Melanopic daylight efficacy ratio	Mean of viable one-minute melEDI/illuminance ratios; indicates melanopic efficacy relative to visual light.	Unit dimensionless Median 0.724 IQR 0.643 to 0.795 Participants 137 Days 687		FDR supported; adjusted p <0.001 Part R-squared 9.6% (6.0% to 17.8%)	Estimate Difference per 1 h: 0.024 (0.017 to 0.031) FDR supported; adjusted p <0.001 Part R-squared 13.0% (6.7% to 20.6%)	Marginal 27.8% (21.3% to 38.1%) Conditional 60.5% (54.6% to 68.2%) Participant-associated 32.7% (24.1% to 40.0%)

Near-eye personal light-exposure metrics and their geographic and photoperiod context						
Definition/relevance		Descriptive summary		Association evidence		Modelled variation
		Overall distribution	Site distribution	Overall site	Civil photoperiod	R ² summary
Timing						
Midpoint of the brightest 10 hours	Centre time of the brightest 10 hours; indexes the main daily circadian light cue.	Unit clock time Median 13:44 IQR 12:48 to 15:00 Participants 141 Days 816		FDR supported; adjusted p <0.001 Part R-squared 6.5% (4.4% to 12.6%)	Estimate Difference per 1 h: 0.054 (-0.045 to 0.153) FDR not supported; adjusted p 0.305 Part R-squared 0.2% (0.0% to 1.8%)	Marginal 6.7% (4.6% to 13.0%) Conditional 22.9% (16.9% to 31.6%) Participant-associated 16.2% (9.6% to 22.4%)
Midpoint of the darkest 10 hours	Centre time of the darkest 10 hours; indexes the main daily darkness cue.	Unit clock time Median 02:54 IQR 02:01 to 03:50 Participants 141 Days 816		FDR supported; adjusted p 0.020 Part R-squared 4.6% (3.0% to 10.5%)	Estimate Difference per 1 h: -0.148 (-0.245 to -0.050) FDR supported; adjusted p 0.004 Part R-squared 2.1% (0.3% to 5.6%)	Marginal 8.6% (5.9% to 15.8%) Conditional 28.2% (22.1% to 36.7%) Participant-associated 19.6% (12.6% to 25.8%)
First light timing above 250 lx meEDI	First waking bright-light exposure; morning timing can advance circadian phase and promote alertness.	Unit clock time Median 09:08 IQR 08:02 to 10:40 Participants 140 Days 727		FDR not supported; adjusted p 0.112 Part R-squared 3.8% (2.3% to 10.2%)	Estimate Difference per 1 h: -0.097 (-0.247 to 0.053) FDR not supported; adjusted p 0.227 Part R-squared 0.5% (0.0% to 2.8%)	Marginal 6.8% (4.4% to 14.4%) Conditional 31.0% (24.6% to 40.5%) Participant-associated 24.2% (16.2% to 31.5%)
Last light timing above 250 lx meEDI	Last bright-light exposure; later timing may delay circadian phase and sleep onset.	Unit clock time Median 18:08 IQR 16:27 to 19:42 Participants 141 Days 687		FDR supported; adjusted p <0.001 Part R-squared 11.6% (8.3% to 18.6%)	Estimate Difference per 1 h: 0.323 (0.186 to 0.460) FDR supported; adjusted p <0.001 Part R-squared 4.5% (1.5% to 8.5%)	Marginal 22.2% (17.0% to 30.2%) Conditional 37.9% (31.6% to 46.3%) Participant-associated 15.8% (9.7% to 22.3%)
Mean timing of exposure above 250 lx meEDI	Average bright-light timing; summarizes the phase of daily circadian stimulation.	Unit clock time Median 13:29 IQR 12:29 to 14:39 Participants 141 Days 742		FDR supported; adjusted p <0.001 Part R-squared 12.7% (9.4% to 19.2%)	Estimate Difference per 1 h: 0.121 (0.028 to 0.213) FDR supported; adjusted p 0.013 Part R-squared 1.2% (0.1% to 3.6%)	Marginal 14.7% (10.7% to 21.9%) Conditional 25.7% (19.8% to 34.2%) Participant-associated 11.0% (5.3% to 17.2%)
Overall distributions are median (interquartile range). Participants is the participant count and Days is the participant-day count. Site and photoperiod part-R ² values can contain overlapping fitted information and must not be summed. Participant-associated R ² is the conditional-minus-marginal contribution associated with the participant random intercept and is undefined for participant-level outcomes. The MDER uses the mean of viable minute-level ratios in both the descriptive summary and geographic-association model cells.						

259 Immediate environments and routines reorganise exposure

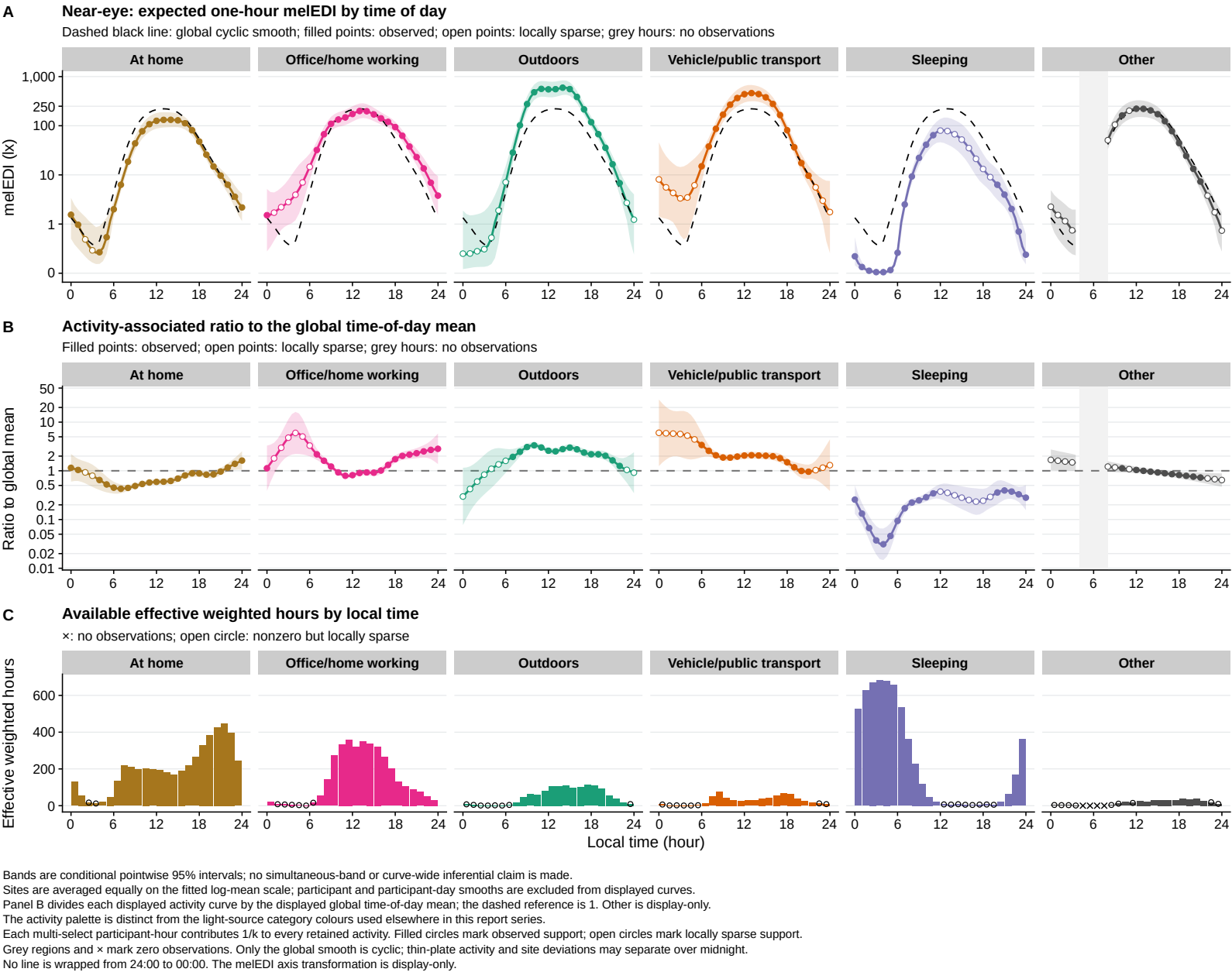
260 Self-reported light source was strongly associated with one-hour near-eye exposure, with an
261 FDR-supported light-source-by-site interaction. Site-average interaction-model estimates
262 were 959.6 lx melanopic EDI for outdoor daylight (95% CI 786.8 to 1,170.2), 198.8 lx for
263 indoor daylight (169.3 to 233.4), 85.1 lx for indoor electric light (71.6 to 101.1) and 23.8 lx for
264 a display screen (17.4 to 32.5). Outdoor daylight was 11.27 times the indoor-electric
265 estimate (8.71 to 14.60) and exceeded the 250-lx daytime recommendation, whereas the
266 indoor-daylight and indoor-electric intervals were below it. During reported sleep, darkness
267 and external-light estimates were 2.24 lx (1.54 to 3.26) and 11.18 lx (8.06 to 15.51), both
268 above the 1-lx bedside upper limit. Outdoor electric light remained imprecise relative to
269 indoor electric light near eye (ratio 1.019, 0.336 to 3.090; FDR-adjusted $p = 0.973$). These
270 contextual averages do not classify people or hours as adherent.

271 External light during sleep in Munich (DE) was estimated at 130.2 lx, 11.65 times the
272 corresponding site average; sleep darkness in Kumasi (GH) was 0.371 lx, 0.166 times its
273 site average. Sparse site-category cells limited some estimates.

274 The primary analysis used a population-mean quasi-Tweedie model with participant-cluster-
275 robust covariance for uncertainty; lag-one residual correlation remained 0.288. Fitted versus
276 observed exact-zero proportions were 39.8% and 27.8%, respectively. An exploratory
277 mixed-effects model quantified variance explained after adding a participant intercept to site-
278 by-light-source fixed terms. Marginal R^2 was 0.796 and conditional R^2 0.876, an 8.0-
279 percentage-point increase in explained variance from the participant intercept. Of the
280 marginal R^2 , light source accounted for 89.3%, site 7.1% and their interaction 3.7%.
281 Separately, an exploratory time-of-day model attributed 60.4% of variation in fitted hourly
282 patterns to global time of day and 24.2% to light-source deviations. Among these two
283 components, light source accounted for 28.6%. The remaining shares were 5.2% for site,
284 7.6% for participant patterns and 2.5% for day-to-day shifts.

285 A sensitivity adjusted the population-average light-source comparisons for each participant's
286 overall category distribution, distinguishing hour-to-hour differences within participants from
287 differences in category composition between participants. The overall category pattern was
288 retained. For external light during sleep, the near-eye ratio relative to indoor electric light
289 changed from 0.449 in the primary additive model to 0.333 (95% CI 0.190 to 0.585), a 25.8%
290 decrease. Thus, between-participant category composition influenced the magnitude, but not
291 the direction, of this contrast. [Supplementary Figure S8](#) shows the temporal, support and
292 site-specific results.

293 Immediate setting combined behaviour and environment, with an FDR-supported activity-by-
294 site interaction. Site-average near-eye estimates were 714.2 lx outdoors (95% CI 561.7 to
295 908.2), 331.2 lx on the road or in a vehicle (274.4 to 399.8), 198.9 lx during office or home
296 work (173.9 to 227.5), 76.4 lx while awake at home (62.0 to 94.1) and 4.28 lx during sleep
297 (3.06 to 5.99). The sleep category included many low or zero observations and occasional
298 high stray-light readings, so its fitted mean need not represent a typical observation. Relative
299 to awake time at home, the ratios were 9.35 outdoors, 4.34 for travel, 2.60 for work and
300 0.056 during sleep. The outdoor estimate exceeded the 250-lx daytime recommendation, the
301 work estimate remained below it, and the sleep estimate exceeded the 1-lx bedside-
302 environment upper limit. Borås (SE) had a fitted outdoor estimate of 3,054 lx, 4.28 times the
303 site average, whereas Tübingen (DE) had 127 lx, 0.18 times the site average. In an
304 exploratory mixed model of five setting categories, marginal R^2 was 0.766 near eye and
305 0.768 at the chest. Adding the participant intercept raised conditional R^2 to 0.858 and 0.859,
306 an explained-variance increment of 9.2 percentage points at both positions. Of the marginal
307 R^2 , immediate setting accounted for 80.7% near eye and 86.1% at the chest, site 12.5% and
308 5.9%, and their interaction 6.9% and 8.0%, respectively. [Figure 3](#) shows the immediate-
309 setting patterns and site-specific estimates. Separately, an exploratory near-eye time-of-day
310 model attributed 52.5% of variation in fitted hourly patterns to time of day and 30.7% to
311 immediate setting. Among these two components, immediate setting accounted for 36.9%.
312 The remaining shares were 5.6% for site, 8.4% for participant patterns and 2.8% for day-to-
313 day shifts.



D Near-eye activity-by-site interaction model
Site-average estimates and site-specific deviation ratios; bold site cells pass the complete near-eye FDR adjustment

	At home	Office/home working	Outdoors	Vehicle/public transport	Sleeping	Other
	P 126 · D 709 · H 5,105	P 124 · D 511 · H 3,624	P 121 · D 480 · H 1,422	P 111 · D 369 · H 818	P 126 · D 714 · H 5,906	P 72 · D 159 · H 391
Site-average estimate	76.4 lx (62.0–94.1) Reference category FDR p (reference)	198.9 lx (173.9–227.5) 2.604× home (2.121–3.198) FDR p <0.001	714.2 lx (561.7–908.2) 9.354× home (6.946–12.595) FDR p <0.001	331.2 lx (274.4–399.8) 4.338× home (3.320–5.668) FDR p <0.001	4.3 lx (3.1–6.0) 0.056× home (0.041–0.076) FDR p <0.001	221.8 lx (97.2–506.4) 2.970× home (1.228–7.182) <i>Additive display only</i>
Borås (SE)	1.74× (0.94–3.22) FDR p 0.152	1.64× (1.22–2.19) FDR p 0.005	4.28× (2.84–6.45) FDR p 0.001	1.67× (0.93–3.00) FDR p 0.159	0.45× (0.22–0.96) FDR p 0.084	<i>Not estimable</i>
Delft (NL)	2.07× (1.30–3.28) FDR p 0.009	1.48× (1.07–2.04) FDR p 0.048	2.32× (1.32–4.09) FDR p 0.014	1.32× (0.86–2.02) FDR p 0.299	1.17× (0.51–2.68) FDR p 0.797	<i>Not estimable</i>
Dortmund (DE)	2.31× (1.14–4.67) FDR p 0.051	1.67× (0.91–3.06) FDR p 0.169	2.00× (1.20–3.34) FDR p 0.025	2.41× (1.44–4.05) FDR p 0.005	4.53× (1.30–15.79) FDR p 0.048	<i>Not estimable</i>
Tübingen (DE)	0.42× (0.28–0.63) FDR p <0.001	0.54× (0.39–0.73) FDR p <0.001	0.18× (0.11–0.30) FDR p <0.001	0.34× (0.21–0.56) FDR p <0.001	0.38× (0.16–0.89) FDR p 0.061	<i>Not estimable</i>
Munich (DE)	1.25× (0.73–2.14) FDR p 0.486	1.54× (1.00–2.36) FDR p 0.102	1.34× (0.72–2.48) FDR p 0.433	0.81× (0.58–1.13) FDR p 0.299	10.47× (4.40–24.89) FDR p <0.001	<i>Not estimable</i>
Madrid (ES)	0.64× (0.46–0.89) FDR p 0.026	1.14× (0.93–1.41) FDR p 0.299	0.97× (0.47–1.98) FDR p 0.953	0.65× (0.33–1.30) FDR p 0.303	0.78× (0.20–3.10) FDR p 0.797	<i>Not estimable</i>
Izmir (TR)	0.80× (0.56–1.16) FDR p 0.314	0.79× (0.58–1.07) FDR p 0.196	0.97× (0.47–1.97) FDR p 0.953	1.14× (0.66–1.96) FDR p 0.732	1.61× (0.90–2.89) FDR p 0.185	<i>Not estimable</i>
San José (CR)	0.78× (0.46–1.31) FDR p 0.428	0.50× (0.33–0.76) FDR p 0.006	<i>Not estimable</i>	0.98× (0.53–1.84) FDR p 0.960	0.50× (0.21–1.21) FDR p 0.196	<i>Not estimable</i>
Kumasi (GH)	0.57× (0.20–1.62) FDR p 0.378	0.67× (0.45–0.99) FDR p 0.088	0.52× (0.33–0.82) FDR p 0.019	0.93× (0.57–1.53) FDR p 0.839	0.16× (0.07–0.42) FDR p 0.001	<i>Not estimable</i>

Site-average estimates and comparisons for the five named categories use the activity-by-site interaction model. Other is the additive display-only estimate and was excluded from the interaction model. Site rows are deviation ratios around each category estimate; parentheses show 95% confidence intervals. Bold site ratios and FDR p-values pass the complete 44-member near-eye family; unsupported cells are not estimable. Five-category interaction frame: 126 participants; 724 participant-days; 16,135 unique participant-hours; 16,875 generated long rows; 16,135 effective weighted hours; nine sites; 4,746 exact-zero participant-hours. Each multi-select hour contributes 1/k. Other-only display support: P 72 · D 159 · H 391. P, participants; D, participant-days; H, unique participant-hours.

314
315 *Figure 3: Personal light exposure by immediate setting across time of day and study sites. The diary variable labelled activity combines behaviour*
316 *and environmental setting. A–C, Exploratory time-of-day analysis. Panel A shows site-average expected one-hour melanopic EDI with clock-*
317 *specific 95% confidence intervals; the dashed black curve is the site-average local-clock smooth. Panel B shows each setting curve relative to*
318 *that smooth, with 1 as the reference. Panel C shows effective weighted hours; open circles indicate locally sparse support and grey regions*
319 *indicate times without observations. D, Site-average category estimates and site-specific deviation ratios from the activity-by-site interaction*
320 *model, with participant-cluster-robust 95% confidence intervals and FDR-adjusted p values. Each multi-select participant-hour contributes weight*
321 *1/k to every retained setting category. The Other category is sparse and display-only. The fitted frame contains 126 participants, 724 participant-*
322 *days, 16,135 unique participant-hours, 16,135 effective weighted hours and 4,746 exact-zero participant-hours. Estimates are observational.*

323 Across models of free versus work day, active versus sedentary daily activity status and
 324 previous-night sleep duration, the site-average reference was a work, sedentary day
 325 following an average 7.8 h of sleep; expected near-eye exposure was 89 lx melanopic EDI
 326 (95% CI 67 to 119). Active versus sedentary status was associated with 2.06 times the
 327 expected exposure (1.54 to 2.76; FDR-adjusted $p < 0.001$). A model constraining the free-
 328 versus-work association across sites estimated 1.45 times higher exposure on free days
 329 (1.14 to 1.85; FDR-adjusted $p = 0.004$). Allowing the association to differ by site gave an
 330 inconclusive site-average ratio of 1.15 (0.93 to 1.44; FDR-adjusted $p = 0.298$), while some
 331 sites differed (interaction $p = 0.017$). Exploratory ratios were FDR-retained in Borås (SE;
 332 1.95, 1.26 to 3.01) and Dortmund (DE; 2.44, 1.42 to 4.22), with FDR-adjusted $p = 0.013$ for
 333 each. The free-versus-work result was also inconclusive under the alternative working
 334 variance and sensitive to omitting either site. Each additional hour of previous-night sleep
 335 duration, centred at 8 h, had an inconclusive ratio of 0.98 (0.85 to 1.12; FDR-adjusted $p =$
 336 0.745). Excluding six participants outside the employment requirement left 15,871 supported
 337 hours from 684 participant-days and 131 participants across all nine sites. The free-versus-
 338 work, active-versus-sedentary and previous-sleep associations remained stable within model
 339 uncertainty; the predictor-by-site interaction for day type also remained FDR-supported.
 340 [Supplementary Table S9](#) and Supplementary Figures [S9](#), [S10](#), [S11](#) and [S12](#) report common,
 341 temporal, position-specific and site-specific results.

342 **Person-level correlates are selective**

343 Self-report analyses did not support general exposure profiling. None of 68 prespecified
 344 near-eye associations between four light-exposure-related behaviour (LEBA) factors and 17
 345 metrics retained FDR support. Unsatisfactory model checks made the sleep-environment
 346 metric unfit for inference, so its estimates were suppressed. Neither average associations
 347 nor site interactions across nine visual-light-sensitivity (VLSQ-8) outcomes retained FDR
 348 support at either position. Supplementary Tables [S10](#), [S11](#) and [S12](#), and Supplementary
 349 Figures [S13](#) and [S14](#), give the full results.

350 Each hour later in corrected midsleep on free days (MCTQ questionnaire) was associated
 351 with first exposure above 250 lx occurring 0.381 h later (95% CI 0.146 to 0.616; FDR-
 352 adjusted $p = 0.003$). Each 10-point shift towards greater morning preference (MEQ
 353 questionnaire) was separately associated with first exposure above 250 lx occurring 0.444 h
 354 earlier (0.189 to 0.699 h earlier; FDR-adjusted $p = 0.001$). The brightest- and darkest-10-
 355 hour midpoints followed the same directions; last exposure and the longest-bright-period
 356 midpoint did not retain support. The distinct questionnaires were not combined.

357 [Supplementary Table S13](#) and [Supplementary Figure S15](#) show adjusted estimates and
358 observed timing distributions.

359 Each additional decade of age was associated with higher near-eye brightest-10-hour mean,
360 time above 1,000 lx and daily dose, with ratios from 1.27 to 1.31. These cross-sectional
361 associations may reflect cohort, occupation, behaviour or other confounding, not ageing
362 itself. Neither near-eye metric associations with biological sex nor near-eye age-by-site or
363 sex-by-site interactions met the FDR criterion. In a separate daily-pattern analysis, near-eye
364 and chest profiles differed by biological sex (FDR-adjusted $p = 0.028$ and 0.029), but the
365 difference did not meet the FDR criterion after restriction to records with complete
366 immediate-setting data and adjustment. This attenuation is compatible with immediate-
367 setting patterning, but restriction and adjustment cannot be disentangled as mechanisms.
368 Gender was recorded separately but not analysed; its category matched biological sex in all
369 but one case. Supplementary Tables [S10](#), [S14](#) and [S15](#), and Supplementary Figures [S16](#)
370 and [S17](#), give the complete results.

371 **Measurement position and robustness constrain** 372 **interpretation**

373 The paired sample of 112 participants and 643 participant-days allowed separate near-eye
374 and chest models to be compared within the same participants and participant-days. The full
375 chest dataset of 154 participants and 902 participant-days broadened coverage beyond the
376 141 participants and 816 participant-days near eye. At several sites, especially San José
377 (CR), offering a chest sensor enabled recruitment among people reluctant to wear the
378 glasses-mounted sensor. This improved feasibility, not the ocular-exposure sample.

379 A parallel analysis of overlapping MeLiDos data reported structured position differences
380 across scale, metric, context, site, day and participant³³. Our separate and common-sample
381 analyses likewise show directional agreement but differences in magnitude or support. Eight-
382 site chest adherence retained daytime and pre-sleep directions and interval conclusions,
383 with calibration and temporal qualifications, but provided no sleep ocular claim or paired test.
384 We neither pooled positions nor inferred equivalence or a universal correction.

385 Non-estimable planned quantities were four sleep-environment variants (main plus three
386 sensitivity variants) at each position in the light-behaviour analysis (LEBA), and one longest-
387 period sensitivity comparison incompatible with the metric representation. We retained these
388 outcomes rather than replacing them with zero or null findings. Qualifications for zeros,

389 residuals, dependence, nonlinear-term overlap, sparse support and influence remain with
390 the affected analyses and in Methods.

391 Discussion

392 The expected 24-hour rhythm was evident across nine sites in seven countries. The
393 advance was a multiscale account of this everyday exposome component: amplitude across
394 sites, people and days, and its reorganisation across immediate settings. The shared local-
395 clock pattern accounted for the largest share of full-model R^2 in the daily-pattern analysis.
396 Separate hourly decompositions showed large light-source and immediate-setting contrasts,
397 with the next-largest shares of variation in fitted hourly patterns after time of day. Participant
398 patterns and participant-day shifts together exceeded site patterns in the daily-pattern
399 model, while site, photoperiod and measured person-level associations were metric-specific.
400 Different outcomes and decompositions make this a qualitative synthesis rather than a single
401 quantitative ranking.

402 The consensus recommendations provide the clearest health-relevant benchmark. Only
403 24.0% of waking minutes reached the daytime range, while 63.3% of pre-sleep and 87.7% of
404 bedside sleep-environment minutes met their upper limits. Modelled free-day adherence was
405 lower during daytime and preceding sleep, varying by site. Pre-sleep adherence was higher
406 but inconclusive under stricter coverage; temporal dependence remained unresolved.
407 Outdoor exposure exceeded the daytime recommendation, but all other conditions remained
408 below it on average. The fitted sleep-category mean exceeded the bedside limit, while
409 87.7% of pooled bedside sleep-environment minutes met it. These quantities describe
410 different aspects of the exposure distribution and should be interpreted together. These
411 pooled fractions, recommendation-window proportions and hourly means are distinct
412 estimands. Together, they identify intervention opportunities and shortfalls. The
413 recommendations principally address healthy adults aged 18 to 55 with regular daytime
414 schedules⁸.

415 This baseline connects mechanistic recommendations with population evidence. Large wrist-
416 sensor cohorts have associated day and night light patterns with psychiatric disorders,
417 incident type 2 diabetes, cardiovascular disease and mortality, and light timing with sleep
418 timing and sleepiness¹⁰⁻¹⁵. Other cohorts have linked evening, nocturnal or strongly seasonal
419 light with sleep in bipolar disorder, obesity, diabetes, hypertension and metabolic health¹⁶⁻¹⁹.
420 These outcomes span sleep and major non-communicable diseases. The exposome frames
421 environmental contributions across the life course^{20,21} but rarely includes time-resolved
422 ocular light. This evidence supports personal-light measurement, not clinical risk
423 classification of our exposure categories. Most cohorts used wrist or bedside illuminance

424 without resolving melanopic exposure at the eye, immediate light source or its multiscale
425 architecture.

426 Exploratory inverse associations of average daytime with sleep and pre-sleep adherence
427 show why the three recommendation windows should not be collapsed into one score. They
428 may reflect schedules, environments or other between-participant differences, not a within-
429 person trade-off or stable phenotype. In exploratory nonlinear models, light source and
430 immediate setting accounted for substantially more variation in fitted hourly patterns than
431 participant-specific patterns: 24.2% versus 7.6%, and 30.7% versus 8.4%, respectively. This
432 finding is consistent with schedules and immediate environments being prominent correlates
433 of hourly exposure, but it is not a causal explanation for the cross-window adherence
434 association. Longitudinal models of dependence and changing routines are needed to test
435 whether improving one part of the pattern displaces or supports another.

436 Location mattered, but not as one north-to-south ranking. Site differences concentrated in
437 daily level, waking bright-light duration and several timing metrics, while dose, spectrum and
438 regularity did not show the same site pattern. Civil photoperiod was associated with a
439 broader set of metrics, with larger day-length associations for bright-light level and dose than
440 for the darkest daily window. Between 14.2 and 16.2 h of civil photoperiod, six of nine
441 metrics showed signs of uncertainty about a photoperiod effect beyond that point. Sites were
442 sampled in distinct calendar periods and covered unequal photoperiod ranges, leaving
443 latitude, season, weather, temperature, culture, mobility and the built environment entangled.
444 Repeated seasonal sampling across more sites is needed to separate these influences.

445 Immediate setting patterned exposure, consistent with environmental opportunity, built
446 settings and behaviour³⁶. Outdoor daylight and being outdoors exceeded indoor electric light
447 and awake time at home, respectively; active days exceeded sedentary days. Site
448 interactions placed outdoors-category exposure higher in Borås (SE) than Tübingen (DE),
449 while the free-versus-work association was most evident in Borås (SE) and Dortmund (DE).
450 Participant intercepts captured remaining between-participant variation, not individual
451 responses or predictions. Future monitoring should preserve local clock time, location,
452 immediate setting, daylight access and site; self-selection, occupation, mobility and weather
453 remain potential confounders.

454 Light-behaviour and visual-sensitivity scores did not meet their FDR criteria, whereas
455 chronotype aligned with selected timing outcomes and age with selected bright-light and
456 dose metrics. This accords with evidence linking circadian preference or timing to daylight
457 exposure, sleep and routine, without identifying causal direction^{37–40}. Metric-level biological-
458 sex comparisons were inconclusive near the eye. The complete available-data curve differed

459 by biological sex, but immediate-setting-complete analyses did not retain this separation.
460 Person-level patterns may concern specific exposure dimensions rather than a general high-
461 versus-low phenotype.

462 Near-eye sensing approximates the incident ocular field during wake but is not a retinal
463 measurement; chest sensing samples a related environmental field. Studies show structured
464 differences by body position, morphology, posture and surrounding light²⁹⁻³¹. The matched
465 sample and parallel placement study show metric- and context-specific MeLiDos differences
466 without a universal correction³³. The chest option broadened recruitment among people
467 unwilling to wear glasses-mounted sensors, especially in San José (CR). Study design must
468 balance biological proximity, burden and recruitment while keeping estimands position-
469 specific.

470 The harmonised protocol combined repeated near-eye sensing with local-clock and
471 contextual information across nine sites and seven countries. A generalised additive model
472 separated the shared 24-hour pattern from site, participant and participant-day layers
473 obscured by daily summaries. Collecting these data demanded substantial effort from
474 participants and study teams. Glasses-mounted loggers required fitting, explanation and
475 daily handling; removal, occlusion, charging, discomfort, weather and social acceptability all
476 affected measurement quality⁴¹⁻⁴³. Despite this burden, the protocol measured closer to the
477 incident ocular field at high temporal and contextual resolution.

478 Several limitations qualify these findings. Modest site samples limit site estimates,
479 interactions and subgroup evaluation. Healthy working-age adults may not represent
480 children, adolescents, older or clinical populations, shift workers, people outside
481 employment, or those with restricted mobility or atypical living conditions. Sparse latitude-by-
482 photoperiod coverage leaves latitude, seasonality, culture, climate and built environment
483 entangled and requires broader seasonal sampling. Retrospective hourly setting and light-
484 source reports enabled harmonisation but could not distinguish fine-grained settings or
485 behaviours within home, work or transport. Very low readings require caution. The
486 manufacturer specifies an operating range starting at 1 lx, and independent laboratory
487 testing began at 2 lx⁴⁴. A scene-based field validation found high overall agreement with
488 laboratory-grade spectral references, but underestimation increased at lower intensities and
489 varied with lighting, time, scene complexity and site⁴⁵. Single-digit values therefore
490 approximate low-light exposure rather than provide precise measurements. Finally, no health
491 outcome was measured.

492 This nine-site reference locates everyday ocular light exposure within the exposome,
493 benchmarks it against physiology-based recommendations, highlights within-site variation

494 among people and days, and identifies temporal, contextual and positional information for
495 future cohorts. Everyday ocular light exposure is structured, measurable and strongly
496 shaped by behaviour and micro-environment, but cannot be reduced to site, latitude or
497 photoperiod. Next steps are broader sampling across photoperiods, latitudes and locations
498 and prospective links to sleep, mental, metabolic, cardiovascular and other outcomes.
499 Causal health benefits or policy effects require interventions and representative longitudinal
500 designs. Potential interventions through clinical advice, occupational health, building design,
501 urban planning or public-health guidance must address the contexts in which people receive
502 light. Time-resolved near-eye measurement offers a practical route from circadian biology to
503 precision prevention and context-sensitive environmental health.

504 Methods

505 Study design and sites

506 MeLiDos is a prospective multicentre observational field study of personal light exposure
507 conducted with a harmonised protocol³⁵. Participants completed an eight-day ambulatory
508 protocol comprising continuous light logging, repeated smartphone assessments and daily
509 diaries under free-living conditions. At an in-person registration visit, participants provided
510 consent, completed baseline questionnaires, received study instructions and were fitted with
511 the loggers. Seven consecutive recording days separated registration and return visits; most
512 sites recorded from Monday to Monday, while Dortmund (DE) recorded from Tuesday to
513 Tuesday. If a return visit was delayed, participants continued until formal study termination
514 and device retrieval. Data were collected between August 2023 and October 2025 at Borås
515 (SE), Delft (NL), Dortmund (DE), Tübingen (DE), Munich (DE), Madrid (ES), Izmir (TR), San
516 José (CR) and Kumasi (GH). The Kumasi (GH) and Izmir (TR) collections are described in
517 site-specific data notes^{46,47}. Sites were an opportunity sample of consortium partners and
518 were not selected to represent national populations. Questionnaire materials were
519 administered in the local language except in Tübingen (DE) and Kumasi (GH), where
520 English was used, and translated materials followed a harmonised translation, review,
521 adjudication, pretesting and documentation process⁴⁸.

522 Participants, recruitment and ethics

523 The registered target population comprised adults aged 18 to 65 years in full-time
524 employment or part-time employment above 80%. Exclusion criteria included diagnosed
525 psychiatric or sleep disorders, shift work in the preceding two months, regular tobacco or
526 recreational drug use, photosensitising medication, visual impairment incompatible with the
527 devices and residence outside the local study area during recording. The retained near-eye
528 sample included nine participants outside the registered age or employment criteria: one
529 participant older than 65 years, two recorded as not employed and seven recorded as
530 marginally employed; one participant met both the age and employment criteria for
531 exclusion. As a near-eye-only eligibility sensitivity, we refitted the additive and predictor-by-
532 site models after excluding participants recorded as not employed or marginally employed.
533 Students and trainees were retained, and age above 65 years was not an exclusion criterion
534 for this sensitivity. The target was at least 15 participants per site, based on prior power
535 calculations and allowing for up to one-third dropout or incomplete recording⁴⁹. Participants
536 were recruited through local advertisements and institutional mailing lists. At some sites,

537 especially San José (CR), reluctance to wear the glasses-mounted near-eye sensor limited
538 recruitment, and the chest option broadened recruitment possibilities.

539 All participants provided written informed consent. Approvals were obtained from the Ethics
540 Committee of the Technical University of Munich for Tübingen (DE; 2023-115-S-KK) and for
541 Munich (DE) and Dortmund (DE; 2024-118-S-SB); the Dortmund (DE) site at the Federal
542 Institute for Occupational Safety and Health (BAuA) operated under this TUM multicentre
543 approval. The other approvals were from the Swedish Ethical Review Authority for Borås
544 (SE; 2024-06792-01), the THUAS Ethics Committee for Delft (NL; positive review dated 29
545 May 2024), the Research Ethics Committee of Universidad San Pablo-CEU for Madrid (ES;
546 834/24/106), the Scientific Research and Publication Ethics Committee of the Faculty of
547 Science and Engineering at İzmir Institute of Technology for Izmir (TR; decision 5/1), the
548 Comité Ético Científico of Universidad de Costa Rica for San José (CR; CEC-279-2025),
549 and the Committee on Human Research, Publications and Ethics of Kwame Nkrumah
550 University of Science and Technology for Kumasi (GH; CHRPE/AP/644/24). The protocol
551 was first trialled with participants at the Tübingen (DE) site, whose feedback informed
552 procedural refinements before implementation at the remaining sites; further details are
553 reported in the published protocol³⁵.

554 **Personal light measurements**

555 Personal light was recorded every 10 s with spectrally sensitive ActLumus loggers (Condor
556 Instruments, São Paulo, Brazil). Eight visible-band channels were processed by the device
557 to generate photopic and alpha-opic quantities, including melanopic EDI⁶. The manufacturer
558 specified a factory-calibrated operating range of 1 to 100,000 lx and 10% precision at 1,000
559 lx. Independent testing against a criterion spectrometer across indoor LED and outdoor
560 daylight conditions covered 2 to 100,000 lx and reported mean photopic error of +3.3% (SD
561 13.4%) for ActLumus⁴⁴. A subsequent scene-based field validation found high overall
562 agreement with laboratory-grade spectral reference instruments, together with systematic
563 underestimation that varied with intensity and setting⁴⁵. Participants wore the near-eye
564 device centrally on non-prescription spectacle frames and, where available, a
565 complementary pendant-mounted device at the chest. The near-eye position measures light
566 closer to the eye but is not a direct retinal measurement. The chest position provided a
567 separate measurement of the local environmental light field. Paired common-sample
568 analyses evaluated how closely it tracked the near-eye measurement as a potential ocular
569 proxy. During reported sleep, devices were placed facing upward on a bedside surface near

570 head level, so both positions characterise the bedside sleep environment. Participants
571 recorded removals and placed devices in an opaque bag during waking non-wear^{43,50}.

572 The placement analysis from the same project used overlapping MeLiDos data and is cited
573 as related methodological evidence, not independent replication³³. No analysis pooled
574 sensor positions or estimated a universal conversion. Common-sample analyses retained
575 the same participants and participant-days or participant-hours at both positions and fitted
576 each position separately.

577 **Contextual and person-level measures**

578 Participants completed the core Consensus Sleep Diary after awakening⁵¹, together with
579 evening reports and paper diaries recording light source and an activity-environment
580 category for each hour. Smartphone assessments of sleepiness and mood were scheduled
581 at 11:00, 14:00, 17:00 and 20:00, and evening questionnaires assessed wellbeing and
582 physical activity⁵²⁻⁵⁴. Smartphone measures and baseline or end-of-study questionnaires
583 used MyCap integrated with REDCap^{55,56}. Participants photographed and uploaded the
584 paper hourly diaries each day.

585 The modified Harvard Light Exposure Assessment categories were indoor electric light,
586 outdoor electric light, indoor daylight, outdoor daylight including shade, an emissive display,
587 darkness during sleep and external light during sleep⁵⁷. The primary light-source analysis
588 used one retained category per participant-hour. The diary variable labelled activity
589 combined behaviour and environmental setting. Its categories distinguished sleep, awake
590 time at home, motorised travel, active travel, office or home work, outdoor work, outdoor
591 leisure and other activities. We therefore refer to this variable as immediate setting in the
592 narrative. For analysis, outdoor travel, outdoor work and outdoor leisure were combined as
593 Outdoors. Reports could contain concurrent categories; duplicates were removed and an
594 hour with k retained labels contributed weight $1/k$ to each, preserving total weight one. Day
595 type was coded as work or free day, daily activity status as sedentary or active, and previous
596 sleep duration was linked from the preceding sleep period.

597 Baseline measures recorded age, biological sex, gender, employment and chronotype; end-
598 of-study measures covered visual light sensitivity, the sleep environment and habitual light-
599 exposure behaviour. Chronotype was represented separately by corrected midsleep on free
600 days from the Munich Chronotype Questionnaire and the Morningness-Eveningness
601 Questionnaire, whose higher scores indicate greater morning preference^{58,59}. Visual
602 sensitivity used the VLSQ-8⁶⁰, sleep environment used the Assessment of Sleep
603 Environment⁶¹, and four Light Exposure Behaviour Assessment (LEBA) factors represented

604 outdoor time, device use in bed, ambient light before bed and morning or daytime light use⁶².
605 Biological sex and gender were recorded as separate variables. Models analysed biological
606 sex, coded Female or Male, but did not analyse gender; its recorded category matched
607 biological sex in all but one case.

608 **Time, sleep/wake status and coverage preparation**

609 Processing preserved both actual time in Coordinated Universal Time and local wall-clock
610 time. Actual time ordered observations and measured elapsed gaps; local time aligned daily
611 patterns across sites. Repeated local minutes during autumn clock changes remained
612 distinct in actual time, and structurally absent spring minutes remained missing. Melanopic
613 EDI values at or above the validated 100,000-lx operating boundary were unavailable
614 analytically but retained in the raw channel with a flag.

615 Diary-defined sleep began at reported sleep preparation and ended at reported wake. Sleep
616 values were retained as the intended bedside record. Declared waking non-wear made the
617 analytical light value unavailable; missing wear labels were not treated as non-wear.
618 Complete 1,440-minute local participant-day grids preserved unsupported periods as
619 missing rather than deleting rows or replacing them with zero. A day was eligible when at
620 least 80% of this combined waking and bedside-sleep record had valid melanopic EDI.
621 Hourly summaries required at least 50% valid minutes in that hour, but an unsupported hour
622 did not erase valid minutes from the daily decision or minute-based metrics. Two otherwise
623 eligible near-eye days and three chest days with finite but entirely zero melanopic EDI were
624 excluded by the defined exact-all-zero screen.

625 **Exposure metrics**

626 Metrics were derived in R⁶³ with LightLogR^{64,65}. The 17 outcomes represented
627 complementary properties of the daily pattern. Three level metrics described the daily
628 geometric mean melanopic EDI and the mean melanopic EDI in the brightest and darkest
629 10-hour windows. Five duration metrics quantified time above 1,000 lx, waking time above
630 250 lx, pre-sleep time below 10 lx, sleep-environment time below 1 lx and the longest
631 continuous period above 250 lx. Five timing metrics located the brightest and darkest
632 windows and the first, last and mean timing above 250 lx. Melanopic EDI dose integrated
633 exposure over time, melanopic daylight efficacy ratio characterised spectral composition
634 relative to visual illuminance, and interdaily stability and intradaily variability characterised
635 regularity and fragmentation. Geometric means were calculated from $\log_{10}(\text{melanopic EDI} +$
636 $0.1 \text{ lx})$, followed by inverse transformation and removal of the same offset⁶⁶. All-observed-

637 zero windows were retained as exact zero within verified numerical tolerance. Metric-specific
638 support rules made only the affected metric unavailable, not the participant-day. A longest-
639 period duration remained a lower bound when missing data could conceal a longer period;
640 exactly identified periods were examined separately.

641 The melanopic daylight efficacy ratio was the arithmetic mean of viable one-minute
642 melanopic EDI to photopic-illuminance ratios on a complete 1,440-position local-clock grid.
643 Both channels had to be finite and strictly positive, and at least 720 positions were required.
644 Repeated autumn minutes were averaged within each channel before division, and absent
645 spring minutes remained missing.

646 **Recommendation-window and adherence analyses**

647 We compared valid one-minute values with the Brown et al. recommendation ranges⁸: at
648 least 250 lx melanopic EDI during daytime, no more than 10 lx during pre-sleep and no more
649 than 1 lx during sleep. Daytime comprised diary-defined wake excluding the three hours
650 before reported sleep, pre-sleep comprised those three hours and sleep comprised the
651 reported sleep interval. Each descriptive numerator was divided by all valid near-eye-record
652 minutes in the corresponding window and pooled across the analysed sample. Daytime and
653 pre-sleep used the worn near-eye record; sleep used the bedside sleep-environment record.

654 The main adherence analysis retained one row per window period and defined adherence as
655 adherent valid minutes divided by all valid minutes in that period. Each cycle linked the sleep
656 interval ending at wake, the ensuing daytime interval excluding pre-sleep, and the following
657 three hours before the next reported sleep. All three inherited the work or free label of the
658 wake-start calendar date, even when a window crossed midnight. Actual diary dates and
659 UTC boundaries established chronology; gaps between diary dates were not bridged.

660 Windows qualified independently, so a cycle need not contain three eligible windows. The
661 any-valid primary sample retained at least one valid comparison; the coverage sensitivity
662 required valid minutes for at least 80% of each expected window interval. Expected duration
663 defined coverage, not the likelihood denominator. Missing minutes were not imputed. Exact
664 0% and 100% periods remained in an endpoint-inflated beta-binomial model. Its mean
665 included the full window-by-site-by-day-type interaction and a participant intercept; extra
666 wholly non-adherent and wholly adherent periods and window-specific dispersion were
667 modelled separately. Population-average adherence integrated the participant distribution
668 and gave each of nine sites equal weight. Separate FDR families covered three window-
669 specific free-minus-work contrasts, three window-specific site interactions, the global
670 interaction, 27 site-specific contrasts against zero, 54 site-minus-average levels and 27 site-

671 effect-minus-average contrasts. The last family's coverage repetition assessed stability
672 without another FDR adjustment. Variance decomposition used balanced 54-cell global and
673 18-cell within-window references; Shapley allocations and R^2 were point-only. Chest models
674 used a separate two-window, eight-site sample and did not test paired placement
675 differences.

676 The exploratory cross-window extension used daytime adherence as the predictor and the
677 preceding sleep or following pre-sleep adherence as outcomes under the same wake-start-
678 date alignment. It retained pairwise-complete observations, not necessarily complete three-
679 window cycles: 1,376 outcome rows comprised 758 daytime-sleep and 618 daytime-pre-
680 sleep pairs from 140 participants and 761 cycles. Its frozen eligibility rules did not include
681 every potential main-model pair. The model separated each participant's cycle-specific
682 daytime deviation from their observed average, adjusting for window, site, day type, their
683 corresponding interactions and the participant's day-type composition. Response-scale
684 contrasts used equal-site and 50:50 work/free weighting. The four within- and between-
685 participant associations formed a separate FDR family. Positive serial dependence remained
686 unresolved and some sensitivity fits were non-estimable, so no within-participant day-level
687 claim was stated. Descriptive profiles used unweighted equal-cycle participant means, with
688 daytime deduplicated within participant and cycle, for 139 complete profiles and 417
689 participant-window points. These are anonymous monitoring-period summaries, not ranks or
690 stable participant types.

691 The main model passed all five applicable endpoint-calibration checks in both samples, but
692 temporal dependence still qualified inference. The 80% coverage restriction retained the
693 daytime and sleep directions and interval conclusions; pre-sleep retained direction but not
694 exclusion of zero. Diary-indexed grouping was a separate sensitivity. Pre-sleep interval
695 conclusions also changed under 70% and 90% coverage and restriction to participants
696 observed on both day types. Actual-date temporal endpoint models failed structural checks.
697 Their beta-binomial fallbacks changed both the response family and random-effects
698 structure, retained boundary cautions and failed all five endpoint-calibration checks in both
699 samples. They did not replace the primary model or resolve its temporal limitation. All nine
700 site-deletion checks and four participant-deletion checks were estimable; one participant
701 deletion and the ordinary-binomial diagnostic failed and supplied no derived inference. The
702 separate chest model retained calibration and temporal qualifications. In the exploratory
703 extension, the 80% restriction preserved all four directions and interval conclusions, but
704 incomplete influence checks and unresolved dependence limited robustness claims.

705 Daily temporal architecture

706 The daily-pattern analysis used supported 30-minute arithmetic means, retaining a bin when
707 at least 15 one-minute values were valid. The outcome was $\log_{10}(\text{melanopic EDI} + 0.1 \text{ lx})$. A
708 generalised additive model fitted with `mgcv::bam()` and fREML combined a cyclic common
709 local-clock smooth, sum-to-zero site curves, participant curves and participant-day random
710 intercepts^{67–70}. A boundary-aware first-order autoregressive working structure prevented
711 residual sequences from crossing participant-day, clock-repeat or gap boundaries. Fitted-
712 curve dispersion was compared on a common 48-bin grid. A Shapley decomposition
713 averaged each component's addition to R^2 across all component orders, thereby distributing
714 the model's explained variance^{71,72}. The decomposition partitioned the full model's row-
715 weighted in-sample R^2 for $\log_{10}(\text{melanopic EDI} + 0.1 \text{ lx})$, defined as the reduction in squared
716 prediction error relative to predicting the fitted-sample mean. The percentages therefore
717 partition full-model R^2 on that transformed response scale. They are not percentages of
718 variance explained in raw melanopic EDI and do not indicate causal or out-of-sample
719 predictive importance. Confidence intervals for dispersion and R^2 -share ratios used 2,000
720 hierarchical cluster resamples while holding the fitted model structure fixed.

721 Geographic and photoperiod analyses

722 We analysed 17 participant-level or participant-day metrics with metric-specific Gaussian or
723 Tweedie exponential-dispersion models^{73,74}. We treated overall site differences, civil
724 photoperiod (day length including civil twilight), absolute latitude and the adequacy of a
725 linear latitude term relative to site as four separate model questions, each adjusted as a
726 complete 17-test FDR family. Participant-day models included participant random intercepts.
727 Mixed-effects models were fitted with `lme4` or `glmmTMB`, and component R^2 values were
728 obtained with `performance` where reported^{75–77}. Site and latitude were not entered as
729 independent fixed predictors in one model because each site has one latitude; site and
730 latitude models instead used the same observations and photoperiod adjustment. Site
731 follow-up contrasts were calculated with `emmeans` only after a supported overall site test and
732 compared each site with a site-average estimate, obtained by giving each included site
733 equal weight⁷⁸.

734 Nonlinear photoperiod analyses fitted each of nine planned metrics separately with a
735 generalised additive model estimated by restricted maximum likelihood, a thin-plate
736 photoperiod smooth with basis dimension six and participant random effects nested within
737 site. The derivative, meaning the slope of the fitted curve, was evaluated with `gratia` at 100

738 equally spaced photoperiod values with 95% intervals⁷⁹. A qualifying descriptive transition
739 required the preceding slope interval to be wholly above zero, the next interval to include
740 zero and every later interval through the recorded maximum to include zero as well. This
741 procedure located a change in the fitted slope but did not test equivalence, a mechanism or
742 a physiological ceiling. Expanded-basis, fixed-site, common-sample and leave-one-site-out
743 analyses assessed stability. The gap-timing-unaware sensitivity dataset still passed the
744 50%-per-hour and 80%-per-day rules, but did not use the timing of remaining missing
745 observations for an additional metric-specific adjustment.

746 **Immediate-context and routine analyses**

747 Light-source and activity-environment analyses modelled the zero-aware one-hour
748 geometric mean melanopic EDI with fixed-power quasi-Tweedie log-mean models and
749 participant-cluster-robust inference^{73,74}. Additive fixed-site models supplied primary category
750 tests. Category-by-site interaction models supplied site-average category estimates, ratios
751 and site-specific context when the interaction was supported. The activity-environment
752 analysis retained fractional $1/k$ weights for concurrent labels. FDR adjustments were applied
753 to the declared category and site-contrast families. Exact-zero calibration, residual
754 dependence, sparse cells and deletion refits were retained as model checks. A secondary
755 population-mean sensitivity added six participant-level category-proportion terms to the light-
756 source model while retaining participant-clustered covariance. This separated hour-to-hour
757 category differences within participants from between-participant differences in category
758 composition; it did not estimate participant random slopes. Exploratory `glmmTMB` Tweedie
759 log-link models fitted by maximum likelihood then added one common participant intercept to
760 site-by-category fixed effects at the near-eye position for light source and separately at both
761 positions for the activity-environment variable, with the power fixed at 1.539919^{76,77}. The
762 activity-environment models retained the five named categories, excluded Other-only hours
763 and used exact $1/k$ weights. Their fixed-predictor variance used the same fractional weights
764 so every participant-hour contributed one unit. Marginal and conditional R^2 used a lognormal
765 distribution-specific variance approximation, and hierarchy-respecting Shapley allocations
766 distributed marginal R^2 across site, category and their interaction. Percentages from the
767 exploratory time-of-day models instead partition variance among fitted linear-predictor
768 values. Because they do not compare this with observed outcome variance, they are not R^2 .
769 Results describe them as variation in fitted hourly patterns, not variance explained in raw
770 exposure. These were model-based point descriptions without bootstrap intervals. No model
771 included random category slopes.

772 The participant-hour analysis modelled each supported participant-hour with a zero-inclusive
773 quasi-Poisson log-link model, fixed site associations and participant-cluster HC3 uncertainty.
774 Free versus work day, active versus sedentary daily activity status and previous-night sleep
775 duration centred at 8 h were entered together. Three-test FDR families were applied
776 separately to pooled associations, predictor-by-site interactions and practical contrasts.

777 **Person-level analyses**

778 Light-related behaviour, visual sensitivity, chronotype, age and measured biological sex
779 were evaluated with site-adjusted metric-specific models. Participant-day outcomes
780 generally used Gaussian or Tweedie mixed models with participant random intercepts;
781 participant-level dynamics outcomes used fixed-effect models. Each analysis retained its
782 complete prespecified false-discovery-rate families, including non-estimable outcomes.
783 Chronotype instruments were modelled separately across five timing outcomes. Age
784 associations were scaled per decade, and female-minus-male contrasts used male as the
785 reference. Average associations and predictor-by-site interactions were distinct questions.

786 The complete biological-sex curve analysis used supported 30-minute arithmetic means
787 transformed as $\log_{10}(\text{melanopic EDI} + 0.1 \text{ lx})$. The generalised additive model included
788 cyclic common and female-specific curves, site curves, participant curves, participant-day
789 random effects and boundary-aware first-order autoregression. The primary test jointly
790 assessed the time-constant and time-varying female-minus-male contributions across the
791 complete 24-hour curve. Clock-specific uncertainty was participant-cluster robust. Level-
792 versus-shape attribution and adjustment for the activity-environment variable were
793 secondary or exploratory analyses.

794 **Multiplicity, uncertainty, model checks and sensitivities**

795 Tests were two-sided. We controlled the false discovery rate with the Benjamini-Hochberg
796 procedure separately within each declared complete family⁸⁰. We report 95% confidence
797 intervals and FDR-adjusted p values for confirmatory families. A result that did not meet an
798 FDR-adjusted criterion was treated as inconclusive rather than as evidence of absence or
799 equivalence. Unless stated otherwise, site-average estimates gave each included site equal
800 weight on the fitted scale before back-transformation. Exploratory analyses retained
801 separate multiplicity families and are labelled as such.

802 Model checks addressed convergence, rank, singularity, residual distribution, zero mass,
803 fitted support, heteroscedasticity, serial dependence, nonlinear-term overlap and participant
804 or site influence. Qualifications were attached to the affected results. Prespecified or

805 structured sensitivities included the separately prepared gap-timing-unaware dataset,
806 common sensor-position samples, alternative support or metric definitions, model-form
807 changes, participant deletion and leave-one-site-out checks. Near-eye and chest models
808 remained separate in every comparison. Analyses were conducted in R, with tidyverse
809 packages supporting data handling and reporting^{63,81}. Exact package versions, formulas,
810 inputs and output files are documented in the Supplementary Information and reproducibility
811 records.

812 Preregistration and deviations

813 The study was preregistered with AsPredicted (#273407)⁸². The final analyses preserve the
814 registered scientific questions but include documented changes to placement priority, state
815 and coverage rules, metric definitions, outcome construction, model families, site structure,
816 multiplicity and sensitivity handling. The main manuscript reports the resulting estimands
817 rather than the historical hypothesis sequence. A complete mapping in the Supplementary
818 Information distinguishes scientific deviations and qualifications from analytical history and
819 technical provenance.

820 Artificial-intelligence assistance

821 During the original manuscript preparation, generative artificial intelligence tools (ChatGPT
822 5.5) were used to support language editing, structural refinement and text condensation. AI
823 assistance was used to improve clarity, coherence, concision and journal fit, but **not** to
824 generate, analyse or interpret primary data.

825 During primary data analysis, AI assistance was used to debug code sections, support figure
826 and table styling, and resolve modelling convergence and singularity issues. Participant
827 comments in their native languages were translated into English with AI assistance. All
828 translations were reviewed and approved or corrected by the respective site authors. AI
829 assistance was **not** used to develop the primary analysis strategy, write analysis scripts or
830 interpret results.

831 During review, an agentic AI assistant (ChatGPT 5.6 Sol) was used to review the analyses
832 for potential issues. This review prompted additional sensitivity analyses to evaluate data-
833 preparation and model-selection choices, expanded model diagnostics, and improved effect-
834 size reporting through Shapley decompositions and more precise language. AI assistance
835 also supported expanded analysis documentation and reproducibility checks.

836 All AI-assisted text was critically reviewed, edited, and approved by the authors, who take
837 full responsibility for the content, accuracy, integrity, and conclusions of the manuscript.

838 Data availability

839 All data used in this study are available through the [MeLiDos Project GitHub organisation](#)
840 and are archived in the [MeLiDos Zenodo community](#). Public site datasets have persistent
841 records for Kumasi (GH), Madrid (ES), Dortmund (DE), Izmir (TR), Tübingen (DE), San José
842 (CR), Munich (DE), Borås (SE) and Delft (NL)^{83–91}. These records and other project
843 resources are indexed through the MeLiDos Project Data Hub⁹² and accessed reproducibly
844 through the documented `melidosData` interface⁹³.

845 Code availability

846 All analysis code is available in the [project repository](#), with rendered reports on [GitHub](#)
847 [Pages](#) and an archive on [Zenodo](#). The archived version will be reconciled with the final
848 manuscript snapshot before submission.

849 References

- 850 1. Czeisler, C. A. & Gooley, J. J. [Sleep and circadian rhythms in humans](#). *Cold Spring*
851 *Harbor Symposia on Quantitative Biology* **72**, 579–597 (2007).
- 852 2. Dijk, D.-J. & Archer, S. N. [Light, sleep, and circadian rhythms: Together again](#). *PLoS*
853 *Biology* **7**, e1000145 (2009).
- 854 3. Roenneberg, T. & Merrow, M. [Entrainment of the human circadian clock](#). *Cold Spring*
855 *Harbor Symposia on Quantitative Biology* **72**, 293–299 (2007).
- 856 4. Blume, C., Garbazza, C. & Spitschan, M. [Effects of light on human circadian](#)
857 [rhythms, sleep and mood](#). *Somnologie* **23**, 147–156 (2019).
- 858 5. Spitschan, M. et al. [Ocular light and optical radiation exposure as a modifiable](#)
859 [environmental determinant of health: Expert consensus on research gaps and](#)
860 [priorities](#). *BMC Medicine* **24**, (2026).
- 861 6. International Commission on Illumination. CIE S 026/E:2018: CIE System for
862 Metrology of Optical Radiation for ipRGC-Influenced Responses to Light. (2018)
863 doi:[10.25039/S026.2018](#).
- 864 7. Brown, T. M. [Melanopic illuminance defines the magnitude of human circadian light](#)
865 [responses under a wide range of conditions](#). *Journal of Pineal Research* **69**, (2020).
- 866 8. Brown, T. M. et al. [Recommendations for daytime, evening, and nighttime indoor light](#)
867 [exposure to best support physiology, sleep, and wakefulness in healthy adults](#).
868 *PLOS Biology* **20**, e3001571 (2022).
- 869 9. Spitschan, M. & Zauner, J. [Light exposure as a modifiable determinant of mental](#)
870 [health](#). *Current Psychiatry Reports* **28**, (2026).
- 871 10. Didikoglu, A. et al. [Associations between light exposure and sleep timing and](#)
872 [sleepiness while awake in a sample of UK adults in everyday life](#). *Proceedings of the*
873 *National Academy of Sciences* **120**, (2023).

- 874 11. Lok, R. et al. [Timing of outdoor light exposure is associated with sleep-wake](#)
875 [consolidation in community-dwelling older men](#). *Frontiers in Sleep* **2**, (2023).
- 876 12. Burns, A. C. et al. [Day and night light exposure are associated with psychiatric](#)
877 [disorders: An objective light study in >85,000 people](#). *Nature Mental Health* **1**, 853–
878 862 (2023).
- 879 13. Windred, D. P. et al. [Personal light exposure patterns and incidence of type 2](#)
880 [diabetes: Analysis of 13 million hours of light sensor data and 670,000 person-years](#)
881 [of prospective observation](#). *The Lancet Regional Health - Europe* **42**, 100943 (2024).
- 882 14. Windred, D. P. et al. [Light exposure at night and cardiovascular disease incidence](#).
883 *JAMA Network Open* **8**, e2539031 (2025).
- 884 15. Windred, D. P. et al. [Brighter nights and darker days predict higher mortality risk: A](#)
885 [prospective analysis of personal light exposure in >88,000 individuals](#). *Proceedings*
886 *of the National Academy of Sciences* **121**, (2024).
- 887 16. Esaki, Y. et al. [Effect of evening light exposure on sleep in bipolar disorder: A](#)
888 [longitudinal analysis for repeated measures in the APPLE cohort](#). *Australian & New*
889 *Zealand Journal of Psychiatry* **55**, 305–313 (2020).
- 890 17. Obayashi, K., Saeki, K. & Kurumatani, N. [Ambient light exposure and changes in](#)
891 [obesity parameters: A longitudinal study of the HEIJO-KYO cohort](#). *The Journal of*
892 *Clinical Endocrinology & Metabolism* **101**, 3539–3547 (2016).
- 893 18. Kim, M. et al. [Light at night in older age is associated with obesity, diabetes, and](#)
894 [hypertension](#). *Sleep* **46**, (2022).
- 895 19. Gubin, D. et al. [Light environment of arctic solstices is coupled with melatonin phase-](#)
896 [amplitude changes and decline of metabolic health](#). *Journal of Pineal Research* **77**,
897 (2024).

- 898 20. Vineis, P. [From john snow to omics: The long journey of environmental epidemiology.](#)
 899 *European Journal of Epidemiology* **33**, 355–363 (2018).
- 900 21. Vermeulen, R., Schymanski, E. L., Barabási, A.-L. & Miller, G. W. [The exposome and](#)
 901 [health: Where chemistry meets biology.](#) *Science* **367**, 392–396 (2020).
- 902 22. Wallace, D. A. et al. [Characteristics of objectively-measured naturalistic light](#)
 903 [exposure patterns in u.s. Adults: A cross-sectional analysis of two cohorts.](#) *Science*
 904 *of The Total Environment* **969**, 178839 (2025).
- 905 23. Savides, T. J., Messin, S., Senger, C. & Kripke, D. F. [Natural light exposure of young](#)
 906 [adults.](#) *Physiology & Behavior* **38**, 571–574 (1986).
- 907 24. Price, L. L. A., Khazova, M. & Udovičić, L. [Assessment of the light exposures of shift-](#)
 908 [working nurses in london and dortmund in relation to recommendations for sleep and](#)
 909 [circadian health.](#) *Annals of Work Exposures and Health* **66**, 447–458 (2021).
- 910 25. Biller, A. M. et al. [Physiologically-relevant light exposure and light behaviour in](#)
 911 [switzerland and malaysia.](#) *Journal of Exposure Science & Environmental*
 912 *Epidemiology* **36**, 409–422 (2025).
- 913 26. Huss, A. et al. [Shedding some light in the dark—a comparison of personal](#)
 914 [measurements with satellite-based estimates of exposure to light at night among](#)
 915 [children in the netherlands.](#) *Environmental Health Perspectives* **127**, 067001 (2019).
- 916 27. Duijnhoven, J. van et al. [Measuring light exposure in daily life: A review of wearable](#)
 917 [light loggers.](#) *Building and Environment* **274**, 112771 (2025).
- 918 28. Hartmeyer, S., Webler, F. & Andersen, M. [Towards a framework for light-dosimetry](#)
 919 [studies: Methodological considerations.](#) *Lighting Research & Technology* **55**, 377–
 920 399 (2022).
- 921 29. De vries, S. W., Mardaljevic, J. & Van duijnhoven, J. CHARACTERISING INTER-
 922 INDIVIDUAL VARIABILITY IN SIMULATED PERSONAL LIGHT EXPOSURE:

- 923 IMPACT OF BODY MORPHOLOGY ON LIGHT-DOSIMETRY. *CIE x051:2025*
 924 *Proceedings of the CIE 2025 Midterm Meeting Vienna, Austria, July 4-11, 2025:*
 925 *Scientific Conference (July 7-9, 2025)* 207–216 (2025)
 926 doi:[10.25039/x051.2025/62xbgg](https://doi.org/10.25039/x051.2025/62xbgg).
- 927 30. Vries, S. W. de, Mardaljevic, J. & Duijnhoven, J. van. [Impact of wear position on](#)
 928 [dosimeter performance: A hybrid measurement-simulation approach to quantify in-](#)
 929 [situ factors](#). *npj Biological Timing and Sleep* **3**, (2026).
- 930 31. Vries, S. W. de, Mardaljevic, J. & Duijnhoven, J. van. [Impact of wear position on](#)
 931 [dosimeter performance: Measurement validity under simulated indoor illumination](#).
 932 *npj Biological Timing and Sleep* **3**, (2026).
- 933 32. Gibaldi, A., Harb, E. N., Wildsoet, C. F. & Banks, M. S. [A child-friendly wearable](#)
 934 [device for quantifying environmental risk factors for myopia](#). *Translational Vision*
 935 *Science & Technology* **13**, 28 (2024).
- 936 33. Zauner, J., Vries, S. W. de, Didikoglu, A., Duijnhoven, J. van & Spitschan, M. Sensor
 937 placement causes outcome-dependent bias in ambulatory light-exposure estimates.
 938 *bioRxiv* <https://doi.org/10.64898/2026.07.28.741277> (2026)
 939 doi:[10.64898/2026.07.28.741277](https://doi.org/10.64898/2026.07.28.741277).
- 940 34. Spitschan, M. et al. [Illuminating the future of wearable light metrology: Overview of](#)
 941 [the MeLiDos project](#). *Measurement* **235**, 114909 (2024).
- 942 35. Guidolin, C. et al. [Protocol for a prospective, multicentre, cross-sectional cohort study](#)
 943 [to assess personal light exposure](#). *BMC Public Health* **24**, (2024).
- 944 36. Biller, A. M., Balakrishnan, P. & Spitschan, M. [Behavioural determinants of](#)
 945 [physiologically-relevant light exposure](#). *Communications Psychology* **2**, (2024).
- 946 37. Roenneberg, T., Pilz, L. K., Zerbini, G. & Winnebeck, E. C. [Chronotype and social](#)
 947 [jetlag: A \(self-\) critical review](#). *Biology* **8**, 54 (2019).

- 948 38. Tortello, C. *et al.* [Chronotype delay and sleep disturbances shaped by the antarctic](#)
949 [polar night](#). *Scientific Reports* **13**, (2023).
- 950 39. Bonatto, F. S. *et al.* [Daylight exposure and mood in real life: Direct association and](#)
951 [mediating role of sleep and routine regularity](#). *Chronobiology International* **41**, 1128–
952 1141 (2024).
- 953 40. Costa Lopes, L. da, Ribeiro da Silva Vallim, J., Tufik, S., Louzada, F. & D’Almeida, V.
954 [Associations between real-life light exposure patterns and sleep behaviour in](#)
955 [adolescents](#). *Journal of Sleep Research* **34**, (2024).
- 956 41. Stefani, O. *et al.* [Wearable light loggers in field conditions: Corneal light](#)
957 [characteristics, user compliance, and acceptance](#). *Clocks & Sleep* **6**, 619–634
958 (2024).
- 959 42. Zauner, J., Biller, A. M. & Spitschan, M. [Impact of light logger and dosimeter](#)
960 [placement on wearability and appeal in real-life settings](#). *Open Research Europe* **5**,
961 293 (2025).
- 962 43. Zauner, J. *et al.* [Auxiliary data, quality assurance and quality control for wearable](#)
963 [light loggers and optical radiation dosimeters](#). *npj Biological Timing and Sleep* **3**,
964 (2026).
- 965 44. Ishihara, A. *et al.* [Performance of wearable light sensors for measuring photopic and](#)
966 [melanopic illuminance under laboratory and free-living conditions](#). *SLEEPJ* **49**,
967 (2025).
- 968 45. Gemici, B. *et al.* Scene-based field validation of wearable light loggers. (2026)
969 doi:[10.48550/arXiv.2606.20719](#).
- 970 46. Agbeshie, G. K. *et al.* [Physiologically relevant real-world light exposure and its](#)
971 [behavioural and environmental determinants in Kumasi, Ghana](#). *Open Research*
972 *Europe* **5**, 300 (2025).

- 973 47. Akgun, S. G. et al. [Physiologically relevant real-world light exposure and its](#)
974 [behavioural and environmental determinants in Izmir, Türkiye](#). *Open Research*
975 *Europe* **6**, 333 (2026).
- 976 48. Walde, P. & Völlm, B. A. [The TRAPD approach as a method for questionnaire](#)
977 [translation](#). *Frontiers in Psychiatry* **14**, (2023).
- 978 49. Zauner, J., Udovicic, L. & Spitschan, M. [Power analysis for personal light exposure](#)
979 [measurements and interventions](#). *PLOS ONE* **19**, e0308768 (2024).
- 980 50. Guidolin, C., Zauner, J., Hartmeyer, S. L. & Spitschan, M. [Collecting, detecting, and](#)
981 [handling non-wear intervals in longitudinal light exposure data](#). *Behavior Research*
982 *Methods* **58**, (2026).
- 983 51. Carney, C. E. et al. [The consensus sleep diary: Standardizing prospective sleep self-](#)
984 [monitoring](#). *Sleep* **35**, 287–302 (2012).
- 985 52. Kaida, K. et al. [Validation of the karolinska sleepiness scale against performance and](#)
986 [EEG variables](#). *Clinical Neurophysiology* **117**, 1574–1581 (2006).
- 987 53. Tsanas, A. et al. [Daily longitudinal self-monitoring of mood variability in bipolar](#)
988 [disorder and borderline personality disorder](#). *Journal of Affective Disorders* **205**, 225–
989 233 (2016).
- 990 54. Sischka, P. E., Costa, A. P., Steffgen, G. & Schmidt, A. F. [The WHO-5 well-being](#)
991 [index validation based on item response theory and the analysis of measurement](#)
992 [invariance across 35 countries](#). *Journal of Affective Disorders Reports* **1**, 100020
993 (2020).
- 994 55. Harris, P. A. et al. [MyCap: A flexible and configurable platform for mobilizing the](#)
995 [participant voice](#). *JAMIA Open* **5**, (2022).
- 996 56. Patridge, E. F. & Bardyn, T. P. [Research electronic data capture \(REDCap\)](#). *Journal*
997 *of the Medical Library Association* **106**, (2018).

- 998 57. Bajaj, A., Rosner, B., Lockley, S. W. & Schernhammer, E. S. [Validation of a light](#)
999 [questionnaire with real-life photopic illuminance measurements: The harvard light](#)
1000 [exposure assessment questionnaire](#). *Cancer Epidemiology, Biomarkers &*
1001 *Prevention* **20**, 1341–1349 (2011).
- 1002 58. Kantermann, T., Sung, H. & Burgess, H. J. [Comparing the morningness-eveningness](#)
1003 [questionnaire and munich ChronoType questionnaire to the dim light melatonin](#)
1004 [onset](#). *Journal of Biological Rhythms* **30**, 449–453 (2015).
- 1005 59. Horne, J. A. & Östberg, O. [A self-assessment questionnaire to determine](#)
1006 [morningness-eveningness in human circadian rhythms](#). *International Journal of*
1007 *Chronobiology* **4**, 97–110 (1976).
- 1008 60. Verriotto, J. D. et al. [New methods for quantification of visual photosensitivity](#)
1009 [threshold and symptoms](#). *Translational Vision Science & Technology* **6**, 18 (2017).
- 1010 61. Grandner, M. A. et al. [Development and initial validation of the assessment of sleep](#)
1011 [environment \(ASE\): Describing and quantifying the impact of subjective](#)
1012 [environmental factors on sleep](#). *International Journal of Environmental Research and*
1013 *Public Health* **19**, 13599 (2022).
- 1014 62. Siraji, M. A. et al. [An inventory of human light exposure behaviour](#). *Scientific Reports*
1015 **13**, (2023).
- 1016 63. R Core Team. R: A language and environment for statistical computing.
1017 <https://www.R-project.org/> (2025).
- 1018 64. Zauner, J., Hartmeyer, S. & Spitschan, M. [LightLogR: Reproducible analysis of](#)
1019 [personal light exposure data](#). **10**, 7601 (2025).
- 1020 65. Hartmeyer, S. & Andersen, M. [Towards a framework for light-dosimetry studies:](#)
1021 [Quantification metrics](#). *Lighting Research & Technology* **56**, 337–365 (2023).

- 1022 66. Zauner, J., Guidolin, C. & Spitschan, M. [How to deal with darkness: Modeling and](#)
 1023 [visualization of zero-inflated personal light exposure data on a logarithmic scale.](#)
 1024 *Journal of Biological Rhythms* **40**, 480–490 (2025).
- 1025 67. Hastie, T. & Tibshirani, R. [Generalized additive models.](#) *Statistical Science* **1**, (1986).
- 1026 68. Hastie, T. & Tibshirani, R. [Generalized additive models for medical research.](#)
 1027 *Statistical Methods in Medical Research* **4**, 187–196 (1995).
- 1028 69. Wood, S. N. [Fast stable restricted maximum likelihood and marginal likelihood](#)
 1029 [estimation of semiparametric generalized linear models.](#) **73**, (2011).
- 1030 70. Pedersen, E. J., Miller, D. L., Simpson, G. L. & Ross, N. [Hierarchical generalized](#)
 1031 [additive models in ecology: An introduction with mgcv.](#) *PeerJ* **7**, e6876 (2019).
- 1032 71. Azen, R. & Budescu, D. V. [The dominance analysis approach for comparing](#)
 1033 [predictors in multiple regression.](#) *Psychological Methods* **8**, 129–148 (2003).
- 1034 72. Grömping, U. [Estimators of relative importance in linear regression based on](#)
 1035 [variance decomposition.](#) *The American Statistician* **61**, 139–147 (2007).
- 1036 73. Jørgensen, B. [Exponential dispersion models.](#) *Journal of the Royal Statistical*
 1037 *Society: Series B (Methodological)* **49**, 127–145 (1987).
- 1038 74. Dunn, P. K. & Smyth, G. K. [Series evaluation of tweedie exponential dispersion](#)
 1039 [model densities.](#) *Statistics and Computing* **15**, 267–280 (2005).
- 1040 75. Bates, D., Mächler, M., Bolker, B. & Walker, S. [Fitting linear mixed-effects models](#)
 1041 [using lme4.](#) **67**, (2015).
- 1042 76. Brooks, M. E. et al. [glmmTMB balances speed and flexibility among packages for](#)
 1043 [zero-inflated generalized linear mixed modeling.](#) **9**, (2017).

- 1044 77. Lüdtke, D., Ben-Shachar, M. S., Patil, I., Waggoner, P. & Makowski, D.
 1045 [Performance: An r package for assessment, comparison and testing of statistical](#)
 1046 [models](#). **6**, 3139 (2021).
- 1047 78. Lenth, R. V. & Piaskowski, J. Emmeans: Estimated marginal means, aka least-
 1048 squares means. <https://doi.org/10.32614/CRAN.package.emmeans> (2026)
 1049 doi:[10.32614/CRAN.package.emmeans](https://doi.org/10.32614/CRAN.package.emmeans).
- 1050 79. Simpson, G. L. [Gratia: An r package for exploring generalized additive models](#). **9**,
 1051 6962 (2024).
- 1052 80. Benjamini, Y. & Hochberg, Y. [Controlling the false discovery rate: A practical and](#)
 1053 [powerful approach to multiple testing](#). *Journal of the Royal Statistical Society Series*
 1054 *B: Statistical Methodology* **57**, 289–300 (1995).
- 1055 81. Wickham, H. et al. [Welcome to the tidyverse](#). **4**, 1686 (2019).
- 1056 82. Zauner, J., Didikoglu, A. & Spitschan, M. [MeLiDos Project – Multi-country](#)
 1057 [determinants of personal light exposure](#). (2026).
- 1058 83. Akuffo, K. O., Agbeshie, G. K., Zauner, J. & Spitschan, M. Personal light exposure
 1059 dataset for kumasi, ghana. <https://doi.org/10.5281/ZENODO.15576731> (2025)
 1060 doi:[10.5281/ZENODO.15576731](https://doi.org/10.5281/ZENODO.15576731).
- 1061 84. Baeza Moyano, D. et al. Personal light exposure dataset for madrid, spain.
 1062 <https://doi.org/10.5281/ZENODO.16834951> (2025) doi:[10.5281/ZENODO.16834951](https://doi.org/10.5281/ZENODO.16834951).
- 1063 85. Broszio, K., Zauner, J. & Spitschan, M. Personal light exposure dataset for dortmund,
 1064 germany. <https://doi.org/10.5281/ZENODO.18111232> (2025)
 1065 doi:[10.5281/ZENODO.18111232](https://doi.org/10.5281/ZENODO.18111232).
- 1066 86. Didikoglu, A. et al. Personal light exposure dataset for izmir, türkiye.
 1067 <https://doi.org/10.5281/ZENODO.16568109> (2025) doi:[10.5281/ZENODO.16568109](https://doi.org/10.5281/ZENODO.16568109).

- 1068 87. Guidolin, C., Zauner, J. & Spitschan, M. Personal light exposure dataset for
 1069 tuebingen, germany. <https://doi.org/10.5281/ZENODO.16895188> (2025)
 1070 doi:[10.5281/ZENODO.16895188](https://doi.org/10.5281/ZENODO.16895188).
- 1071 88. Sancho-Salas, A., von-Breymann, H., Zauner, J. & Spitschan, M. Personal light
 1072 exposure dataset for san pedro, san josé, costa rica.
 1073 <https://doi.org/10.5281/ZENODO.17289456> (2026) doi:[10.5281/ZENODO.17289456](https://doi.org/10.5281/ZENODO.17289456).
- 1074 89. Hilden, S., Lee, S., Zauner, J. & Spitschan, M. Personal light exposure dataset for
 1075 munich, germany. <https://doi.org/10.5281/ZENODO.16893901> (2025)
 1076 doi:[10.5281/ZENODO.16893901](https://doi.org/10.5281/ZENODO.16893901).
- 1077 90. Nilsson Tengelin, M., Svensson, I., Källberg, S., Zauner, J. & Spitschan, M. Personal
 1078 light exposure dataset for borås, sweden.
 1079 <https://doi.org/10.5281/ZENODO.18925834> (2026) doi:[10.5281/ZENODO.18925834](https://doi.org/10.5281/ZENODO.18925834).
- 1080 91. Aerts, S. *et al.* Personal light exposure dataset for delft, the netherlands.
 1081 <https://doi.org/10.5281/ZENODO.17979893> (2025) doi:[10.5281/ZENODO.17979893](https://doi.org/10.5281/ZENODO.17979893).
- 1082 92. Zauner, J. *et al.* MeLiDos project data hub.
 1083 <https://doi.org/10.5281/ZENODO.19678543> (2026) doi:[10.5281/ZENODO.19678543](https://doi.org/10.5281/ZENODO.19678543).
- 1084 93. Zauner, J. & Spitschan, M. melidosData: Load data from the MeLiDos field study.
 1085 <https://melidosproject.github.io/melidosData/> (2026).

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1119 Competing interests

1120 M.S. declares the following potential competing interests between 2021 and 2025: academic
1121 roles as a member of the Board of Directors of the Society of Light, Rhythms, and Circadian
1122 Health, Chair of Joint Technical Committee 20 of the International Commission on
1123 Illumination, member of the Daylight Academy, and Chair of the Research Data Alliance
1124 Working Group Optical Radiation and Visual Experience Data; remunerated roles as
1125 Speaker of the Steering Committee of the Daylight Academy, ad hoc reviewer for the Health
1126 and Digital Executive Agency of the European Commission and the Swedish Research
1127 Council, Associate Editor for *LEUKOS*, examiner for the University of Manchester, Flinders
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1137 EP23159999.4A, "System and method for corneal-plane physiologically-relevant light
1138 logging with an application to personalized light interventions related to health and well-
1139 being." M.S. declares that these roles and relationships had no influence on the work
1140 presented here.

1141 J.Z. declares the following potential competing interests between 2021 and 2025: academic
1142 roles as a member of Joint Technical Committee 20 of the International Commission on
1143 Illumination, member of the Research Data Alliance Working Group Optical Radiation and
1144 Visual Experience Data, and Speaker of group 2, melanopic effects of light, of the Technical
1145 Scientific Committee of the German Society of Lighting Technology and Design;
1146 remunerated roles as examiner for the Swiss Lighting Society; teacher for the German
1147 Society of Lighting Technology and Design, Munich University of Applied Sciences and the
1148 Technical University of Applied Sciences Rosenheim; associated partner at 3lpi lighting
1149 design + engineering, Munich; tool and three-dimensional-model designer for Zumtobel
1150 Lighting GmbH; and course designer for Munich University of Applied Sciences and Virtual
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1158 A.D. declares academic roles as a member of Joint Technical Committee 20 of the
1159 International Commission on Illumination and Division Reporter DR6-50 for the “4th
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1162 company active in the lighting sector. All other authors declare no competing interests.

1163 **Supplementary Information**

1164 **Study support and exposure definitions**

1165 **Supplementary Table S1. Descriptive sample flow**

Descriptive sample flow and analytical support		
Stage	Quantity	Count
Participant roster		
Available normalized participant metadata	Participants	191
Primary near-eye sensor position		
At least 80% complete before all-zero screen	Participant-days	818
Exact all-zero days excluded	Participant-days	2
Final descriptive dataset	Participants	141
Final descriptive dataset	Participant-days	816
Final descriptive dataset	One-minute real observations	1,175,160
Complementary chest sensor position		
At least 80% complete before all-zero screen	Participant-days	905
Exact all-zero days excluded	Participant-days	3
Final descriptive dataset	Participants	154
Final descriptive dataset	Participant-days	902
Final descriptive dataset	One-minute real observations	1,298,880
Paired main subset		
Paired main subset	Participants	112
Paired main subset	Paired participant-days	643
Counts use their stated denominator. An excluded participant-day is an all-zero day, not necessarily an excluded participant. The paired row reports paired participant-days, not minute-level observation pairs.		

1166
1167 **Supplementary Table S1.** Participant, participant-day, and one-minute observation counts across the roster, sensor-specific completeness and all-zero
1168 screens, final descriptive datasets, and paired main subset. Each row names its denominator explicitly. An excluded participant-day is not necessarily an
1169 excluded participant.

1170 **Supplementary Table S2. Full near-eye metric dictionary and distributions**

Metric descriptive summary (near eye)													
Metric ¹	Unit	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)	Scaling ²	Distribution ³
Duration													
Time above 1,000 lx meEDI Bright-light exposure duration; relevant to daytime alerting and circadian entrainment.	HH:MM	00:41 (00:11, 01:33) <i>01:10 ± 01:28</i> N=141; d=816	01:18 (00:31, 03:24) <i>02:00 ± 01:54</i> N=13; d=78	01:18 (00:29, 02:12) <i>01:33 ± 01:28</i> N=13; d=78	01:03 (00:20, 02:32) <i>01:48 ± 02:14</i> N=18; d=107	00:40 (00:07, 01:25) <i>01:06 ± 01:22</i> N=26; d=150	00:39 (00:10, 01:46) <i>01:09 ± 01:18</i> N=10; d=60	00:27 (00:06, 00:53) <i>00:38 ± 00:47</i> N=23; d=129	00:30 (00:06, 01:04) <i>00:44 ± 00:51</i> N=17; d=101	00:19 (00:09, 00:43) <i>00:32 ± 00:33</i> N=6; d=32	00:43 (00:17, 01:14) <i>00:58 ± 00:57</i> N=15; d=81	Symlog (base 10; threshold 1)	
Time above 250 lx meEDI during wake Waking time in recommended daytime light; relevant to alertness, entrainment, and subsequent sleep.	HH:MM	02:25 (00:51, 04:37) <i>03:00 ± 02:29</i> N=141; d=737	04:06 (02:30, 06:15) <i>04:21 ± 02:24</i> N=13; d=73	03:29 (02:03, 05:16) <i>03:39 ± 02:25</i> N=13; d=64	03:42 (02:05, 05:58) <i>04:05 ± 02:51</i> N=18; d=90	02:01 (00:47, 04:20) <i>02:52 ± 02:39</i> N=26; d=141	02:38 (00:55, 04:19) <i>02:57 ± 02:19</i> N=10; d=54	02:43 (00:54, 04:43) <i>03:02 ± 02:19</i> N=23; d=113	01:30 (00:41, 03:26) <i>02:09 ± 01:57</i> N=17; d=94	01:39 (00:33, 02:40) <i>01:48 ± 01:21</i> N=6; d=30	01:14 (00:25, 02:37) <i>01:45 ± 01:46</i> N=15; d=78	Symlog (base 10; threshold 1)	
Time below 10 lx meEDI before sleep Low-light time before bed; limits evening melatonin suppression and circadian delay.	HH:MM	01:53 (01:01, 02:35) <i>01:51 ± 01:02</i> N=139; d=655	02:16 (01:30, 02:39) <i>02:05 ± 00:53</i> N=13; d=67	01:50 (01:21, 02:52) <i>01:59 ± 00:57</i> N=12; d=50	01:42 (00:53, 02:17) <i>01:41 ± 00:58</i> N=18; d=84	01:52 (01:04, 02:25) <i>01:52 ± 01:06</i> N=26; d=132	01:43 (00:42, 02:32) <i>01:45 ± 01:10</i> N=10; d=52	01:43 (00:58, 02:50) <i>01:50 ± 01:05</i> N=23; d=105	01:31 (00:49, 02:15) <i>01:38 ± 01:00</i> N=17; d=82	02:16 (01:15, 02:39) <i>01:56 ± 01:00</i> N=6; d=25	02:30 (01:28, 02:52) <i>02:09 ± 00:57</i> N=14; d=58	Symlog (base 10; threshold 1)	
Time below 1 lx meEDI during sleep Darkness during sleep; supports nocturnal melatonin and an undisturbed sleep environment.	HH:MM	07:08 (05:58, 08:15) <i>07:04 ± 01:57</i> N=141; d=778	07:08 (06:03, 08:18) <i>07:12 ± 01:46</i> N=13; d=76	07:14 (05:56, 08:06) <i>07:01 ± 01:50</i> N=13; d=68	07:08 (05:56, 08:16) <i>07:05 ± 01:50</i> N=18; d=95	06:54 (05:51, 07:46) <i>06:47 ± 01:30</i> N=26; d=150	05:52 (04:41, 07:03) <i>05:58 ± 01:58</i> N=10; d=60	08:06 (07:10, 09:10) <i>08:02 ± 01:57</i> N=23; d=118	06:45 (05:08, 08:15) <i>06:40 ± 02:07</i> N=17; d=99	06:56 (06:06, 07:36) <i>06:39 ± 01:43</i> N=6; d=31	07:31 (06:31, 09:00) <i>07:35 ± 02:15</i> N=15; d=81	Symlog (base 10; threshold 1)	
Longest period above 250 lx meEDI Longest sustained bright-light bout; captures continuity of daytime circadian stimulation.	HH:MM	00:38 (00:17, 01:12) <i>00:55 ± 01:00</i> N=141; d=816	00:57 (00:28, 01:27) <i>01:12 ± 01:02</i> N=13; d=78	00:48 (00:23, 01:20) <i>01:09 ± 01:15</i> N=13; d=78	01:00 (00:21, 01:40) <i>01:20 ± 01:31</i> N=18; d=107	00:35 (00:13, 01:06) <i>00:50 ± 00:54</i> N=26; d=150	00:38 (00:17, 01:32) <i>01:01 ± 01:02</i> N=10; d=60	00:38 (00:16, 01:04) <i>00:45 ± 00:36</i> N=23; d=129	00:34 (00:14, 00:55) <i>00:43 ± 00:41</i> N=17; d=101	00:28 (00:16, 00:43) <i>00:30 ± 00:20</i> N=6; d=32	00:23 (00:15, 00:49) <i>00:38 ± 00:45</i> N=15; d=81	Symlog (base 10; threshold 1)	
Dynamics													
Interdaily stability Day-to-day regularity of the light–dark pattern; higher regularity supports circadian stability.	—	0.308 (0.248, 0.38) <i>0.318 ± 0.095</i> N=141; d=816	0.311 (0.272, 0.367) <i>0.315 ± 0.078</i> N=13; d=78	0.258 (0.239, 0.389) <i>0.288 ± 0.094</i> N=13; d=78	0.264 (0.219, 0.319) <i>0.274 ± 0.074</i> N=18; d=107	0.329 (0.263, 0.424) <i>0.344 ± 0.107</i> N=26; d=150	0.244 (0.218, 0.277) <i>0.252 ± 0.058</i> N=10; d=60	0.352 (0.321, 0.455) <i>0.368 ± 0.092</i> N=23; d=129	0.278 (0.254, 0.394) <i>0.323 ± 0.104</i> N=17; d=101	0.342 (0.315, 0.473) <i>0.37 ± 0.118</i> N=6; d=32	0.293 (0.255, 0.354) <i>0.299 ± 0.062</i> N=15; d=81	Identity	
Intradaily variability Within-day fragmentation of light exposure; higher values indicate less consolidated light–dark input.	—	1.253 (0.93, 1.502) <i>1.229 ± 0.389</i> N=141; d=816	0.997 (0.818, 1.388) <i>1.088 ± 0.343</i> N=13; d=78	1.193 (0.96, 1.389) <i>1.238 ± 0.345</i> N=13; d=78	1.182 (0.928, 1.49) <i>1.177 ± 0.407</i> N=18; d=107	1.154 (0.801, 1.442) <i>1.125 ± 0.354</i> N=26; d=150	1.48 (0.869, 1.642) <i>1.332 ± 0.477</i> N=10; d=60	1.251 (0.999, 1.497) <i>1.251 ± 0.413</i> N=23; d=129	1.359 (0.95, 1.546) <i>1.307 ± 0.415</i> N=17; d=101	1.399 (1.282, 1.483) <i>1.355 ± 0.256</i> N=6; d=32	1.324 (1.029, 1.595) <i>1.344 ± 0.413</i> N=15; d=81	Identity	

1171

Metric descriptive summary (near eye)													
Metric ¹	Unit	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)	Scaling ²	Distribution ³
Exposure history													
melEDI dose Intensity–duration-weighted melanopic exposure; summarizes cumulative non-visual retinal light input.	klx·h	4.96 (1.936, 12.313) <i>10.358 ± 16.216</i> N=141; d=761	9.296 (3.892, 21.895) <i>19.923 ± 27.784</i> N=13; d=71	10.495 (3.855, 17.395) <i>15.881 ± 20.395</i> N=13; d=74	6.751 (2.469, 14.435) <i>13.809 ± 23.711</i> N=18; d=97	3.862 (1.411, 10.217) <i>7.719 ± 9.847</i> N=26; d=137	7.11 (1.897, 15.789) <i>10.055 ± 11.237</i> N=10; d=55	3.317 (1.703, 6.29) <i>6.115 ± 8.79</i> N=23; d=125	4.35 (1.791, 10.625) <i>7.49 ± 8.898</i> N=17; d=96	2.195 (1.339, 4.117) <i>3.281 ± 2.573</i> N=6; d=30	6.045 (2.49, 14.907) <i>10.016 ± 10.799</i> N=15; d=76	Symlog (base 10; threshold 1)	
Level													
Mean melEDI Geometric average of daily melEDI values, including zeros; summarizes overall exposure while reducing peak influence.	lx	5.154 (2.831, 9.225) <i>7.587 ± 9.496</i> N=141; d=816	7.569 (4.55, 11.613) <i>9.171 ± 7.111</i> N=13; d=78	6.659 (4.18, 11.674) <i>8.274 ± 6.558</i> N=13; d=78	7.704 (3.081, 13.924) <i>12.017 ± 18.865</i> N=18; d=107	5.232 (3.224, 8.188) <i>7.221 ± 6.33</i> N=26; d=150	9.013 (4.011, 15.089) <i>12.102 ± 12.013</i> N=10; d=60	3.505 (1.469, 6.211) <i>4.166 ± 3.272</i> N=23; d=129	5.857 (3.726, 7.986) <i>7.778 ± 6.946</i> N=17; d=101	4.325 (3.585, 6.77) <i>5.248 ± 2.85</i> N=6; d=32	2.144 (0.744, 4.042) <i>3.019 ± 2.896</i> N=15; d=81	Symlog (base 10; threshold 1)	
Brightest 10 h mean Mean of the brightest 10 hours; reflects the strength of the main daytime light episode.	lx	110.566 (41.513, 243.212) <i>214.866 ± 508.56</i> N=141; d=816	252.689 (104.514, 461.953) <i>399.302 ± 518.102</i> N=13; d=78	175.974 (78.625, 299.106) <i>262.206 ± 329.201</i> N=13; d=78	156.184 (67.783, 339.731) <i>422.54 ± 1,209.616</i> N=18; d=107	100.057 (33.255, 209.904) <i>179.372 ± 241.53</i> N=26; d=150	130.78 (54.044, 259.726) <i>201.947 ± 194.042</i> N=10; d=60	101.102 (24.355, 218.457) <i>136.312 ± 134.07</i> N=23; d=129	83.24 (53.17, 151.786) <i>120.172 ± 101.545</i> N=17; d=101	68.143 (45.331, 132.458) <i>96.735 ± 72.311</i> N=6; d=32	46.66 (9.242, 103.175) <i>82.489 ± 115.07</i> N=15; d=81	Symlog (base 10; threshold 1)	
Darkest 10 h mean Mean of the darkest 10 hours; lower values during the biological night favour melatonin preservation and sleep.	lx	0.103 (0.02, 0.253) <i>0.243 ± 0.513</i> N=141; d=816	0.08 (0.016, 0.191) <i>0.183 ± 0.299</i> N=13; d=78	0.1 (0.035, 0.23) <i>0.2 ± 0.332</i> N=13; d=78	0.14 (0.036, 0.282) <i>0.223 ± 0.299</i> N=18; d=107	0.149 (0.074, 0.309) <i>0.273 ± 0.465</i> N=26; d=150	0.325 (0.119, 0.71) <i>0.515 ± 0.653</i> N=10; d=60	0.022 (0, 0.08) <i>0.054 ± 0.075</i> N=23; d=129	0.152 (0.052, 0.481) <i>0.475 ± 0.882</i> N=17; d=101	0.133 (0.055, 0.309) <i>0.411 ± 1.044</i> N=6; d=32	0.01 (0, 0.092) <i>0.059 ± 0.104</i> N=15; d=81	Symlog (base 10; threshold 1)	
Spectrum													
Melanopic daylight efficacy ratio Mean of viable one-minute melEDI/illuminance ratios; indicates melanopic efficacy relative to visual light.	—	0.724 (0.643, 0.795) <i>0.724 ± 0.116</i> N=137; d=687	0.788 (0.725, 0.872) <i>0.796 ± 0.162</i> N=13; d=74	0.743 (0.699, 0.796) <i>0.743 ± 0.077</i> N=13; d=65	0.781 (0.675, 0.853) <i>0.768 ± 0.118</i> N=18; d=97	0.65 (0.592, 0.728) <i>0.667 ± 0.098</i> N=26; d=144	0.724 (0.652, 0.773) <i>0.72 ± 0.093</i> N=10; d=55	0.668 (0.629, 0.721) <i>0.673 ± 0.07</i> N=22; d=83	0.747 (0.634, 0.834) <i>0.735 ± 0.122</i> N=17; d=91	0.719 (0.661, 0.761) <i>0.706 ± 0.067</i> N=6; d=30	0.755 (0.729, 0.82) <i>0.758 ± 0.097</i> N=12; d=48	Identity	

Metric descriptive summary (near eye)													
Metric ¹	Unit	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)	Scaling ²	Distribution ³
Timing													
Midpoint of the brightest 10 hours Centre time of the brightest 10 hours; indexes the main daily circadian light cue.	HH:MM clock time	13:44 (12:48, 15:00) 13:56 ± 01:51 N=141; d=816	13:16 (12:28, 14:10) 13:20 ± 01:33 N=13; d=78	14:12 (13:23, 15:18) 14:22 ± 01:51 N=13; d=78	13:52 (12:54, 15:04) 14:00 ± 01:42 N=18; d=107	13:38 (12:47, 14:44) 13:52 ± 01:42 N=26; d=150	14:54 (12:50, 16:03) 14:31 ± 02:14 N=10; d=60	13:44 (13:06, 15:20) 14:10 ± 01:51 N=23; d=129	14:10 (13:32, 15:21) 14:25 ± 01:37 N=17; d=101	12:34 (11:49, 14:02) 12:55 ± 01:44 N=6; d=32	12:49 (12:06, 13:39) 13:03 ± 01:58 N=15; d=81	Circular clock	
Midpoint of the darkest 10 hours Centre time of the darkest 10 hours; indexes the main daily darkness cue.	HH:MM clock time	02:54 (02:01, 03:50) 02:58 ± 01:41 N=141; d=816	02:02 (01:25, 02:48) 02:03 ± 01:07 N=13; d=78	02:55 (02:01, 03:27) 02:50 ± 02:05 N=13; d=78	02:32 (01:45, 03:22) 02:37 ± 01:19 N=18; d=107	03:08 (02:11, 04:10) 03:16 ± 01:34 N=26; d=150	02:37 (01:54, 04:17) 03:10 ± 02:00 N=10; d=60	03:39 (02:43, 04:34) 03:34 ± 01:49 N=23; d=129	03:31 (02:33, 04:14) 03:26 ± 01:25 N=17; d=101	01:45 (01:05, 03:01) 02:10 ± 01:27 N=6; d=32	02:33 (02:04, 03:17) 02:36 ± 01:36 N=15; d=81	Circular clock	
First light timing above 250 lx melEDI First waking bright-light exposure; morning timing can advance circadian phase and promote alertness.	HH:MM clock time	09:08 (08:02, 10:40) 09:26 ± 02:22 N=140; d=727	07:57 (06:56, 08:33) 08:03 ± 01:32 N=13; d=74	09:04 (08:18, 09:52) 09:17 ± 01:56 N=13; d=68	08:36 (07:10, 10:13) 08:45 ± 02:21 N=18; d=93	09:43 (08:13, 10:53) 09:45 ± 02:21 N=25; d=133	09:36 (07:58, 10:53) 09:39 ± 03:12 N=10; d=56	09:32 (08:40, 11:21) 10:05 ± 02:18 N=23; d=118	10:01 (09:06, 11:52) 10:26 ± 02:28 N=17; d=95	08:11 (07:26, 09:57) 08:36 ± 01:30 N=6; d=29	08:43 (08:01, 10:10) 09:19 ± 01:52 N=15; d=61	Circular clock	
Last light timing above 250 lx melEDI Last bright-light exposure; later timing may delay circadian phase and sleep onset.	HH:MM clock time	18:08 (16:27, 19:42) 18:03 ± 02:36 N=141; d=687	18:41 (17:38, 19:38) 18:39 ± 01:38 N=13; d=73	18:30 (17:37, 19:59) 18:37 ± 02:08 N=13; d=62	19:34 (18:04, 20:42) 19:16 ± 02:28 N=18; d=89	17:37 (15:35, 19:09) 17:18 ± 02:44 N=26; d=121	19:50 (18:37, 20:50) 19:33 ± 02:11 N=10; d=53	17:48 (16:46, 19:19) 18:01 ± 02:21 N=23; d=106	18:45 (17:02, 19:38) 18:23 ± 02:15 N=17; d=89	15:45 (13:20, 16:38) 15:22 ± 02:38 N=6; d=28	15:54 (14:34, 17:25) 15:46 ± 02:13 N=15; d=66	Circular clock	
Mean timing of exposure above 250 lx melEDI Average bright-light timing; summarizes the phase of daily circadian stimulation.	HH:MM clock time	13:29 (12:29, 14:39) 13:31 ± 01:52 N=141; d=742	13:20 (12:29, 14:24) 13:21 ± 01:25 N=13; d=72	13:52 (13:13, 15:06) 14:02 ± 01:41 N=13; d=71	13:44 (12:28, 14:55) 13:51 ± 01:56 N=18; d=93	13:17 (12:17, 14:31) 13:24 ± 01:45 N=26; d=138	14:04 (12:58, 15:25) 14:09 ± 02:22 N=10; d=55	13:43 (12:55, 14:29) 13:47 ± 01:33 N=23; d=117	13:58 (12:54, 14:58) 14:04 ± 01:39 N=17; d=94	11:30 (10:28, 12:38) 11:26 ± 01:36 N=6; d=31	12:17 (11:04, 13:04) 12:11 ± 01:36 N=15; d=71	Circular clock	
Median (25th percentile, 75th percentile), <i>circular or arithmetic mean ± standard deviation</i> , N=participants; d=participant-days with a finite value for that metric.													
¹ The brief meaning and relevance notes describe established physiological constructs; this descriptive table does not estimate individual health effects.													
² Scaling describes the distribution column only; printed values remain on their stated scale. Symlog uses base 10 with a linear region through 1 in the stated unit; Identity is linear; Circular clock unwraps values around the metric-specific clock centre.													
³ Red lines indicate site medians. Symlog and circular-clock rows are transformed only for plotting.													

1173

1174 **Supplementary Table S2.** Definitions, exact support, median (interquartile range) summaries, and density displays for the 17 near-eye personal light-exposure metrics overall and by site. Values use three decimal places where 1175 applicable. Metrics are distinct summaries and should not be interpreted as interchangeable outcomes.

1176 **Supplementary Table S3. Descriptive recommendation-range fractions**

Site ¹	Minutes in the recommended range: ²				Share of all eligible real minutes: ³			
	Daytime	Pre-sleep	Sleep ⁴	Total ⁵	Daytime	Pre-sleep	Sleep	Unclassified ⁶
Overall	24.0% 137,792 / 573,712	63.3% 81,894 / 129,390	87.7% 336,052 / 383,366	51.2% 555,738 / 1,086,468	51.6% 606,215 / 1,175,160	11.9% 140,152 / 1,175,160	32.6% 383,607 / 1,175,160	3.8% 45,186 / 1,175,160
Borås (SE)	33.2% 19,225 / 57,952	69.9% 8,914 / 12,759	93.4% 33,152 / 35,489	57.7% 61,291 / 106,200	54.0% 60,642 / 112,260	12.2% 13,704 / 112,260	31.7% 35,549 / 112,260	2.1% 2,365 / 112,260
Delft (NL)	29.6% 14,760 / 49,852	66.7% 7,181 / 10,774	86.4% 29,915 / 34,621	54.4% 51,856 / 95,247	47.0% 52,755 / 112,260	10.9% 12,283 / 112,260	30.9% 34,682 / 112,260	11.2% 12,540 / 112,260
Dortmund (DE)	32.4% 22,631 / 69,900	59.6% 9,749 / 16,361	87.5% 42,829 / 48,928	55.6% 75,209 / 135,189	48.1% 74,094 / 154,080	11.3% 17,409 / 154,080	31.8% 48,928 / 154,080	8.9% 13,649 / 154,080
Tübingen (DE)	22.5% 25,082 / 111,396	61.8% 15,623 / 25,283	87.1% 61,030 / 70,029	49.2% 101,735 / 206,708	55.1% 119,150 / 216,120	12.4% 26,881 / 216,120	32.4% 70,089 / 216,120	0.0% 0 / 216,120
Munich (DE)	25.0% 10,681 / 42,685	60.6% 5,963 / 9,833	71.7% 21,485 / 29,951	46.2% 38,129 / 82,469	53.0% 45,831 / 86,400	12.3% 10,618 / 86,400	34.7% 29,951 / 86,400	0.0% 0 / 86,400
Madrid (ES)	24.2% 21,067 / 87,030	62.7% 12,766 / 20,366	95.9% 58,374 / 60,891	54.8% 92,207 / 168,287	48.4% 89,948 / 185,880	11.7% 21,684 / 185,880	32.8% 60,891 / 185,880	7.2% 13,357 / 185,880
Izmir (TR)	17.8% 12,708 / 71,562	55.1% 9,221 / 16,736	79.0% 39,936 / 50,556	44.6% 61,865 / 138,854	51.5% 74,926 / 145,440	12.3% 17,883 / 145,440	34.8% 50,556 / 145,440	1.4% 2,075 / 145,440
San José (CR)	13.8% 3,266 / 23,640	64.6% 3,124 / 4,839	87.1% 12,438 / 14,275	44.0% 18,828 / 42,754	54.9% 25,285 / 46,080	11.5% 5,320 / 46,080	31.0% 14,275 / 46,080	2.6% 1,200 / 46,080
Kumasi (GH)	14.0% 8,372 / 59,695	75.2% 9,353 / 12,439	95.5% 36,893 / 38,626	49.3% 54,618 / 110,760	54.5% 63,584 / 116,640	12.3% 14,370 / 116,640	33.2% 38,686 / 116,640	0.0% 0 / 116,640

¹ Near eye participant and participant-day sample sizes are reported in the participant table; every percentage cell gives its exact minute denominator.

² Recommended ranges follow Brown et al. (2022): Daytime ≥250 lx melanopic EDI, Pre-sleep ≤10 lx melanopic EDI, and Sleep ≤1 lx melanopic EDI. Daytime, Pre-sleep, and Sleep identify the recommendation windows. Each grey numerator/denominator gives minutes within the applicable recommendation range over all valid one-minute observations in that window. Sleep describes the bedside sleep environment.

³ Each grey relevant / all count gives minutes in the displayed diary state over all eligible real minutes before classification by diary state or measurement availability.

⁴ During diary-defined sleep the bedside sensor describes the sleep environment rather than direct ocular exposure.

⁵ Total pools all classified valid minutes.

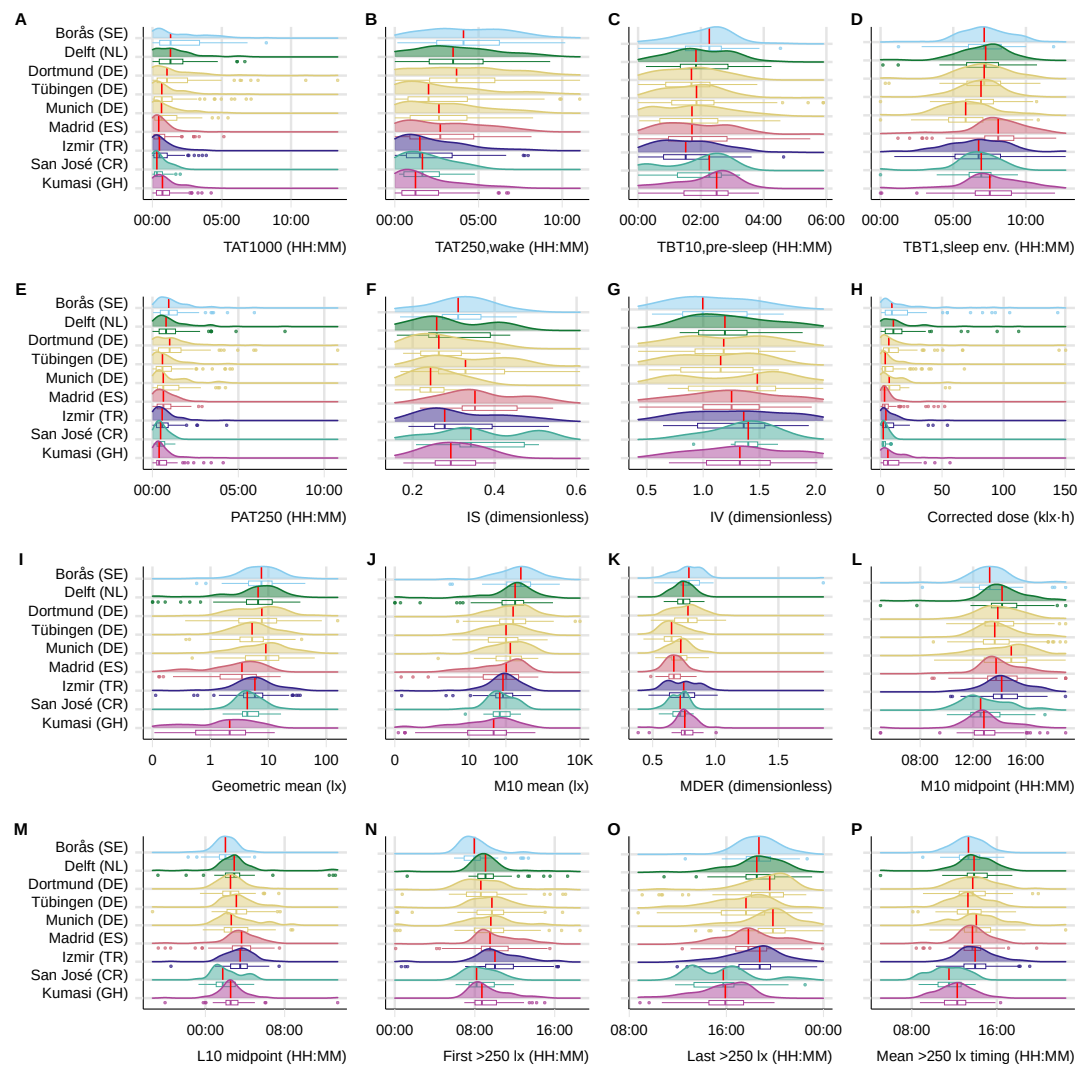
⁶ Unclassified time can result from missing diary state, missing or removed melEDI measurements, or both.

1177

1178 **Supplementary Table S3.** Fractions of valid near-eye minutes meeting each Brown et al. recommendation range, with explicit numerators and denominators.
1179 Daytime, Pre-sleep, and Sleep identify the recommendation windows. The Sleep summaries use near-eye illuminance recorded during sleep and describe the
1180 bedside sleep environment rather than verified ocular exposure. These pooled-minute summaries are descriptive and differ from the modelled
1181 recommendation-window-period estimates in main Table 2.

1182 **Supplementary Figure S1. Near-eye metric distributions**

1183

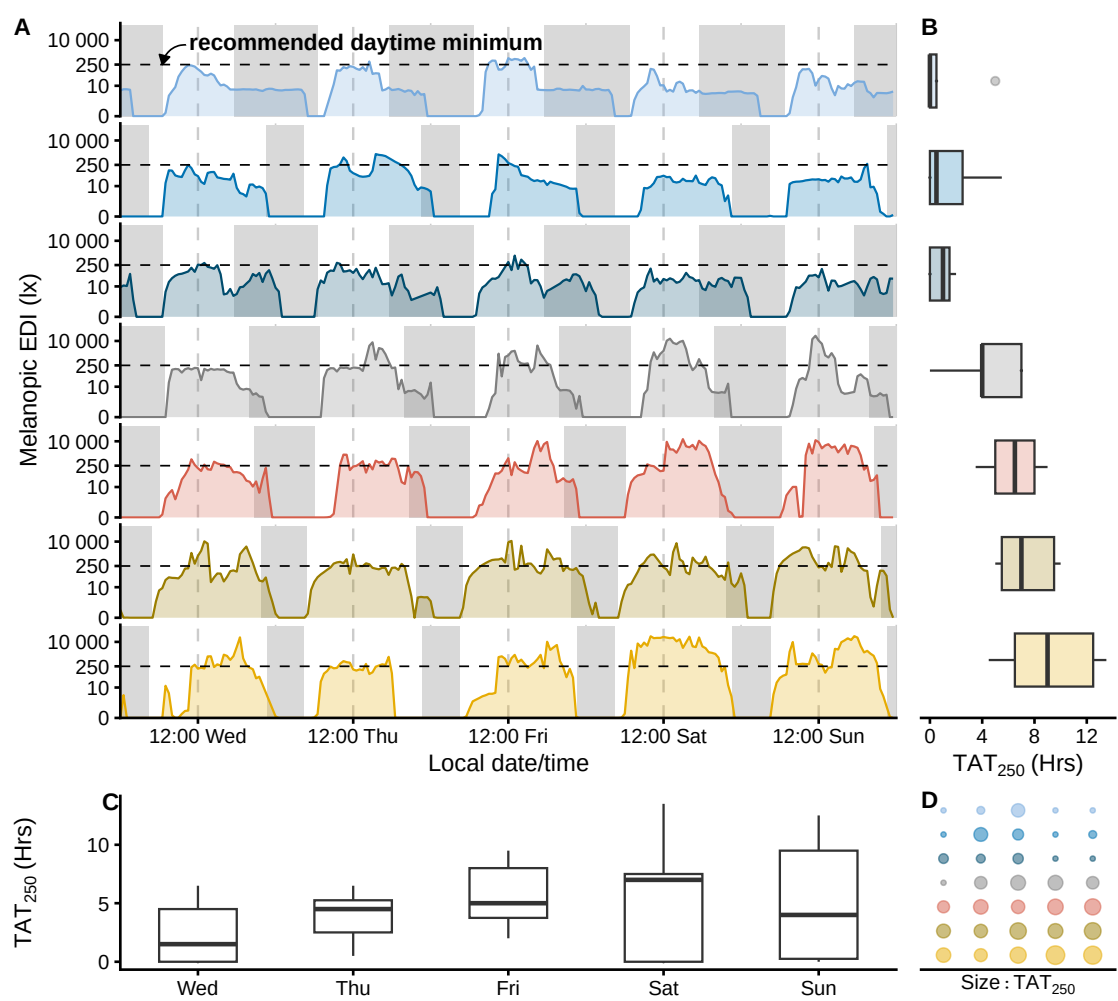


1184

1185 Supplementary Figure S1. Detailed near-eye metric distributions by country-coded study
1186 site.

1187 **Supplementary Figure S2. Worked metric derivation**

1188



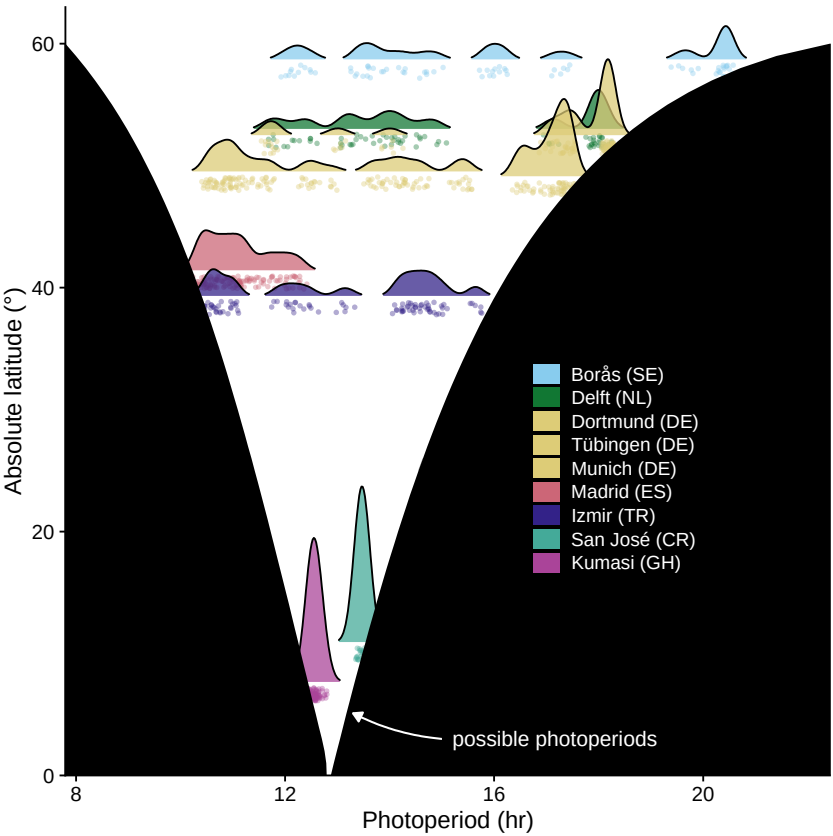
TAT = time above threshold during daytime; subscripts give melEDI thresholds in lx.
Lines and TAT250 values use the same stored 30-minute near-eye samples.

1189

1190 Supplementary Figure S2. Examples linking 30-minute time series to the daily time-
1191 above-250-lx metric.

1192 **Supplementary Figure S3. Latitude and civil-photoperiod**
1193 **measurement range**

1194



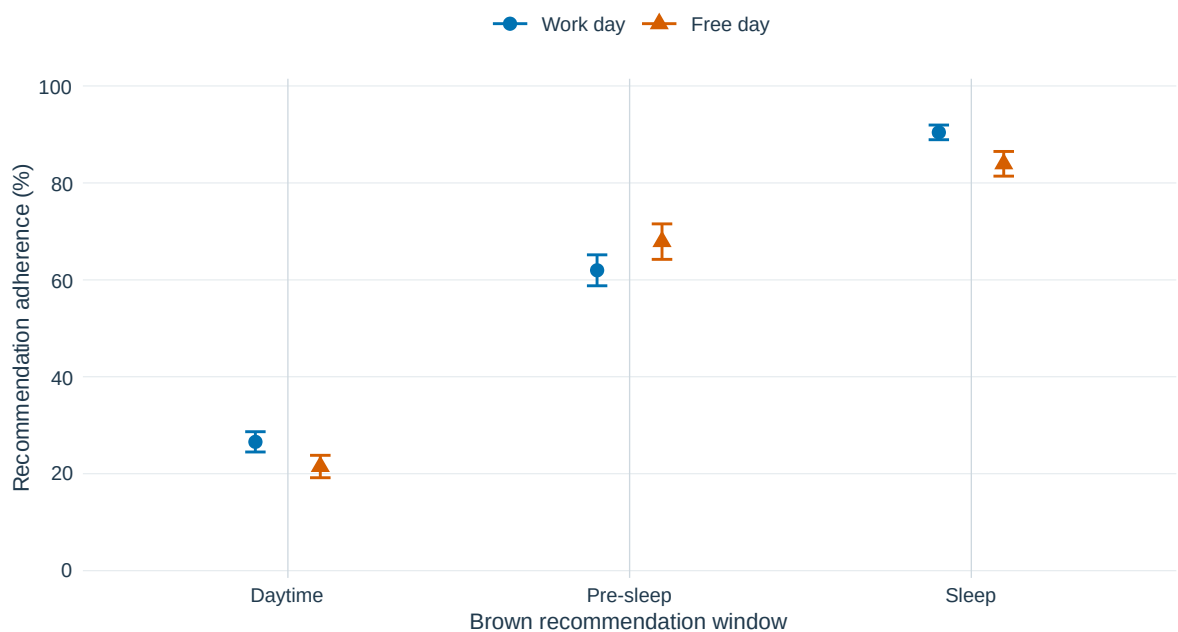
1195
1196 Supplementary Figure S3. Observed civil photoperiod and theoretical bounds by absolute
1197 latitude.

1198 **Recommendation adherence**

1199 Daytime, Pre-sleep, and Sleep identify the Brown et al. recommendation windows. In the
1200 modelled comparisons, the wake-start date assigns the work or free label to the preceding
1201 sleep interval, ensuing daytime interval and following pre-sleep window. Each window
1202 qualifies independently. These modelled comparisons are distinct from the pooled-minute
1203 fractions in Supplementary Table S3.

1204 **Supplementary Figure S4. Site-average recommendation adherence**

1205



1206 Equal weight for nine sites. Bars are 95% confidence intervals.

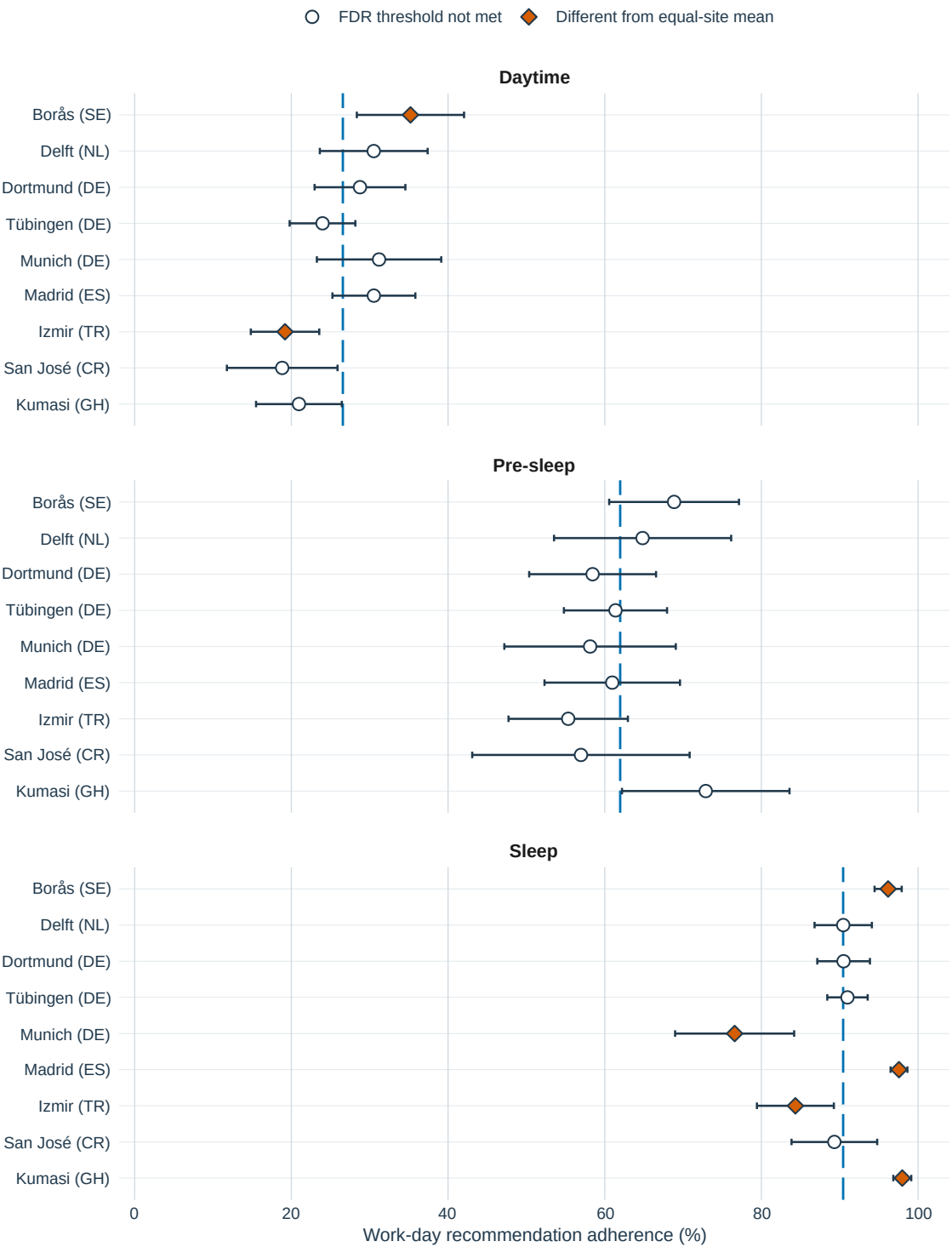
1207 **Supplementary Figure S4. Site-average recommendation adherence on work and free days.**

1208 Daytime, Pre-sleep and Sleep identify the Brown et al. recommendation windows. Each
1209 point gives equal weight to nine sites; bars are 95% confidence intervals. The wake-start
1210 date labels the preceding sleep, daytime and following pre-sleep windows. Sleep describes
1211 the device-recorded bedside environment, not ocular exposure or a whole-bedroom
1212 measurement. The pre-sleep contrast remains positive but its interval includes zero under
1213 the 80% coverage restriction; temporal dependence remains unresolved. The coverage
1214 sensitivity is reported in the Results and Methods.

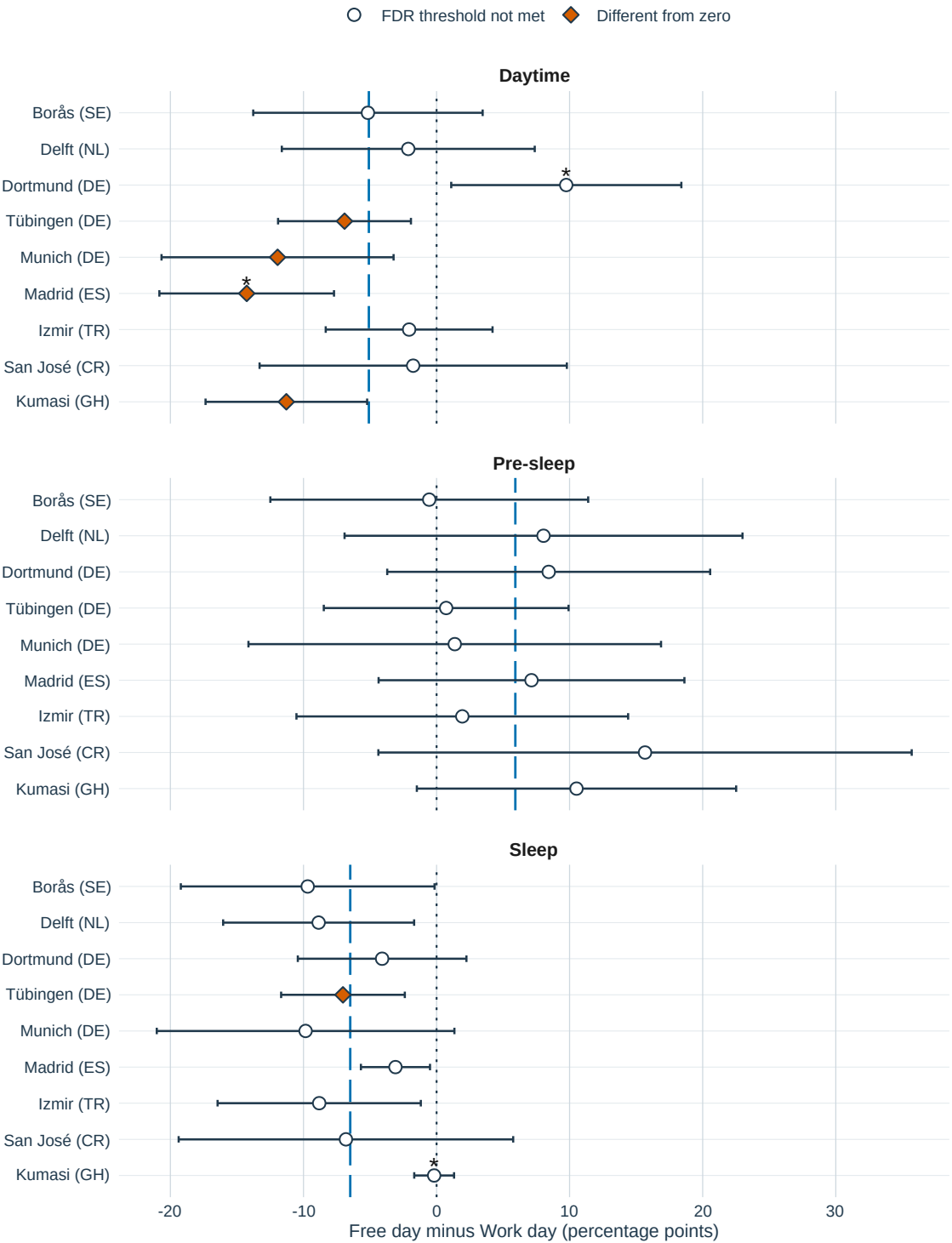
1215 **Supplementary Figure S5. Site-specific adherence and day-type**
1216 **differences**

1217

1218 **A**



1219



* Different from the equal-site Free-minus-Work effect (separate FDR family).
Dotted line: zero. Long-dashed blue line: equal-site effect.

1222 Supplementary Figure S5. Site-specific recommendation adherence, shown as two separate
1223 images. A, Work-day adherence, with filled diamonds identifying site-minus-average
1224 differences that met their FDR criterion. B, Free-minus-work differences, with filled diamonds
1225 identifying differences from zero and asterisks identifying departures from the window-
1226 specific equal-site difference in a separate 27-comparison FDR family. Daytime, Pre-sleep
1227 and Sleep identify the Brown et al. recommendation windows. Points are estimates and bars
1228 are 95% confidence intervals. Each equal-site reference includes the indexed site. Sites are
1229 components of one pooled observational model, not independent replications or causal
1230 effects of location. Day type follows the wake-start date; sleep is the preceding bedside
1231 sleep-environment window. Temporal dependence remains unresolved, and the pre-sleep
1232 coverage qualification applies.

1233 **Supplementary Table S4. Exploratory cross-window recommendation-adherence associations**

Exploratory cross-window recommendation-adherence associations				
Outcome window	Difference per +10 percentage points Daytime adherence, percentage points (95% CI)	FDR-adjusted p	FDR-correction set/family	Interpretation for recommendation adherence
Within participant, linked cycle				
Sleep	-0.26 (-0.95 to 0.43)	0.453	Four prespecified primary cross-window tests	No retained evidence that a cycle with 10 percentage points higher Daytime adherence had different Sleep adherence; the day-level claim is withheld because serial dependence remains unresolved.
Pre-sleep	-1.09 (-2.75 to 0.56)	0.258	Four prespecified primary cross-window tests	No retained evidence that a cycle with 10 percentage points higher Daytime adherence had different Pre-sleep adherence; the day-level claim is withheld because serial dependence remains unresolved.

1234

Table continued

Exploratory cross-window recommendation-adherence associations				
Outcome window	Difference per +10 percentage points Daytime adherence, percentage points (95% CI)	FDR-adjusted p	FDR-correction set/family	Interpretation for recommendation adherence
Between participants, observed monitoring-period average				
Sleep	-2.52 (-3.55 to -1.48)	<0.001	Four prespecified primary cross-window tests	Participants with 10 percentage points higher average Daytime adherence had 2.52 percentage points lower average Sleep adherence over the monitoring period.
Pre-sleep	-3.59 (-5.94 to -1.25)	0.005	Four prespecified primary cross-window tests	Participants with 10 percentage points higher average Daytime adherence had 3.59 percentage points lower average Pre-sleep adherence over the monitoring period.
All four tests form one FDR-correction set. Site-standardized estimates give nine sites equal weight and use 50:50 Work/Free weighting. The primary sample contained 1,376 target-window rows from 140 participants and 761 linked cycles. Within-participant claims are withheld because temporal dependence remains unresolved. Between-participant estimates describe observed monitoring-period averages, not rankings, stable traits, or causal effects. Confidence intervals are not multiplicity adjusted.				

1235

1236 **Supplementary Table S4.** Exploratory within-participant and between-participant associations across Brown et al. recommendation windows. Results are

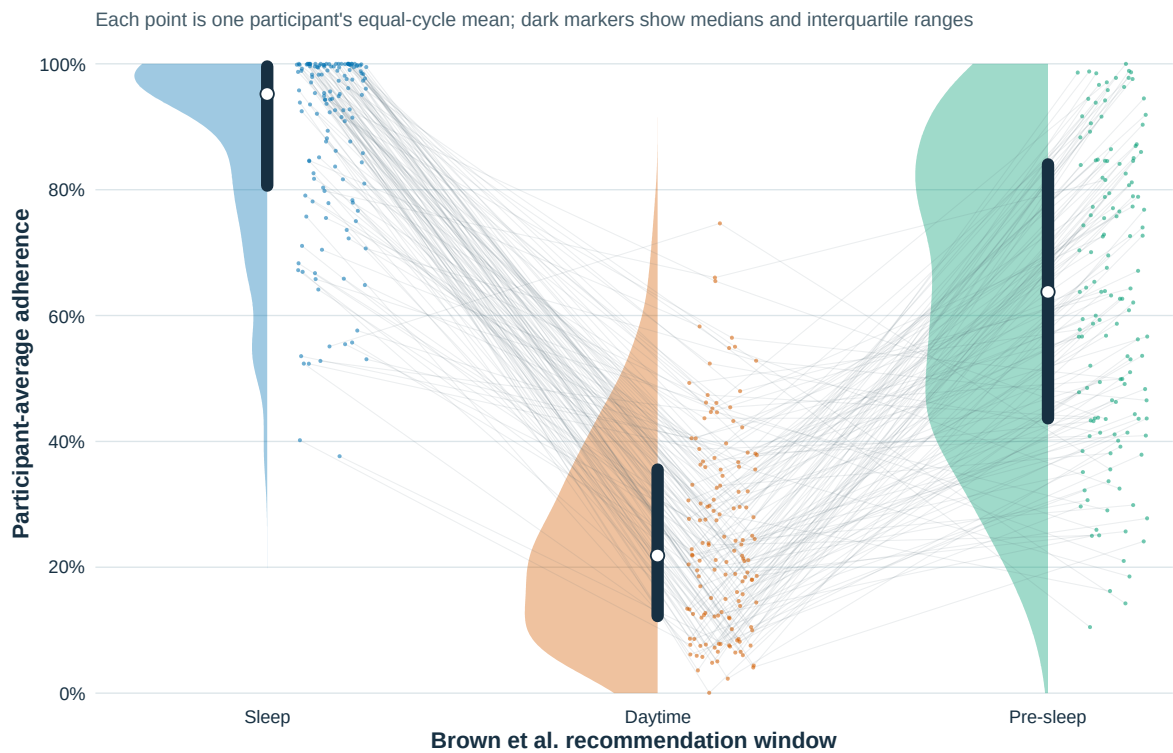
1237 percentage-point differences per 10 percentage points higher Daytime adherence. The four tests form one FDR family. The within-participant day-level claim is

1238 withheld because serial dependence remains unresolved. Between-participant results describe observed monitoring-period averages and do not establish

1239 rankings, stable traits, or causal effects.

1240 **Supplementary Figure S6. Anonymous participant**
1241 **recommendation-adherence profiles**

1242



1243

1244 Supplementary Figure S6. Participant-average recommendation adherence across the
1245 Brown et al. recommendation windows. Daytime, Pre-sleep, and Sleep are the window
1246 labels. Each colored point is one anonymous participant's equal-cycle mean within that
1247 window. Faint lines connect the same anonymous participant across the three window
1248 summaries; they are not time trajectories or ranks. Half-violin shapes show the window
1249 distributions, and dark bars with white points show interquartile ranges and medians.
1250 Because each window has its own Brown et al. threshold and direction, absolute
1251 percentages should be interpreted within window.

1252 **Daily architecture**

1253 **Supplementary Table S5. Near-eye fitted-curve dispersion and full-model R² allocation**

Near-eye fitted-curve dispersion and conditional Shapley allocation			
	Fitted-curve result (95% CI)	Conditional Shapley result (95% CI)	Share (95% CI)
Component summaries			
Shared local-clock curve	Not applicable	0.611 (0.550 to 0.657)	78.9% (73.6% to 82.6%)
Site pattern	0.100 (0.041 to 0.139)	0.016 (0.007 to 0.026)	2.0% (0.9% to 3.5%)
Participant pattern	0.180 (0.126 to 0.214)	0.100 (0.079 to 0.125)	12.9% (10.1% to 16.4%)
Participant-day shift	0.019 (0.010 to 0.024)	0.048 (0.035 to 0.064)	6.2% (4.4% to 8.6%)
Participant pattern + day shift	0.200 (0.140 to 0.230)	Not separately allocated	Not applicable
Participant-to-site comparisons			
Participant / site	1.797 (1.157 to 4.332)	6.41 (3.85 to 15.14)	Not applicable
(Participant + day) / site	1.991 (1.291 to 4.767)	9.49 (5.77 to 21.81)	90.5% (85.2% to 95.6%)

Full-model in-sample R² was 0.775 (0.739 to 0.804). Fitted-curve variation is in squared log₁₀(melEDI + 0.1 lx) prediction units; curve and Shapley ratios are unitless. Conditional Shapley values allocate in-sample R² with the shared local-clock curve as the mandatory baseline. Intervals are 95% percentile hierarchical cluster-bootstrap intervals from 2,000 replicates. Dispersion and Shapley credit are different estimands and must not be added or interpreted causally.

1254

1255 **Supplementary Table S5.** Near-eye fitted-curve dispersion and Shapley allocation of full-model in-sample R². Fitted-curve variation is in squared
1256 log₁₀(melanopic EDI + 0.1 lx) prediction units; dispersion and R²-share ratios are unitless. The allocation includes the shared local-clock curve in every
1257 component model as the baseline. Intervals are 95% hierarchical cluster-bootstrap percentile intervals from 2,000 replicates and are conditional on the fitted
1258 models. Dispersion and R² allocation describe different estimands and should not be added or interpreted causally.

1259 **Supplementary Table S6. Chest fitted-curve dispersion and full-model R² allocation**

Complementary chest fitted-curve dispersion and conditional Shapley allocation

	Fitted-curve result (95% CI)	Conditional Shapley result (95% CI)	Share (95% CI)
Component summaries			
Shared local-clock curve	Not applicable	0.583 (0.523 to 0.629)	79.1% (73.6% to 82.8%)
Site pattern	0.101 (0.049 to 0.116)	0.015 (0.008 to 0.024)	2.1% (1.1% to 3.3%)
Participant pattern	0.148 (0.100 to 0.199)	0.087 (0.071 to 0.110)	11.9% (9.5% to 15.3%)
Participant-day shift	0.035 (0.020 to 0.039)	0.051 (0.040 to 0.067)	7.0% (5.3% to 9.4%)
Participant pattern + day shift	0.183 (0.127 to 0.228)	Not separately allocated	Not applicable
Participant-to-site comparisons			
Participant / site	1.465 (0.973 to 3.179)	5.69 (3.72 to 10.68)	Not applicable
(Participant + day) / site	1.809 (1.219 to 3.736)	9.02 (6.01 to 17.03)	90.0% (85.7% to 94.5%)

Full-model in-sample R² was 0.737 (0.703 to 0.766). Fitted-curve variation is in squared log₁₀(meIEDI + 0.1 lx) prediction units; curve and Shapley ratios are unitless. Conditional Shapley values allocate in-sample R² with the shared local-clock curve as the mandatory baseline. Intervals are 95% percentile hierarchical cluster-bootstrap intervals from 2,000 replicates. Dispersion and Shapley credit are different estimands and must not be added or interpreted causally.

1260

1261 **Supplementary Table S6.** Complementary chest fitted-curve dispersion and Shapley allocation of full-model in-sample R². The quantities, bootstrap intervals
1262 and interpretation follow Supplementary Table S5; chest and near-eye measurements remain distinct sensor-position estimands.

1263 Geographic and civil-photoperiod context

1264 Supplementary Figure S7. Geographic support and nonlinear 1265 photoperiod classifications

1266 **A**

	Near eye				Chest			
	Overall site	Photoperiod	Latitude	Site versus linear latitude	Overall site	Photoperiod	Latitude	Site versus linear latitude
Interdaily stability	–	–	–	–	–	–	–	–
Intradaily variability	–	–	–	–	–	–	–	–
Mean melEDI	✓	✓	✓	✓	✓	✓	–	✓
Brightest 10 h mean	✓	✓	✓	–	✓	✓	–	✓
Darkest 10 h mean	✓	✓	–	✓	✓	✓	✓	✓
Time above 1,000 lx melEDI	–	✓	–	–	✓	✓	✓	✓
Time above 250 lx melEDI during wake	✓	✓	✓	–	✓	✓	–	✓
Time below 10 lx melEDI before sleep	✓	✓	–	✓	–	✓	–	–
Time below 1 lx melEDI during sleep	–	–	–	✓	✓	–	✓	✓
Longest continuous period above 250 lx melEDI	–	✓	–	–	–	✓	–	–
Midpoint of the brightest 10 hours	✓	–	✓	✓	✓	–	✓	✓
Midpoint of the darkest 10 hours	✓	✓	–	✓	✓	✓	✓	✓
Mean timing of exposure above 250 lx melEDI	✓	✓	✓	✓	✓	✓	✓	✓
First light timing above 250 lx melEDI	–	–	–	–	✓	–	✓	✓
Last light timing above 250 lx melEDI	✓	✓	✓	✓	✓	✓	✓	–
melEDI dose	–	✓	–	–	✓	✓	–	✓
Melanopic daylight efficacy ratio	✓	✓	✓	✓	✓	–	✓	–

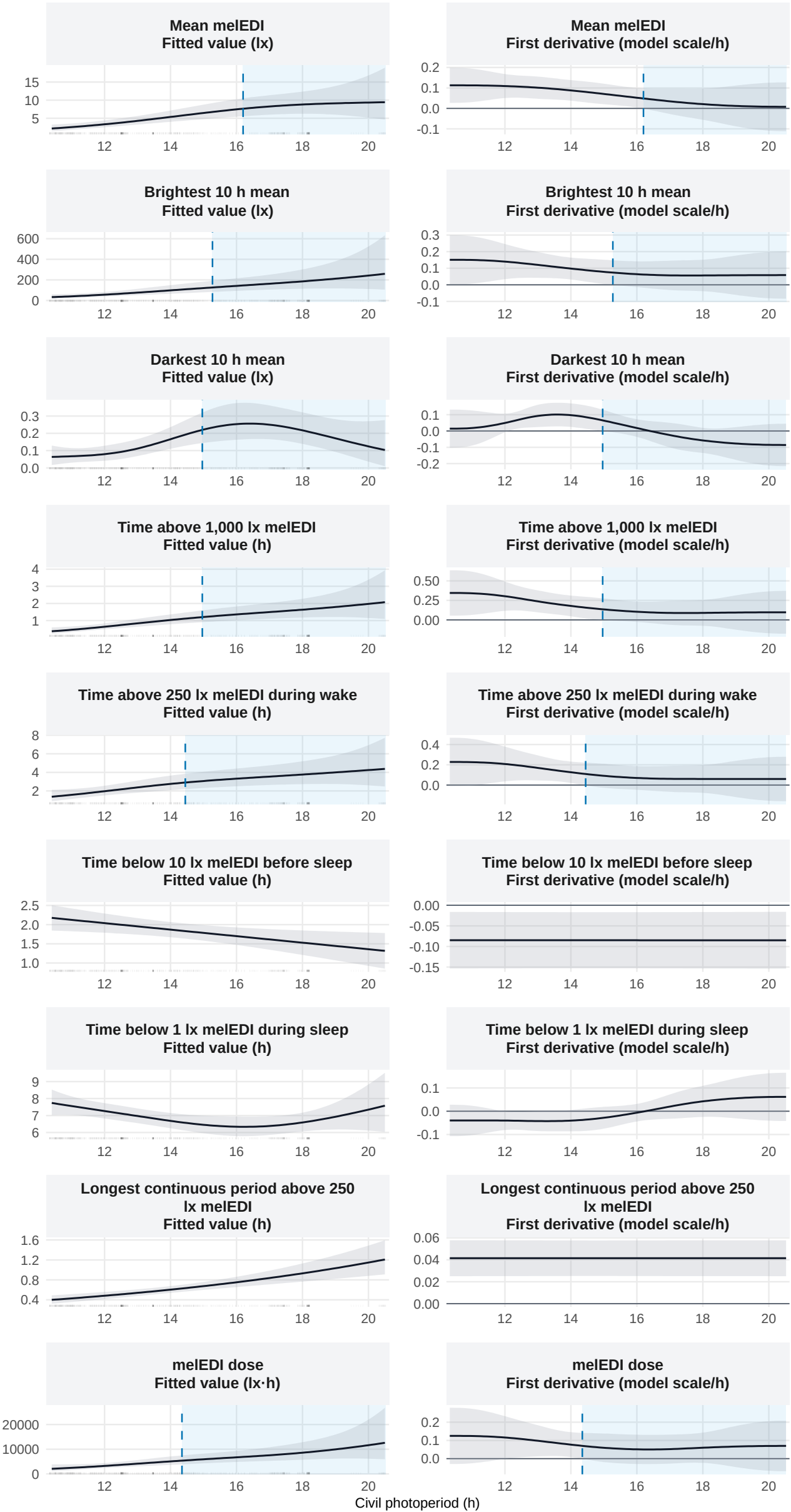
FDR-adjusted result Not supported Supported

1267

1268 Supplementary Figure S7. Geographic and civil-photoperiod associations across near-eye
1269 personal light-exposure metrics. A, Evidence retained after FDR adjustment for overall site,
1270 civil photoperiod, latitude, and site-versus-linear-latitude adequacy. B, Nonlinear civil-
1271 photoperiod associations and their fitted slopes. A qualifying transition denotes an increase
1272 followed by a sustained near-flat tail under the specified slope rule. These descriptive
1273 associations do not identify a mechanism, and no physiological or environmental ceiling was
1274 identified.

Near-eye — primary

Each row pairs the response-scale fitted metric value (left) with the model-scale first derivative used for classification (right). Grey ribbons are unconditional pointwise 95% intervals; dashed lines and blue tails mark the derivative-defined plateau pattern.



1277 **Supplementary Table S7. Geographic and photoperiod associations**

		Photoperiod		Latitude		Site-versus-latitude FDR-adjusted p	Exact fitted sample
		Photoperiod association (95% CI)	Photoperiod FDR-adjusted p	Latitude association per 10° (95% CI)	Latitude FDR-adjusted p		
Dynamics							
Interdaily stability	0.228	×0.98 [0.94–1.02]	0.401	×1.01 [0.96–1.06]	0.903	0.175	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Intradaily variability	0.483	-0.02 [-0.06–0.02]	0.327	-0.04 [-0.08–0.01]	0.139	0.723	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Level							
Mean melEDI	<0.001	×1.18 [1.11–1.26]	<0.001	×1.15 [1.07–1.25]	0.002	0.032	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Brightest 10 h mean	0.011	×1.24 [1.13–1.36]	<0.001	×1.24 [1.11–1.39]	0.001	0.325	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Darkest 10 h mean	<0.001	×1.09 [1.03–1.14]	0.002	×1.04 [0.97–1.11]	0.400	<0.001	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$

1278

Table continued

	Overall site FDR-adjusted p	Photoperiod		Latitude		Site-versus-latitude FDR-adjusted p	Exact fitted sample
		Photoperiod association (95% CI)	Photoperiod FDR-adjusted p	Latitude association per 10° (95% CI)	Latitude FDR-adjusted p		
Duration							
Time above 1,000 lx melEDI	0.071	×1.21 [1.13–1.29]	<0.001	×1.03 [0.94–1.12]	0.769	0.053	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Time above 250 lx melEDI during wake	0.007	×1.14 [1.07–1.20]	<0.001	×1.12 [1.04–1.20]	0.013	0.053	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 737$
Time below 10 lx melEDI before sleep	0.049	-0.13 [-0.20--0.07] h	<0.001	0.00 [-0.09–0.09] h	0.962	0.040	$n_{\text{participants}} = 139$; $n_{\text{participant-days}} = 655$
Time below 1 lx melEDI during sleep	0.058	×0.99 [0.97–1.01]	0.227	×1.00 [0.97–1.02]	0.903	0.043	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 778$
Longest continuous period above 250 lx melEDI	0.398	×1.12 [1.07–1.19]	<0.001	×1.06 [0.99–1.13]	0.139	0.626	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$

	Overall site FDR- adjusted p	Photoperiod		Latitude		Site- versus- latitude FDR- adjusted p	Exact fitted sample
		Photoperiod association (95% CI)	Photoperiod FDR- adjusted p	Latitude association per 10° (95% CI)	Latitude FDR- adjusted p		
Timing							
Midpoint of the brightest 10 hours	<0.001	0.05 [-0.05–0.15] h	0.305	0.19 [0.06–0.32] h	0.013	0.010	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Midpoint of the darkest 10 hours	0.020	-0.15 [-0.25--0.05] h	0.004	0.12 [-0.00–0.24] h	0.114	0.043	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Mean timing of exposure above 250 lx melEDI	<0.001	0.12 [0.03–0.21] h	0.013	0.33 [0.21–0.46] h	<0.001	<0.001	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 742$
First light timing above 250 lx melEDI	0.112	-0.10 [-0.25–0.05] h	0.227	0.02 [-0.17–0.21] h	0.903	0.077	$n_{\text{participants}} = 140$; $n_{\text{participant-days}} = 727$
Last light timing above 250 lx melEDI	<0.001	0.32 [0.19–0.46] h	<0.001	0.46 [0.27–0.64] h	<0.001	<0.001	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 687$

1280

	Overall site FDR- adjusted p	Photoperiod		Latitude		Site- versus- latitude FDR- adjusted p	Exact fitted sample
		Photoperiod association (95% CI)	Photoperiod FDR- adjusted p	Latitude association per 10° (95% CI)	Latitude FDR- adjusted p		
Exposure history							
melEDI dose	0.209	×1.28 [1.17–1.39]	<0.001	×1.02 [0.91–1.13]	0.903	0.156	<i>n</i> _{participants} = 141; <i>n</i> _{participant-days} = 761
Spectrum							
Melanopic daylight efficacy ratio	<0.001	0.02 [0.02–0.03]	<0.001	-0.01 [-0.02--0.00]	0.013	0.010	<i>n</i> _{participants} = 137; <i>n</i> _{participant-days} = 702
This summary combines the model-level decisions, photoperiod and latitude associations, 95% CIs, and exact fitted samples. Each displayed p-value is FDR-adjusted within its explicitly labelled complete 17-test family; bold values meet the FDR-adjusted p < 0.050 rule. Full raw p-values and test statistics remain in the detailed tables below. Exact fitted samples use italic n with participant and participant-day subscripts. For the 15 participant-day models, the participant-day count equals the number of fitted observations. The two participant-level dynamics models use one fitted observation per participant and retain the participant-day count as contributing repeated-day support. All models include nine sites.							

1281

1282 **Supplementary Table S7.** Overall-site, civil-photoperiod and absolute-latitude associations for the 17 near-eye personal light-exposure metrics. Each displayed
1283 *p* value is FDR-adjusted within its stated 17-test family. Site and latitude were evaluated in separate models because absolute latitude is fixed within and
1284 confounded with study site.

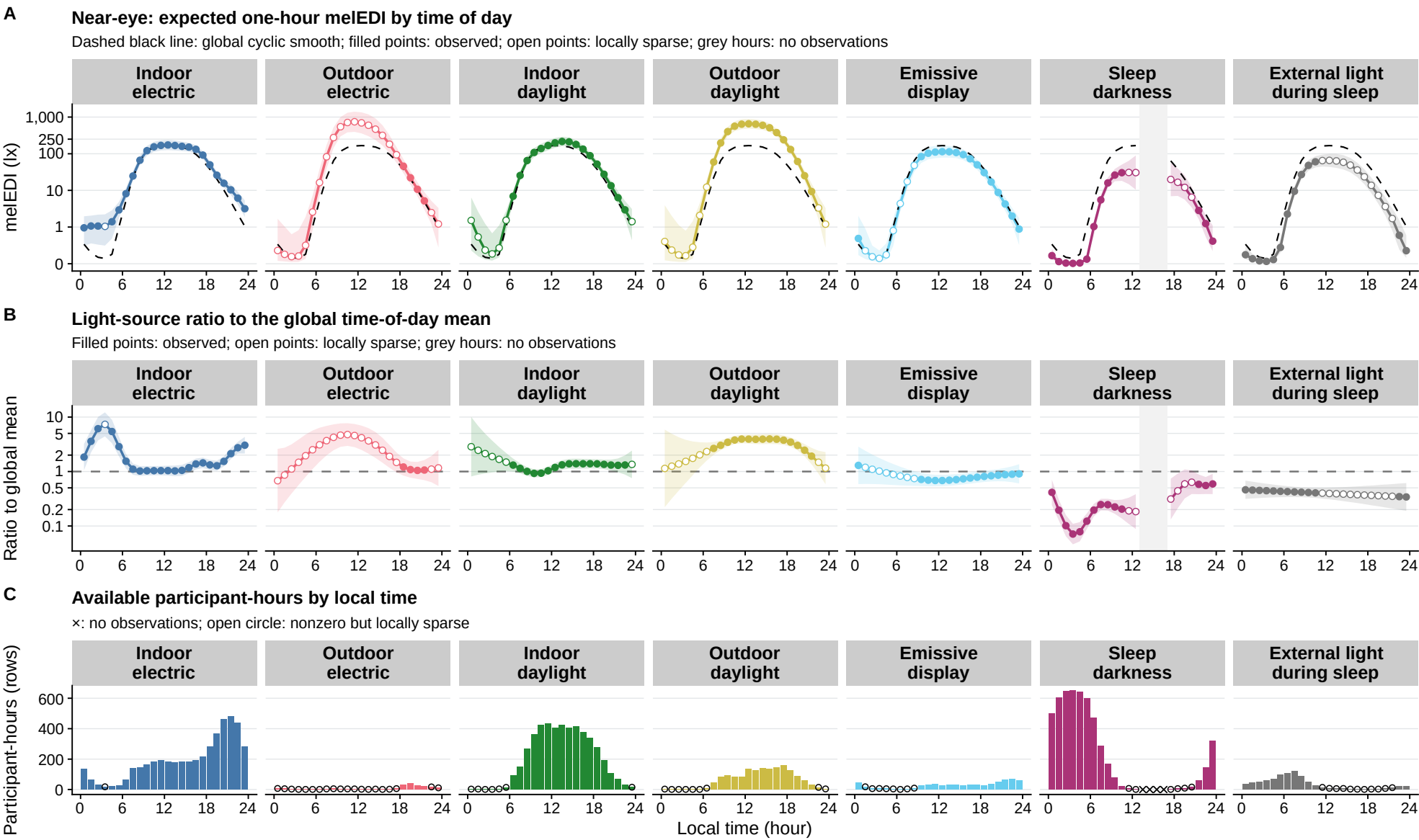
	Qualifying transition	Qualifying transition bracket (h)	Endpoint derivative	Interpretation
Light level				
Mean melEDI	Yes	16.10–16.20	0.008 (95% CI -0.111 to 0.127)	Detected increase followed by a sustained zero-compatible tail
Brightest 10 h mean	Yes	15.17–15.27	0.059 (95% CI -0.083 to 0.201)	Detected increase followed by a sustained zero-compatible tail
Darkest 10 h mean	Yes	14.86–14.96	-0.086 (95% CI -0.216 to 0.044)	Detected increase followed by a sustained zero-compatible tail
Duration and continuous period				
Time above 1,000 lx melEDI	Yes	14.86–14.96	0.098 (95% CI -0.176 to 0.371)	Detected increase followed by a sustained zero-compatible tail
Time above 250 lx melEDI during wake	Yes	14.35–14.45	0.060 (95% CI -0.159 to 0.279)	Detected increase followed by a sustained zero-compatible tail
Time below 10 lx melEDI before sleep	No	—	-0.085 (95% CI -0.154 to -0.016)	No preceding detected increase
Time below 1 lx melEDI during sleep	No	—	0.062 (95% CI -0.042 to 0.165)	No preceding detected increase
Longest continuous period above 250 lx melEDI	No	—	0.041 (95% CI 0.025 to 0.058)	Increase remained detected at the recorded maximum
Exposure history				
melEDI dose	Yes	14.24–14.35	0.070 (95% CI -0.069 to 0.209)	Detected increase followed by a sustained zero-compatible tail
A ‘Yes’ is the formal derivative-defined classification: the immediately preceding point has an interval wholly above zero, the next contains zero, and every later interval through the recorded maximum remains zero-compatible. Endpoint derivatives and 95% confidence intervals are on each model’s linear-predictor scale, in model-scale units per hour of civil photoperiod.				

1286

1287 **Supplementary Table S8.** Derivative-based classifications for nine near-eye personal light-exposure metrics across the observed civil-photoperiod range. A

1288 qualifying transition required the preceding fitted-slope interval to be wholly above zero, the next interval to include zero and every later interval through the

1289 recorded maximum to remain zero-compatible. These classifications are descriptive and do not identify a physiological or environmental ceiling.



Bands are conditional pointwise 95% intervals; no simultaneous-band or curve-wide inferential claim is made. Site, participant, and participant-day smooths remain in the fitted model but are excluded from the displayed curves. Panel B divides each displayed category mean by the displayed global time-of-day mean; the dashed reference is 1. In panels A and B, filled circles mark observed support; open circles mark nonzero but locally sparse support (<20 participant-hours, <5 participants, or <3 sites at that category-hour). Grey regions and × mark zero observations. The melEDI axis transformation is display-only. Exploratory fixed-power Tweedie mean GAM ($p = 1.539919$; log link).

D

Near-eye light-source-by-site interaction model

Site-average estimates and site-specific deviation ratios; bold site cells pass the complete near-eye FDR adjustment

	Indoor electric	Outdoor electric	Indoor daylight	Outdoor daylight	Emissive display	Sleep darkness	External light during sleep
	P 138 · D 695 · H 4,629	P 54 · D 118 · H 203	P 138 · D 692 · H 4,804	P 135 · D 512 · H 1,557	P 88 · D 235 · H 664	P 130 · D 712 · H 5,225	P 66 · D 233 · H 853
Site-average estimate	85.1 lx 95% CI 71.6–101.1 1.000× reference	9.4 lx 95% CI 6.9–12.8 0.110× indoor 95% CI 0.078–0.156	198.8 lx 95% CI 169.2–233.4 2.335× indoor 95% CI 1.877–2.905	959.6 lx 95% CI 786.8–1170.2 11.274× indoor 95% CI 8.709–14.595	23.8 lx 95% CI 17.4–32.5 0.280× indoor 95% CI 0.197–0.398	2.2 lx 95% CI 1.5–3.3 0.026× indoor 95% CI 0.018–0.039	11.2 lx 95% CI 8.1–15.5 0.131× indoor 95% CI 0.091–0.189
Borås (SE)	1.43× (1.02–1.99) FDR p 0.083	Not estimable	1.61× (1.12–2.32) FDR p 0.029	3.07× (1.92–4.90) FDR p <0.001	0.31× (0.16–0.57) FDR p 0.002	1.52× (0.39–5.90) FDR p 0.704	0.70× (0.21–2.36) FDR p 0.707
Delft (NL)	1.83× (1.32–2.54) FDR p 0.002	Not estimable	1.46× (0.93–2.29) FDR p 0.190	1.58× (0.89–2.80) FDR p 0.214	7.17× (4.00–12.86) FDR p <0.001	0.94× (0.49–1.80) FDR p 0.894	Not estimable
Dortmund (DE)	1.26× (0.69–2.29) FDR p 0.595	Not estimable	1.62× (0.83–3.17) FDR p 0.263	1.48× (0.91–2.40) FDR p 0.210	2.86× (1.66–4.93) FDR p 0.002	0.89× (0.45–1.75) FDR p 0.801	9.22× (2.50–34.03) FDR p 0.004
Tübingen (DE)	0.56× (0.38–0.82) FDR p 0.009	20.42× (3.95–105.59) FDR p 0.002	1.04× (0.72–1.50) FDR p 0.866	0.89× (0.51–1.56) FDR p 0.776	0.25× (0.14–0.44) FDR p <0.001	0.78× (0.32–1.92) FDR p 0.710	1.59× (0.71–3.59) FDR p 0.406
Munich (DE)	1.70× (0.63–4.57) FDR p 0.425	Not estimable	1.00× (0.65–1.55) FDR p 0.983	0.94× (0.55–1.61) FDR p 0.866	Not estimable	6.60× (2.11–20.69) FDR p 0.005	11.65× (7.19–18.85) FDR p <0.001
Madrid (ES)	1.48× (1.15–1.90) FDR p 0.007	2.68× (1.56–4.61) FDR p 0.002	0.83× (0.57–1.19) FDR p 0.436	0.65× (0.33–1.26) FDR p 0.332	3.68× (2.24–6.03) FDR p <0.001	1.59× (0.30–8.32) FDR p 0.710	Not estimable
Izmir (TR)	0.69× (0.52–0.92) FDR p 0.029	1.85× (0.84–4.07) FDR p 0.222	0.86× (0.65–1.13) FDR p 0.422	1.12× (0.65–1.94) FDR p 0.776	1.32× (0.70–2.47) FDR p 0.540	0.97× (0.43–2.18) FDR p 0.952	1.59× (0.92–2.76) FDR p 0.190
San José (CR)	0.54× (0.38–0.77) FDR p 0.003	Not estimable	0.72× (0.49–1.05) FDR p 0.189	0.46× (0.21–1.01) FDR p 0.115	1.32× (0.65–2.66) FDR p 0.595	0.60× (0.25–1.42) FDR p 0.390	Not estimable
Kumasi (GH)	0.58× (0.38–0.88) FDR p 0.029	6.24× (2.28–17.06) FDR p 0.002	0.49× (0.26–0.92) FDR p 0.065	0.50× (0.36–0.68) FDR p <0.001	1.27× (0.40–4.07) FDR p 0.776	0.17× (0.06–0.49) FDR p 0.004	Not estimable

Site-average estimates and comparisons use the light-source-by-site interaction model.

Site rows are deviation ratios around each category estimate; parentheses show 95% confidence intervals.

Bold site ratios and FDR p-values pass the complete 53-member near-eye family; unsupported cells are not estimable.

Near-eye frame: 17,935 participant-hours; 140 participants; 801 participant-days; nine sites.

Each accepted participant-hour has one primary light-source category; 1/k multi-label weighting does not apply.

Panels A–C exclude site, participant, and participant-day smooths from displayed curves; panel D uses stored interaction-model estimands.

P, participants; D, participant-days; H, participant-hours. Country-coded labels and row order provide non-colour site cues.

1295

1296 Supplementary Figure S8. Near-eye light exposure by reported light source across time of day and study site. A–C, Exploratory time-of-day analysis. Panel A

1297 shows expected one-hour melanopic EDI with clock-specific 95% confidence intervals; the dashed line is the site-average local-clock smooth. Panel B shows

1298 each light-source curve relative to that smooth, with 1 as the reference. Panel C shows available participant-hours; open circles indicate locally sparse support

1299 and grey gaps indicate no observations. D, Site-average category estimates and site-specific deviation ratios from the light-source-by-site interaction model,

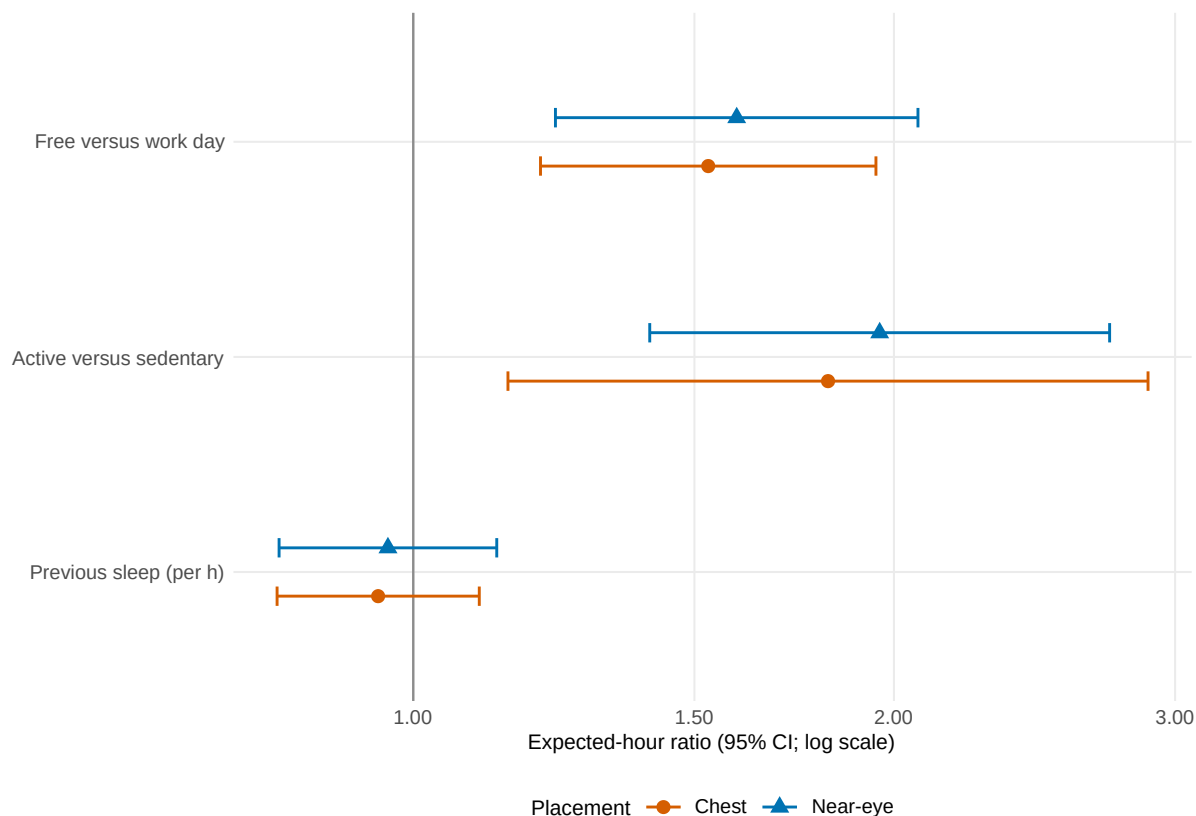
1300 with participant-cluster-robust 95% confidence intervals, FDR-adjusted p values, and exact support. The analysis includes 17,935 participant-hours from 140

1301 participants, covering 801 participant-days and nine sites. Panels A–C and panel D answer different questions and are not numerically interchangeable.

1302 Estimates are observational. During reported sleep, measurements describe the bedside sleep environment.

Hourly routine analyses

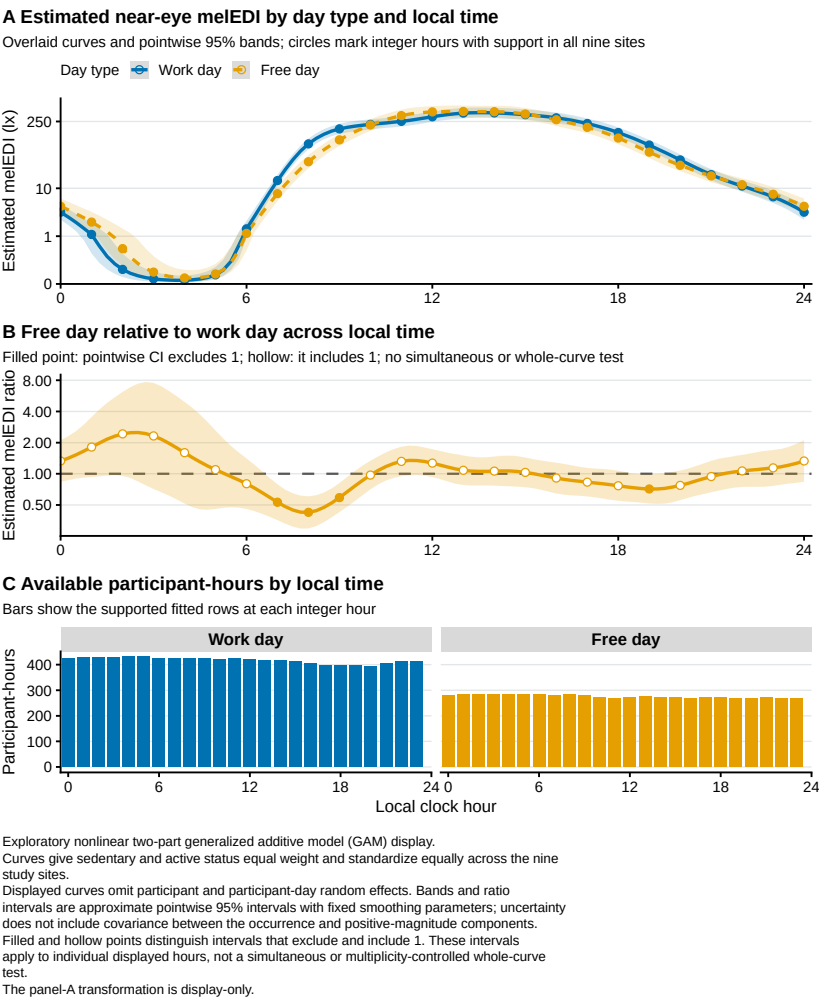
Supplementary Figure S9. Paired sensor-position hourly associations



Supplementary Figure S9. Near-eye and complementary chest associations estimated separately for the same participants and participant-days. The sets of supported participant-hours differ between sensor positions, so the display is not an observation-level identity comparison and does not test equivalence.

1312 **Supplementary Figure S10. Free-versus-Work hourly timing**

1313

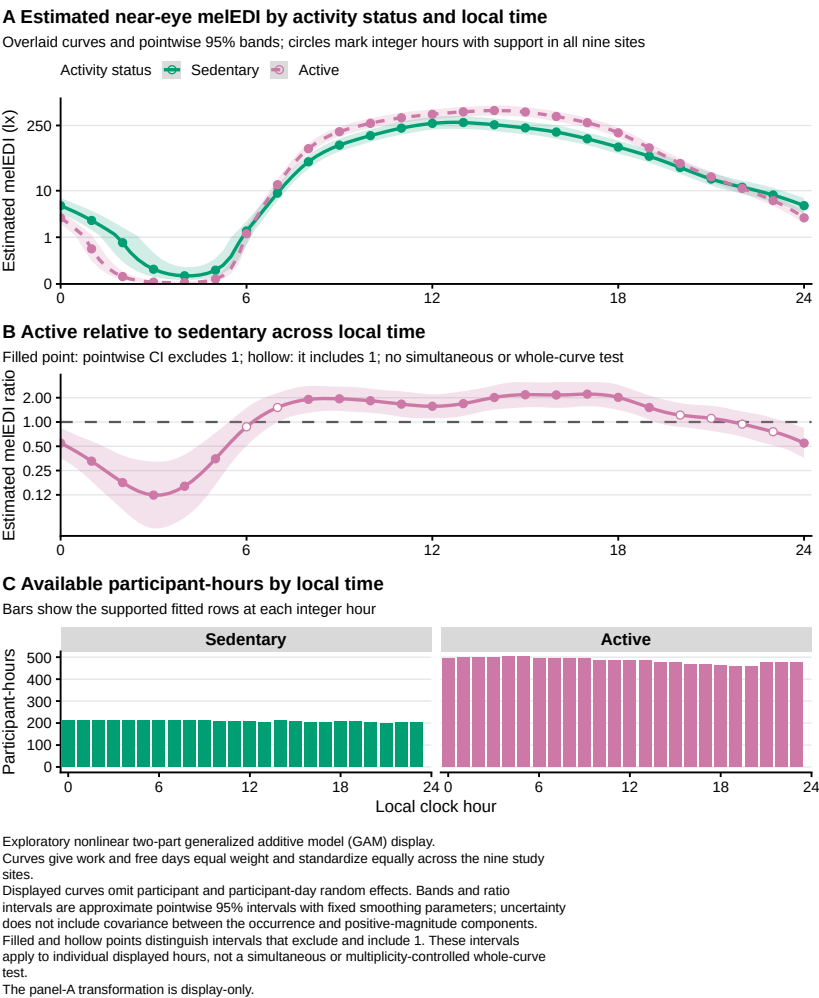


1314

1315 Supplementary Figure S10. Exploratory Free-versus-Work hourly fitted exposure and
1316 response-scale ratio curves with 95% confidence intervals and participant-hour support.

1317 **Supplementary Figure S11. Active-versus-Sedentary hourly timing**

1318



1319

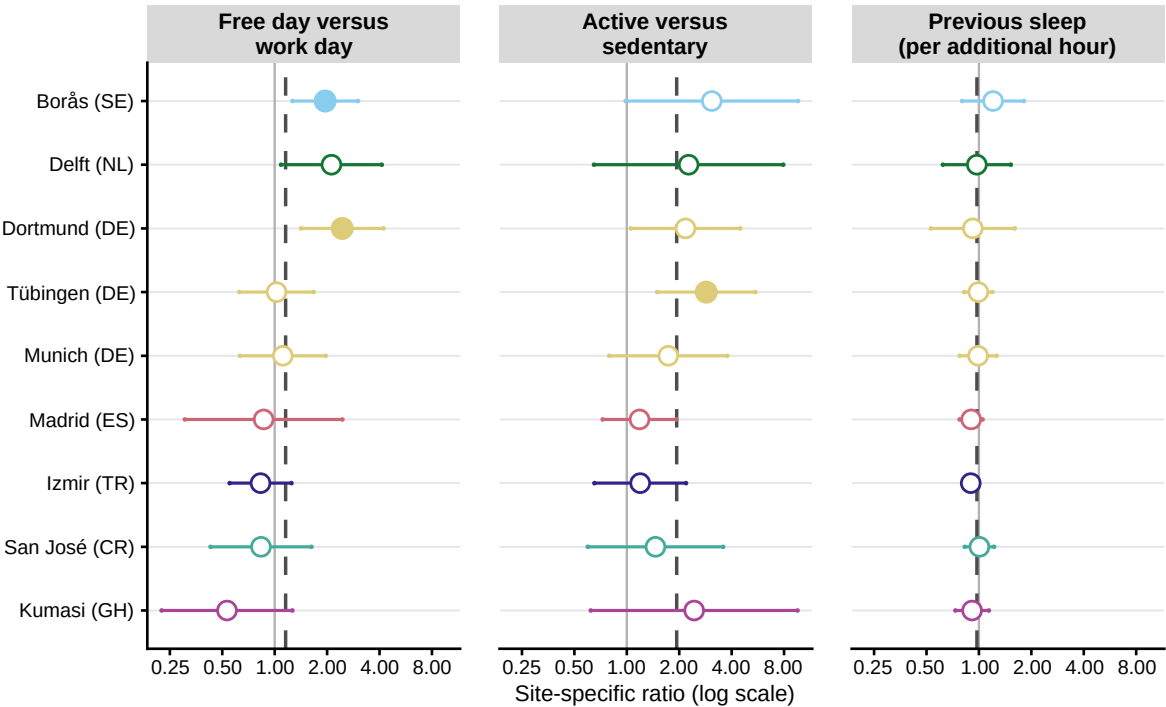
1320 Supplementary Figure S11. Exploratory Active-versus-Sedentary hourly fitted exposure and
1321 response-scale ratio curves with 95% confidence intervals and participant-hour support.

1322 **Supplementary Figure S12. Site-specific hourly routine**
1323 **associations**

1324

Site-specific mean hourly near-eye meEDI ratios

Dashed: site-average estimate from this model; filled: retained after nine-site FDR adjustments; open: not retained



Bars are participant-cluster HC3 pointwise 95% confidence intervals from the current predictor-by-site interaction model. The grey line at 1 is the site-specific association null. The dashed line is the site-average geometric mean of the nine ratios in this same model. Filled points pass a separate nine-site FDR adjustment within that predictor; this does not test deviation from the site-average estimate or the overall predictor-by-site interaction.

1325

1326 Supplementary Figure S12. Site-specific estimates from the predictor-by-site interaction
1327 model. Filled and hollow points distinguish associations that did and did not meet the 0.050
1328 criterion after a separate nine-site FDR adjustment within each predictor. The 95%
1329 confidence intervals are unadjusted; dashed lines show site-average estimates calculated
1330 with equal weight for each of the nine sites. Sites are components of one pooled model, not
1331 independent replications or causal effects of location.

1332 **Supplementary Table S9. Participant-hour routine associations**

Predictor	Ratio (95% CI)	FDR-adjusted p
Free day versus work day	1.45 × (1.14–1.85)	0.004
Active versus sedentary	2.06 × (1.54–2.76)	<0.001
Previous sleep duration (per hour)	0.98 × (0.85–1.12)	0.745
Directions are free versus work day, active versus sedentary day status, and the ratio per one additional hour of previous sleep. Participant-cluster-robust 95% CIs use HC3 with 137 participants. The three p-values form one FDR-adjusted family; the null ratio is 1.		

1333

1334 **Supplementary Table S9.** Common-association estimates from the participant-hour analysis of Free versus Work day, Active versus Sedentary daily activity
1335 status and each additional hour of previous-night sleep duration. Ratios and participant-cluster-robust 95% confidence intervals come from the fixed-site model;
1336 the three *p* values form one FDR-adjusted family. Predictor-by-site differences are shown in Supplementary Figure S12.

Person-level evidence synthesis						
	Outcome or analysis scale	Result	FDR-correction set	FDR decision	Exact sample	Interpretation and qualification
Light-exposure behaviour and awareness factors	68 factor-by-metric cells across four participant-level factors	No cell retained FDR support; four sleep-environment cells were non-estimable	68 near-eye factor-by-metric tests	None retained	Metric-specific near-eye samples in Supplementary Table S11	Broad inconclusive result; non-retention is not evidence of no association
Visual light sensitivity	Nine near-eye metrics per one VLSQ-8 SD	Largest displayed ratio shift was melanopic EDI dose, ×0.846 (0.716 to 0.999)	Nine near-eye metric tests	None retained; adjusted p = 0.160 for this row	141 participants, 761 participant-days, nine sites for the displayed row	Broad inconclusive result
Corrected midsleep on Free days	Timing outcomes per one-hour later MSFsc	M10 midpoint +0.205 h (0.045 to 0.366); L10 midpoint +0.276 h (0.118 to 0.435); first light above 250 lx +0.381 h (0.146 to 0.616)	Five MCTQ timing outcomes, separate from MEQ	These three rows retained	139 to 140 participants; 722 to 810 participant-days; nine sites	Selective timing associations
Morningness-Eveningness Questionnaire	Timing outcomes per 10 points greater morning preference	M10 midpoint -0.278 h (-0.451 to -0.105); L10 midpoint -0.314 h (-0.487 to -0.142); first light above 250 lx -0.444 h (-0.699 to -0.189)	Five MEQ timing outcomes, separate from MCTQ	These three rows retained	140 to 141 participants; 727 to 816 participant-days; nine sites	Selective timing associations

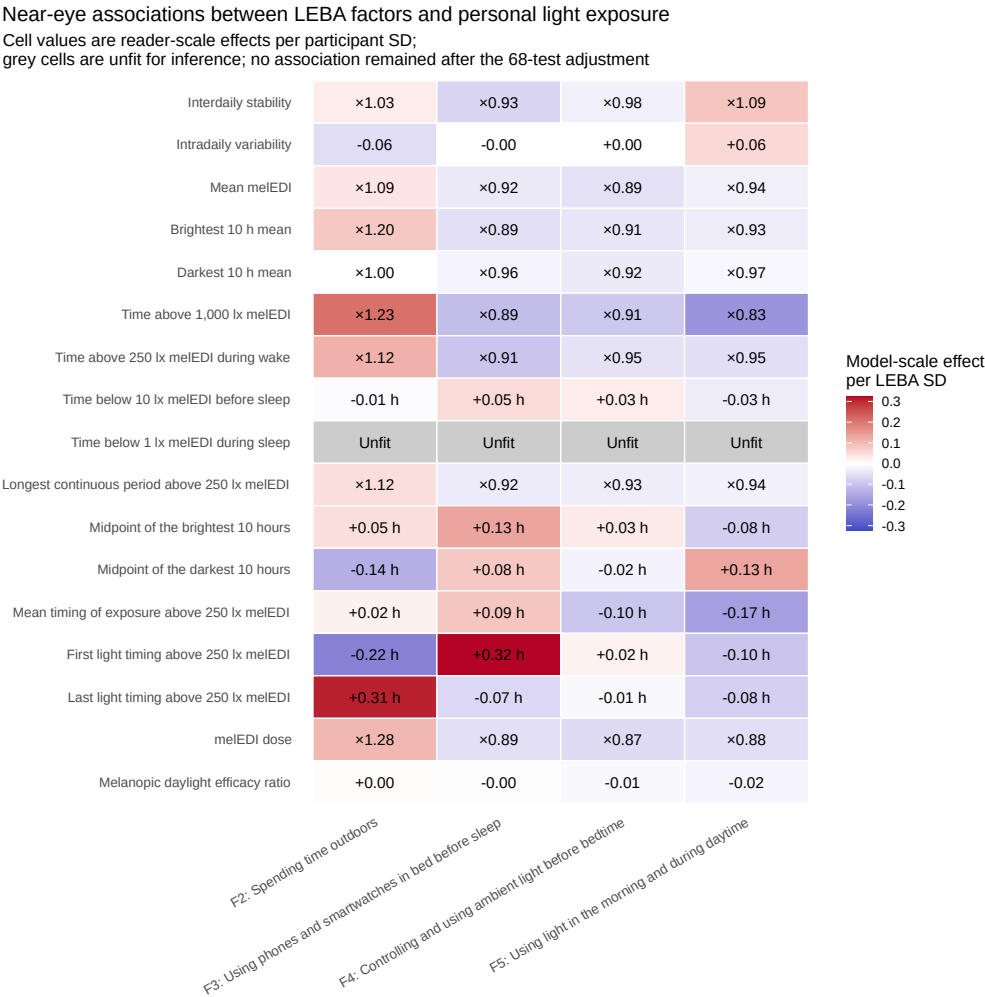
Person-level evidence synthesis						
	Outcome or analysis scale	Result	FDR-correction set	FDR decision	Exact sample	Interpretation and qualification
Age	Near-eye daily metrics per 10 years	Brightest 10 h geometric mean ×1.31 (1.09 to 1.58); time above 1,000 lx ×1.27 (1.11 to 1.45); melanopic EDI dose ×1.31 (1.09 to 1.56)	Near-eye age-metric tests	Three rows retained	141 participants; 761 to 816 participant-days; nine sites	Selective age associations; complementary chest results remain separate
Metric-level biological sex	Near-eye daily metrics, Female minus Male	No near-eye metric-level result retained FDR support	Near-eye biological-sex metric tests	None retained	Metric-specific samples in Supplementary Table S14	Biological sex and gender were recorded separately; gender was not analysed
Biological-sex-specific daily curve	Complete 24-hour curve and activity-complete sensitivity	The all-available near-eye complete-curve test was supported, but separation attenuated and was not retained in the activity-complete near-eye analysis	Global complete-curve tests and separately prespecified sensitivity decisions	Analysis-specific global decisions	All-available near eye: 141 participants, 816 participant-days, 37,756 observations, nine sites	Attenuation-first interpretation; confidence intervals apply at the displayed clock times

Results that did not meet the applicable FDR criterion are inconclusive rather than evidence of no association. The two chronotype instruments were analysed separately. Biological sex and gender were recorded separately; gender was not analysed.

1340
1341 **Supplementary Table S10.** Summary of person-level results in the near-eye analyses. Each row identifies the analysis-specific FDR-correction set or decision
1342 structure applied to that result. Results that did not meet the applicable FDR criterion are inconclusive rather than evidence of no association. Four light-
1343 behaviour sleep-environment cells were non-estimable. The two chronotype instruments were analysed separately. Biological sex and gender were recorded
1344 separately; gender was not analysed. The complete available-data curve difference was attenuated and not retained in the activity-complete near-eye analysis.

1345 **Supplementary Figure S13 and Table S11. Light-exposure**
1346 **behaviour and awareness**

1347



1348

1349 Supplementary Figure S13. Near-eye associations per one participant-level standard
1350 deviation of each Light Exposure Behaviour Assessment factor. None of the 68 tests
1351 retained FDR-adjusted support. Four sleep-environment cells were not estimable and are
1352 labelled Unfit with estimates suppressed. Lack of adjusted support is inconclusive rather
1353 than proof of no association.

	F2	F3	F4	F5
Interdaily stability	OR 1.028 (0.956–1.106) FDR-adjusted p = 0.658	OR 0.931 (0.868–0.999) FDR-adjusted p = 0.260	OR 0.975 (0.906–1.050) FDR-adjusted p = 0.679	OR 1.089 (1.011–1.174) FDR-adjusted p = 0.202
Intradaily variability	-0.057 (-0.124–+0.009) FDR-adjusted p = 0.351	-0.001 (-0.068–+0.065) FDR-adjusted p = 0.978	+0.001 (-0.068–+0.069) FDR-adjusted p = 0.978	+0.057 (-0.013–+0.127) FDR-adjusted p = 0.362
Mean melEDI	×1.092 (0.965–1.235) FDR-adjusted p = 0.379	×0.917 (0.813–1.034) FDR-adjusted p = 0.379	×0.891 (0.787–1.009) FDR-adjusted p = 0.313	×0.941 (0.827–1.070) FDR-adjusted p = 0.585
Brightest 10 h mean	×1.204 (1.008–1.439) FDR-adjusted p = 0.260	×0.888 (0.745–1.058) FDR-adjusted p = 0.384	×0.906 (0.756–1.087) FDR-adjusted p = 0.545	×0.931 (0.772–1.123) FDR-adjusted p = 0.658
Darkest 10 h mean	×0.995 (0.905–1.094) FDR-adjusted p = 0.958	×0.956 (0.871–1.049) FDR-adjusted p = 0.577	×0.921 (0.838–1.013) FDR-adjusted p = 0.351	×0.970 (0.879–1.070) FDR-adjusted p = 0.716
Time above 1,000 lx melEDI	×1.234 (1.084–1.405) FDR-adjusted p = 0.124	×0.895 (0.785–1.020) FDR-adjusted p = 0.362	×0.911 (0.797–1.043) FDR-adjusted p = 0.390	×0.831 (0.725–0.953) FDR-adjusted p = 0.169
Time above 250 lx melEDI during wake	×1.122 (1.006–1.252) FDR-adjusted p = 0.260	×0.907 (0.814–1.010) FDR-adjusted p = 0.351	×0.953 (0.852–1.065) FDR-adjusted p = 0.614	×0.946 (0.843–1.061) FDR-adjusted p = 0.586
Time below 10 lx melEDI before sleep	-0.005 h (-0.135–+0.124 h) FDR-adjusted p = 0.958	+0.052 h (-0.073–+0.177 h) FDR-adjusted p = 0.614	+0.028 h (-0.104–+0.159 h) FDR-adjusted p = 0.781	-0.031 h (-0.167–+0.105 h) FDR-adjusted p = 0.781
Time below 1 lx melEDI during sleep	Unfit for inference Estimate and p-values suppressed from the reader display	Unfit for inference Estimate and p-values suppressed from the reader display	Unfit for inference Estimate and p-values suppressed from the reader display	Unfit for inference Estimate and p-values suppressed from the reader display
Longest continuous period above 250 lx melEDI	×1.117 (1.007–1.238) FDR-adjusted p = 0.260	×0.924 (0.835–1.023) FDR-adjusted p = 0.375	×0.933 (0.840–1.037) FDR-adjusted p = 0.393	×0.945 (0.847–1.053) FDR-adjusted p = 0.557

Associations are reported per one participant-level SD of the LEBA score with 95% CIs. FDR-adjusted p is bold only at alpha = 0.050 across all 68 tests. Sleep-environment cells are unfit for inference and suppressed.

1356

1357 *Supplementary Table S11 (continued).*

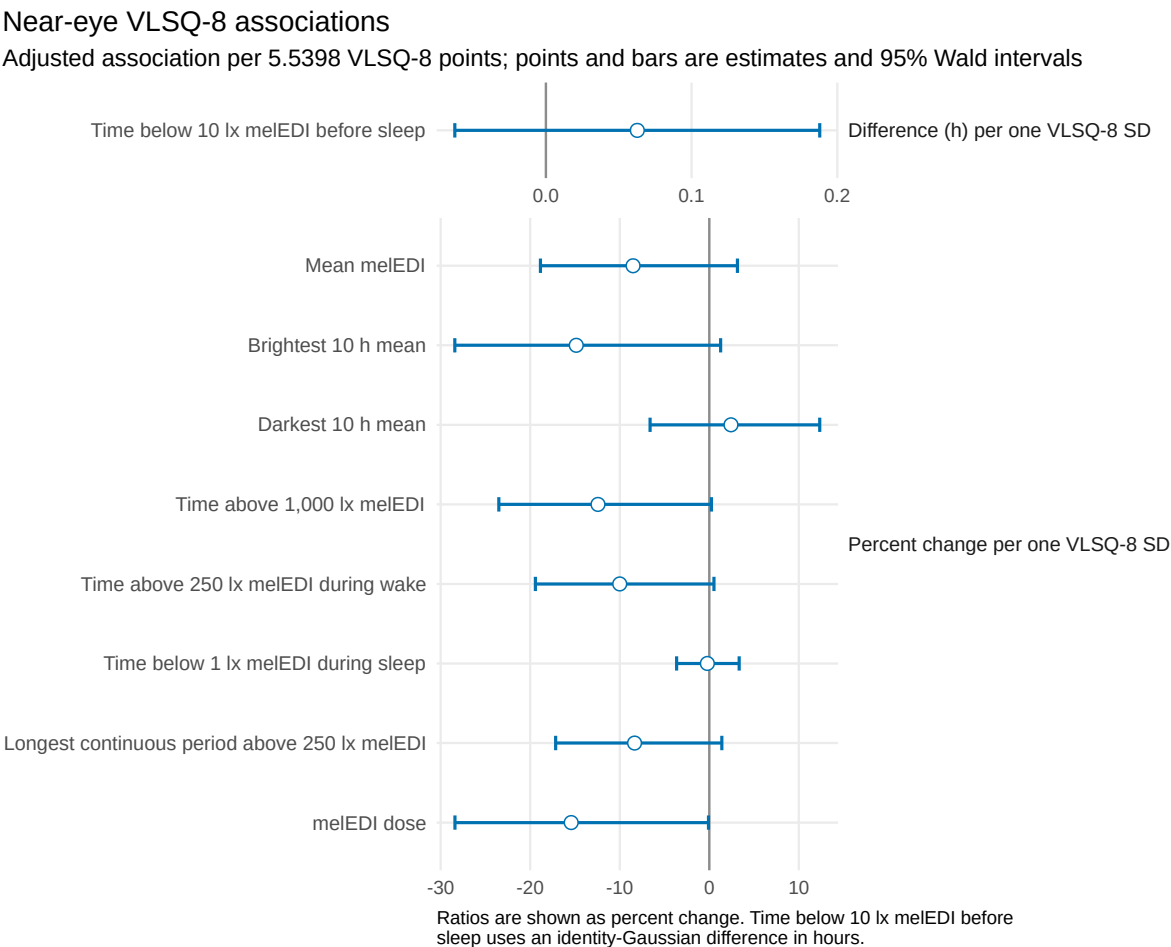
	F2	F3	F4	F5
Midpoint of the brightest 10 hours	+0.049 h (-0.132–+0.229 h) FDR-adjusted p = 0.747	+0.133 h (-0.042–+0.309 h) FDR-adjusted p = 0.375	+0.030 h (-0.153–+0.213 h) FDR-adjusted p = 0.868	-0.083 h (-0.270–+0.105 h) FDR-adjusted p = 0.601
Midpoint of the darkest 10 hours	-0.136 h (-0.317–+0.044 h) FDR-adjusted p = 0.375	+0.080 h (-0.097–+0.258 h) FDR-adjusted p = 0.594	-0.020 h (-0.204–+0.164 h) FDR-adjusted p = 0.935	+0.132 h (-0.056–+0.320 h) FDR-adjusted p = 0.379
Mean timing of exposure above 250 lx melEDI	+0.020 h (-0.152–+0.192 h) FDR-adjusted p = 0.935	+0.086 h (-0.081–+0.253 h) FDR-adjusted p = 0.558	-0.097 h (-0.267–+0.074 h) FDR-adjusted p = 0.510	-0.166 h (-0.339–+0.008 h) FDR-adjusted p = 0.301
First light timing above 250 lx melEDI	-0.219 h (-0.489–+0.052 h) FDR-adjusted p = 0.362	+0.324 h (+0.062–+0.585 h) FDR-adjusted p = 0.179	+0.019 h (-0.252–+0.291 h) FDR-adjusted p = 0.958	-0.097 h (-0.380–+0.185 h) FDR-adjusted p = 0.679
Last light timing above 250 lx melEDI	+0.308 h (+0.043–+0.573 h) FDR-adjusted p = 0.202	-0.066 h (-0.328–+0.196 h) FDR-adjusted p = 0.760	-0.011 h (-0.280–+0.259 h) FDR-adjusted p = 0.958	-0.081 h (-0.361–+0.200 h) FDR-adjusted p = 0.743
melEDI dose	×1.278 (1.079–1.514) FDR-adjusted p = 0.124	×0.885 (0.748–1.048) FDR-adjusted p = 0.379	×0.872 (0.732–1.038) FDR-adjusted p = 0.375	×0.881 (0.736–1.054) FDR-adjusted p = 0.379
Melanopic daylight efficacy ratio	+0.004 (-0.011–+0.019) FDR-adjusted p = 0.747	-0.004 (-0.019–+0.011) FDR-adjusted p = 0.747	-0.010 (-0.026–+0.005) FDR-adjusted p = 0.380	-0.020 (-0.035–+0.004) FDR-adjusted p = 0.169

Associations are reported per one participant-level SD of the LEBA score with 95% CIs. FDR-adjusted p is bold only at alpha = 0.050 across all 68 tests. Sleep-environment cells are unfit for inference and suppressed.

1358

1359 **Supplementary Figure S14 and Table S12. Visual light sensitivity**

1360



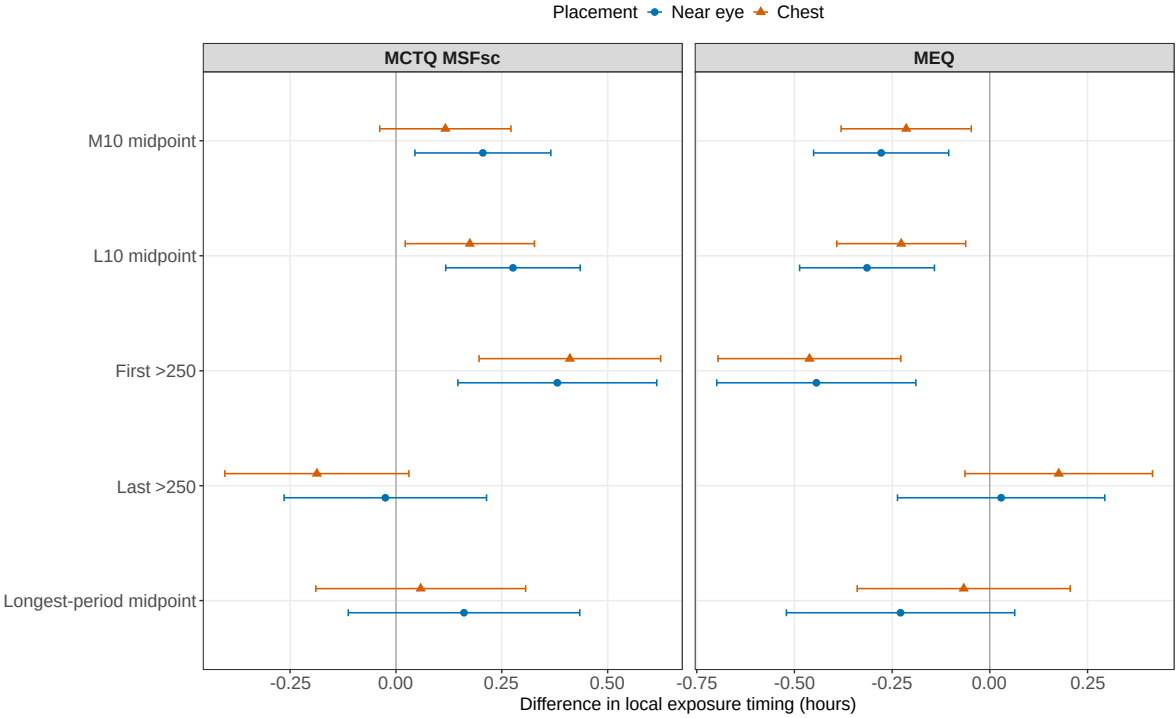
1361

1362 Supplementary Figure S14. Near-eye site-average associations between the Visual Light
1363 Sensitivity Questionnaire-8 and nine personal light-exposure metrics. Associations compare
1364 scores separated by one participant-level standard deviation. Horizontal bars are 95% Wald
1365 confidence intervals. None of the nine associations retained FDR-adjusted support.

Metric	Scale	Association per VLSQ-8 SD (95% CI)	FDR-adjusted p	Participants / participant-days / sites
Mean melEDI	Ratio (percentage change)	×0.915 (0.811–1.031)	0.221	141 / 816 / 9
Brightest 10 h mean	Ratio (percentage change)	×0.851 (0.716–1.013)	0.160	141 / 816 / 9
Darkest 10 h mean	Ratio (percentage change)	×1.024 (0.934–1.123)	0.689	141 / 816 / 9
Time above 1,000 lx melEDI	Ratio (percentage change)	×0.876 (0.765–1.002)	0.160	141 / 816 / 9
Time above 250 lx melEDI during wake	Ratio (percentage change)	×0.900 (0.806–1.005)	0.160	141 / 737 / 9
Time below 10 lx melEDI before sleep	Difference in hours	+0.063 h (-0.062–+0.188 h)	0.422	139 / 655 / 9
Time below 1 lx melEDI during sleep	Ratio (percentage change)	×0.998 (0.964–1.033)	0.903	141 / 778 / 9
Longest continuous period above 250 lx melEDI	Ratio (percentage change)	×0.917 (0.828–1.014)	0.167	141 / 816 / 9
melEDI dose	Ratio (percentage change)	×0.846 (0.716–0.999)	0.160	141 / 761 / 9
Associations compare scores separated by one participant-level SD (5.540 VLSQ-8 points). The hour-scale row is an absolute difference; ratio rows can be read as percentage changes. Intervals are two-sided 95% Wald CIs. FDR adjustment is across all nine metrics; adjusted p-values would be bold at 0.050.				

1368 **Supplementary Figure S15 and Table S13. Chronotype and timing**

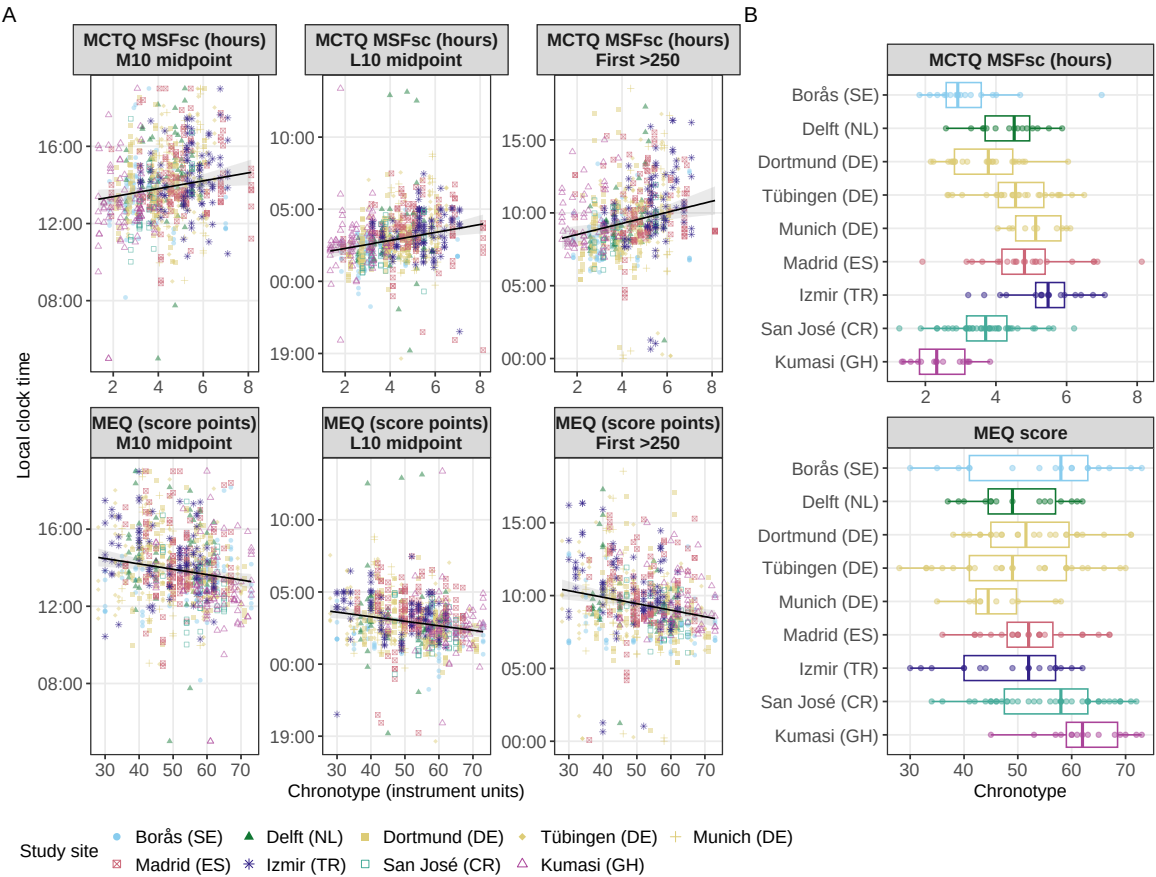
1369 **A**



1370

1371 **B**

Primary near-eye chronotype and light-exposure timing
FDR-supported associations (left) and descriptive chronotype distributions by site (right)



1372

1373 Supplementary Figure S15. Chronotype and timing of personal light exposure. A, Study-site-
1374 adjusted associations of sleep-timing-based corrected midsleep on Free days and
1375 questionnaire-based morningness-eveningness with the analysed timing metrics. Points are
1376 estimates and bars are 95% confidence intervals; the two chronotype instruments remain
1377 separate. B, Observed participant-day timing values across chronotype and country-coded
1378 study sites for the samples used in the corresponding models. Each point represents one
1379 participant-day in a fitted sample. The observed panel is descriptive and does not replace
1380 the adjusted models or estimate independent site effects. Near-eye estimates provide
1381 ocular-exposure evidence; chest estimates, where shown, are complementary non-ocular
1382 evidence.

Metric	Instrument	Signed association, h (95% CI)	FDR-adjusted p	Fitted sample: participants / days / observations / hours / sites
Midpoint of the brightest 10 hours	MCTQ MSFsc	+0.205 (+0.045 to +0.366)	0.017	140 / 810 / 810 / 18711.7 h / 9
Midpoint of the brightest 10 hours	MEQ	-0.278 (-0.451 to -0.105)	0.002	141 / 816 / 816 / 18851.0 h / 9
Midpoint of the darkest 10 hours	MCTQ MSFsc	+0.276 (+0.118 to +0.435)	0.003	140 / 810 / 810 / 18711.7 h / 9
Midpoint of the darkest 10 hours	MEQ	-0.314 (-0.487 to -0.142)	0.001	141 / 816 / 816 / 18851.0 h / 9
First light timing above 250 lx meLEDI	MCTQ MSFsc	+0.381 (+0.146 to +0.616)	0.003	139 / 722 / 722 / 16716.3 h / 9
First light timing above 250 lx meLEDI	MEQ	-0.444 (-0.699 to -0.189)	0.001	140 / 727 / 727 / 16832.5 h / 9
Last light timing above 250 lx meLEDI	MCTQ MSFsc	-0.025 (-0.264 to +0.214)	0.811	140 / 683 / 683 / 15900.8 h / 9
Last light timing above 250 lx meLEDI	MEQ	+0.029 (-0.236 to +0.294)	0.814	141 / 687 / 687 / 15995.0 h / 9
Midpoint of the longest continuous period above 250 lx meLEDI	MCTQ MSFsc	+0.161 (-0.113 to +0.434)	0.284	131 / 478 / 478 / 11325.5 h / 9
Midpoint of the longest continuous period above 250 lx meLEDI	MEQ	-0.228 (-0.521 to +0.064)	0.135	132 / 482 / 482 / 11419.8 h / 9
MCTQ associations are per one-hour later MSFsc; MEQ associations are per 10 points greater morning preference. Positive associations indicate later timing and negative associations earlier. Intervals are 95% Wald CIs. FDR-adjusted p-values are bold when FDR-adjusted p < 0.050 within the instrument-specific five-outcome family.				

Age distributions and associations retained after FDR adjustment

Near-eye evidence is primary and chest evidence complementary; placements are not pooled

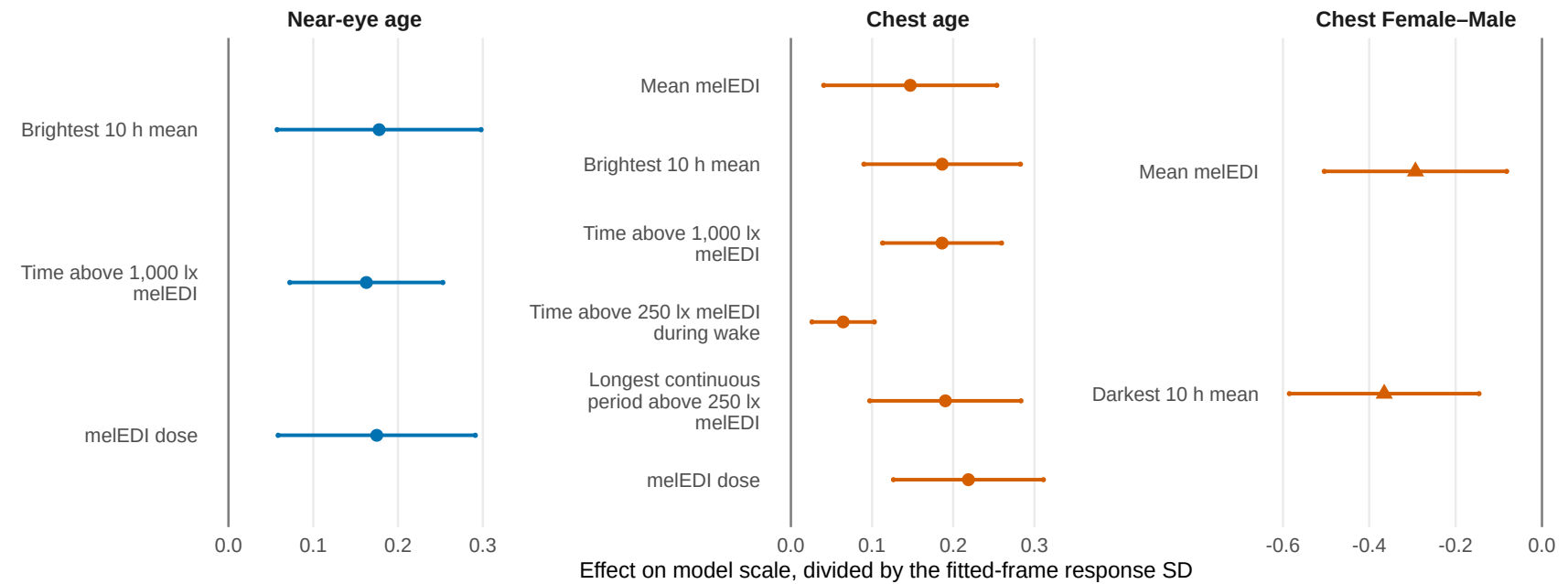
A Participant age distribution in each fitted placement sample

One point per participant; sites follow submitted-manuscript order and colours (near eye n = 141; chest n = 154)



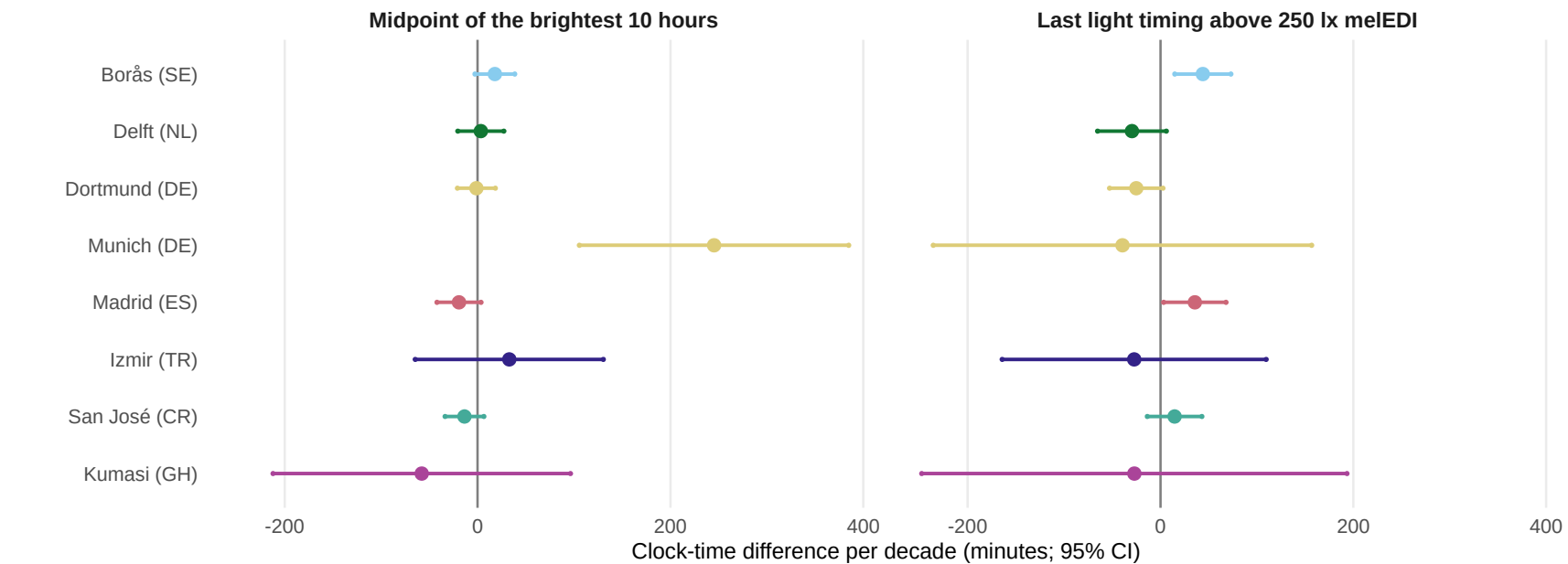
B All main associations retained after FDR adjustment

Standardized fitted-frame effects and 95% confidence intervals; 11 retained across separate 17-metric FDR families



C Site-specific age estimates for two retained age-by-site interactions

Chest placement; site-specific estimates are descriptive components of the two omnibus age-by-site interactions, not separate site-level tests



Main-effect standardization is for display only; practical estimates, exact samples, and FDR-adjusted p-values are reported in the accompanying table.

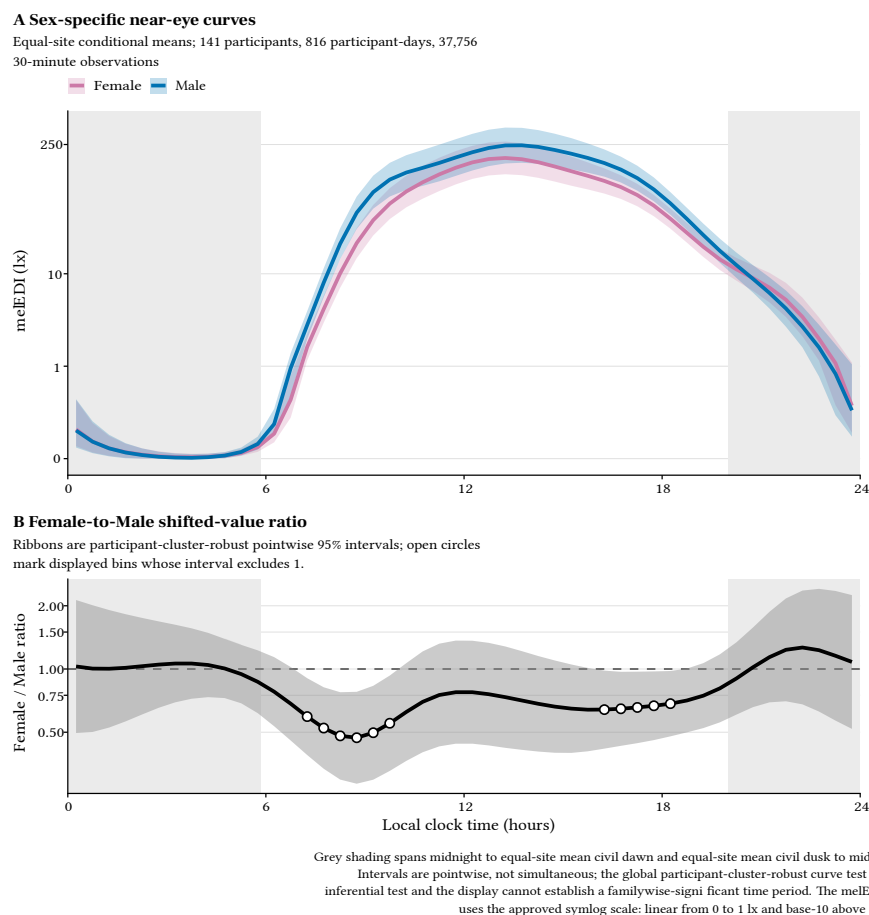
1391 *Supplementary Table S14. Age and biological sex*

Placement and role	Predictor	Metric	Practical association (95% CI)	FDR-adjusted p	Sample
Near eye (primary)	Age, per 10 years	Brightest 10 h mean	1.31× (1.09–1.58)	0.018	141 participants; 816 participant-days/observations
Near eye (primary)	Age, per 10 years	Time above 1,000 lx melEDI	1.27× (1.11–1.45)	0.009	141 participants; 816 participant-days/observations
Near eye (primary)	Age, per 10 years	melEDI dose	1.31× (1.09–1.56)	0.018	141 participants; 761 participant-days/observations
Chest (complementary)	Age, per 10 years	Mean melEDI	1.16× (1.04–1.30)	0.017	154 participants; 902 participant-days/observations
Chest (complementary)	Age, per 10 years	Brightest 10 h mean	1.35× (1.15–1.57)	<0.001	154 participants; 902 participant-days/observations
Chest (complementary)	Age, per 10 years	Time above 1,000 lx melEDI	1.31× (1.18–1.46)	<0.001	154 participants; 902 participant-days/observations
Chest (complementary)	Age, per 10 years	Time above 250 lx melEDI during wake	1.16× (1.06–1.26)	0.004	154 participants; 818 participant-days/observations
Chest (complementary)	Age, per 10 years	Longest continuous period above 250 lx melEDI	1.18× (1.09–1.27)	<0.001	154 participants; 902 participant-days/observations
Chest (complementary)	Age, per 10 years	melEDI dose	1.40× (1.21–1.61)	<0.001	154 participants; 851 participant-days/observations
Chest (complementary)	Measured biological sex, Female minus Male	Mean melEDI	0.74× (0.60–0.92)	0.048	154 participants; 902 participant-days/observations
Chest (complementary)	Measured biological sex, Female minus Male	Darkest 10 h mean	0.76× (0.65–0.90)	0.016	154 participants; 902 participant-days/observations

Associations are site-adjusted estimates per 10-year age increase or Female minus Male. Ratios are back-transformed to the practical scale. Each FDR-adjusted p-value belongs to its sensor-position- and predictor-specific complete 17-test family; bold values meet the FDR-adjusted p < 0.050 rule. Samples give the exact fitted participants and participant-day observations for these retained daily-response results.

1393 **Supplementary Figure S17 and Table S15. Biological-sex-specific**
1394 **daily curves**

1395



1396

1397 Supplementary Figure S17. Near-eye biological-sex-specific fitted melanopic equivalent
1398 daylight illuminance curves and Female-to-Male shifted-value ratio. Ribbons and contrast
1399 intervals are participant-cluster-robust 95% confidence intervals at the displayed clock times.
1400 Grey shading gives site-average civil-night context with each site weighted equally and is not
1401 a model covariate. Biological sex, coded Female or Male, was analysed; gender was
1402 recorded separately and was not analysed.

1403 *Supplementary Table S15. Biological-sex-specific daily patterns and global tests*

	Role	Test	F statistic	Numerator df	Denominator df	Raw p	FDR-adjusted p	Support
Near eye	Primary	Complete Female-minus-Male 24-hour curve	20.946	9.692	131	0.028	0.028	Supported
Chest	Complementary	Complete Female-minus-Male 24-hour curve	4.876	1.003	152	0.029	0.029	Supported

Primary near-eye fitted sample: 141 participants, 816 participant-days, 37,756 30-minute observations, and 9 sites. Complementary chest fitted sample: 154 participants, 902 participant-days, 41,842 30-minute observations, and 8 sites.

Each row is one joint complete-curve test. Raw p is bold at raw p < 0.050; FDR-adjusted p is bold independently at adjusted p < 0.050. Each labelled global family contains one test, so the stored raw and adjusted values coincide.

1404

1405 **Measurement position, robustness, and reproducibility**

1406 Near-eye and complementary chest measurements remain separate sensor-position
1407 estimands. Common-sample analyses use the same participants and participant-days at
1408 both positions and fit the positions separately. They do not pool positions, test equivalence,
1409 or establish a universal correction. Detailed model checks, sensitivities, preregistration
1410 deviations, exact formulas, package versions, and source documentation are provided in the
1411 analysis reports and reproducibility package.